Quantum Key Distribution

Lecture notes

(based on R. Wolf, *Quantum Key Distribution*)

Zayn

Contents

1 Cryptographic background **3**
1.1 Classical cryptography . . . 3
1.2 Information-theoretic security . . . 3

2 Mathematical tools **4**
2.1 Hilbert spaces and operators . . . 4
2.2 Quantum states . . . 5
2.3 Quantum channels . . . 6
2.4 Measurements . . . 8
2.5 Composite systems and entanglement . . . 8
2.6 The no-cloning theorem . . . 10
2.7 Purification . . . 10
2.8 Distance measures . . . 11

3 Information and entropies **13**
3.1 Classical entropies . . . 13
3.2 Quantum entropies . . . 15
3.3 One-shot entropies . . . 18
3.4 Entropic uncertainty relations . . . 20

4 Quantum key distribution protocols **21**
4.1 Setting . . . 21
4.2 Prepare-and-measure protocols . . . 22
4.3 Entanglement-based protocols . . . 24
4.4 From prepare-and-measure to entanglement-based protocols . . . 26
4.5 Classical post-processing . . . 26

5 Security analysis **29**
5.1 Definition of security . . . 29
5.2 Assumptions . . . 31
5.3 Eavesdropping strategies . . . 31
5.4 Security of BB84 . . . 34
5.5 Modern techniques . . . 37
5.6 Finite-key analysis . . . 38

6 Device-independent QKD **40**
6.1 Black boxes and Bell inequalities . . . 40
6.2 DIQKD protocols and security . . . 41
6.3 Loopholes . . . 44

7 Practical QKD **44**
7.1 Practical challenges . . . 44
7.2 Detector side channels and MDI QKD . . . 45
7.3 Continuous-variable QKD . . . 47
7.4 Experimental status . . . 50

A Basic probability theory **50**
A.1 Definitions . . . 50

B Calderbank–Shor–Steane codes **51**
B.1 Classical linear codes 51
B.2 Quantum error correction 52

1 Cryptographic background

1.1 Classical cryptography

Definition 1.1.1: A *symmetric encryption scheme* consists of finite sets $\mathcal{K}$ (keys), $\mathcal{M}$ (messages), $\mathcal{C}$ (ciphertexts) and maps

$$\mathrm{Enc} : \mathcal{K} \times \mathcal{M} \to \mathcal{C}, \qquad \mathrm{Dec} : \mathcal{K} \times \mathcal{C} \to \mathcal{M}$$

such that $\mathrm{Dec}_k(\mathrm{Enc}_k(m)) = m$ for all $k \in \mathcal{K}$ and $m \in \mathcal{M}$. Alice and Bob share the key k; the adversary Eve has access to the ciphertext.

Remark 1.1.2 (Kerckhoffs' principle): The security of a scheme must not rely on the secrecy of Enc and Dec: Eve is assumed to know the full protocol, and the only secret is the key.

Remark 1.1.3: In *public-key* (asymmetric) schemes such as RSA, encryption uses a public key and decryption a private key. Their security is *computational*: it relies on the assumed hardness of a mathematical problem (e.g., factoring). Shor's algorithm factors integers in polynomial time on a quantum computer, so such schemes are not secure against a quantum adversary. Moreover, ciphertexts stored today can be decrypted once the hardness assumption fails.

1.2 Information-theoretic security

Definition 1.2.1: Let M, K be independent random variables on $\mathcal{M}$ and $\mathcal{K}$, and $C := \mathrm{Enc}_K(M)$. The scheme is *perfectly secret* if

$$\Pr[M = m \mid C = c] = \Pr[M = m] \qquad \text{for all } m \in \mathcal{M},\ c \in \mathcal{C} \text{ with } \Pr[C = c] > 0,$$

i.e., M and C are independent.

Example 1.2.2 (One-time pad): Let $\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0,1\}^n$, $\mathrm{Enc}_k(m) := m \oplus k$ and $\mathrm{Dec}_k(c) := c \oplus k$, where $\oplus$ denotes bitwise addition modulo 2, and let K be uniformly distributed.

Lemma 1.2.3: The one-time pad is perfectly secret.

Proof. Since K is independent of M and uniform, for all m, c,

$$\Pr[C = c \mid M = m] = \Pr[K = m \oplus c \mid M = m] = \Pr[K = m \oplus c] = 2^{-n}.$$

Hence $\Pr[C = c] = \sum_m \Pr[M = m] \Pr[C = c \mid M = m] = 2^{-n}$, and by Bayes' rule

$$\Pr[M = m \mid C = c] = \frac{\Pr[C = c \mid M = m] \Pr[M = m]}{\Pr[C = c]} = \Pr[M = m]. \qquad \square$$

Theorem 1.2.1: Shannon

If a scheme is perfectly secret for a message distribution with full support on $\mathcal{M}$, then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof. Fix c with $\Pr[C = c] > 0$ and set $\mathcal{M}(c) := \{\mathrm{Dec}_k(c) \mid k \in \mathcal{K}\}$. Every message m with $\Pr[M = m \mid C = c] > 0$ lies in $\mathcal{M}(c)$, because $C = c$ and $M = m$ force $m = \mathrm{Dec}_K(c)$. If $|\mathcal{K}| < |\mathcal{M}|$, then $|\mathcal{M}(c)| \leq |\mathcal{K}| < |\mathcal{M}|$, so there is some $m \notin \mathcal{M}(c)$. For this m we have $\Pr[M = m \mid C = c] = 0 \neq \Pr[M = m]$, a contradiction. $\square$

Remark 1.2.4: A key of the one-time pad must not be reused: if $c = m \oplus k$ and $c' = m' \oplus k$, then $c \oplus c' = m \oplus m'$ is known to Eve. Information-theoretically secure encryption therefore requires as much fresh secret key as there is message. This is the *key distribution problem.*

Remark 1.2.5: *Quantum key distribution* (QKD) solves the key distribution problem: given an insecure quantum channel and an authenticated (but public) classical channel, Alice and Bob generate a shared key whose secrecy is guaranteed by the laws of quantum mechanics alone, without computational assumptions. Authentication itself requires a short pre-shared key, so QKD is, strictly speaking, a *key expansion* protocol.

2 Mathematical tools

2.1 Hilbert spaces and operators

Definition 2.1.1: A *Hilbert space* $\mathcal{H}$ is a complex vector space with an *inner product* $\mathcal{H}\times\mathcal{H}\to\mathbb{C}$, $(\varphi,\psi)\mapsto\langle\varphi|\psi\rangle$, satisfying for all $\varphi,\psi,\chi\in\mathcal{H}$ and $\lambda\in\mathbb{C}$

(i) $\langle\varphi|\psi+\lambda\chi\rangle=\langle\varphi|\psi\rangle+\lambda\langle\varphi|\chi\rangle$ (linearity in the second argument),

(ii) $\langle\varphi|\psi\rangle=\overline{\langle\psi|\varphi\rangle}$,

(iii) $\langle\varphi|\varphi\rangle\geq 0$, with equality if and only if $\varphi=0$,

which is complete with respect to the norm $\|\psi\|:=\sqrt{\langle\psi|\psi\rangle}$. We write vectors as $|\psi\rangle$ and $\langle\psi|$ for the functional $|\varphi\rangle\mapsto\langle\psi|\varphi\rangle$.

Remark 2.1.2: By (i) and (ii), the inner product is antilinear in the first argument: $\langle\lambda\varphi|\psi\rangle=\bar\lambda\langle\varphi|\psi\rangle$. Throughout, Hilbert spaces are finite-dimensional unless stated otherwise (the exception is Section 7.3). Then $\mathcal{H}\cong\mathbb{C}^d$ and completeness is automatic.

Lemma 2.1.3 (Cauchy–Schwarz): $|\langle\varphi|\psi\rangle|^2\leq\langle\varphi|\varphi\rangle\langle\psi|\psi\rangle$ for all $\varphi,\psi\in\mathcal{H}$, with equality if and only if φ and ψ are linearly dependent.

Proof. If $\psi=0$ there is nothing to show. Otherwise let $\lambda:=\langle\psi|\varphi\rangle/\langle\psi|\psi\rangle$. By (iii),

$$0\leq\langle\varphi-\lambda\psi|\varphi-\lambda\psi\rangle=\langle\varphi|\varphi\rangle-\frac{|\langle\psi|\varphi\rangle|^2}{\langle\psi|\psi\rangle},$$

with equality if and only if $\varphi=\lambda\psi$. □

Definition 2.1.4: An *orthonormal basis* of $\mathcal{H}$ is a family $\{|i\rangle\}_{i=1}^d$ with $\langle i|j\rangle=\delta_{i,j}$ spanning $\mathcal{H}$. For $\mathcal{H}=\mathbb{C}^2$ (a *qubit*) we use the *computational basis* (Z-basis) $\{|0\rangle,|1\rangle\}$ and the *Hadamard basis* (X-basis)

$$|\pm\rangle:=\tfrac{1}{\sqrt2}\big(|0\rangle\pm|1\rangle\big).$$

Definition 2.1.5: Let $\mathcal{B}(\mathcal{H}_A,\mathcal{H}_B)$ denote the linear maps $\mathcal{H}_A\to\mathcal{H}_B$ and $\mathcal{B}(\mathcal{H}):=\mathcal{B}(\mathcal{H},\mathcal{H})$. The *adjoint* $A^\dagger\in\mathcal{B}(\mathcal{H}_B,\mathcal{H}_A)$ of $A\in\mathcal{B}(\mathcal{H}_A,\mathcal{H}_B)$ is defined by $\langle\varphi|A\psi\rangle=\langle A^\dagger\varphi|\psi\rangle$. An operator $A\in\mathcal{B}(\mathcal{H})$ is

(i) *Hermitian* if $A=A^\dagger$,

(ii) *unitary* if $A^\dagger A=AA^\dagger=\mathbb{I}$,

(iii) *positive semidefinite*, $A\geq 0$, if $\langle\psi|A|\psi\rangle\geq 0$ for all $|\psi\rangle\in\mathcal{H}$,

(iv) a *projector* if $A=A^\dagger=A^2$.

A map $V\in\mathcal{B}(\mathcal{H}_A,\mathcal{H}_B)$ with $V^\dagger V=\mathbb{I}_A$ is an *isometry*. We write $A\geq B$ if $A-B\geq 0$. The *trace* is $\operatorname{Tr}A:=\sum_i\langle i|A|i\rangle$ for any orthonormal basis $\{|i\rangle\}$.

Lemma 2.1.6: (i) $(AB)^\dagger=B^\dagger A^\dagger$ and $(|\varphi\rangle\langle\psi|)^\dagger=|\psi\rangle\langle\varphi|$.

(ii) A Hermitian operator has real eigenvalues and an orthonormal eigenbasis; $A \geq 0$ if and only if A is Hermitian with non-negative eigenvalues.

(iii) The eigenvalues of a projector are 0 and 1.

(iv) $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$; in particular, the trace does not depend on the basis.

Proof. (i) $\langle\varphi|AB\psi\rangle = \langle A^\dagger\varphi|B\psi\rangle = \langle B^\dagger A^\dagger\varphi|\psi\rangle$; similarly for the outer product. (ii) is the spectral theorem; note that $\langle\psi|A|\psi\rangle \in \mathbb{R}$ for all ψ forces $A = A^\dagger$ by polarization. (iii) If $P|v\rangle = \lambda|v\rangle$, then $\lambda|v\rangle = P|v\rangle = P^2|v\rangle = \lambda^2|v\rangle$, so $\lambda \in \{0,1\}$. (iv) $\operatorname{Tr}(AB) = \sum_{i,j} A_{ij}B_{ji} = \operatorname{Tr}(BA)$; a change of basis $A \mapsto U^\dagger AU$ leaves the trace invariant by cyclicity. □

Example 2.1.7 (Pauli matrices):

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By direct computation they are Hermitian and traceless, $\sigma_j^2 = \mathbb{I}$ (hence unitary), and

$$XY = iZ, \qquad YZ = iX, \qquad ZX = iY, \qquad \sigma_j\sigma_k = -\sigma_k\sigma_j \ (j \neq k).$$

Together with $\mathbb{I}$ they form a basis of $\mathcal{B}(\mathbb{C}^2)$, orthogonal with respect to $(A,B) \mapsto \operatorname{Tr}(A^\dagger B)$. The *Hadamard* gate $H := \frac{1}{\sqrt{2}}(X+Z)$ is Hermitian and unitary, satisfies $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, and $HZH = X$, $HXH = Z$, $HYH = -Y$.

Definition 2.1.8: The *tensor product* $\mathcal{H}_A \otimes \mathcal{H}_B$ of Hilbert spaces with orthonormal bases $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$ is the Hilbert space with orthonormal basis $\{|i\rangle_A \otimes |j\rangle_B\}_{i,j}$ and inner product $\big(\langle\varphi| \otimes \langle\chi|\big)\big(|\psi\rangle \otimes |\omega\rangle\big) = \langle\varphi|\psi\rangle\langle\chi|\omega\rangle$; in particular $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim \mathcal{H}_A \cdot \dim \mathcal{H}_B$. We abbreviate $|i\rangle_A \otimes |j\rangle_B = |i\rangle_A|j\rangle_B = |ij\rangle_{AB}$. For $A \in \mathcal{B}(\mathcal{H}_A)$, $B \in \mathcal{B}(\mathcal{H}_B)$, $(A\otimes B)(|\psi\rangle\otimes|\varphi\rangle) := A|\psi\rangle \otimes B|\varphi\rangle$.

Definition 2.1.9: The *partial trace* $\operatorname{Tr}_B : \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B) \to \mathcal{B}(\mathcal{H}_A)$ is the linear map defined by

$$\operatorname{Tr}_B\big(|a\rangle\langle a'| \otimes |b\rangle\langle b'|\big) := \langle b'|b\rangle\, |a\rangle\langle a'|.$$

Lemma 2.1.10: (i) $\operatorname{Tr}_B M = \sum_j (\mathbb{I}_A \otimes \langle j|_B) M (\mathbb{I}_A \otimes |j\rangle_B)$ for any orthonormal basis $\{|j\rangle_B\}$.

(ii) $\operatorname{Tr}\big(\operatorname{Tr}_B(M)\, A\big) = \operatorname{Tr}\big(M(A\otimes\mathbb{I}_B)\big)$ for all $A \in \mathcal{B}(\mathcal{H}_A)$, and Tr_B is the unique linear map with this property.

(iii) $\operatorname{Tr}_A \operatorname{Tr}_B M = \operatorname{Tr}_B \operatorname{Tr}_A M = \operatorname{Tr} M$.

(iv) $\operatorname{Tr}_B\big((A\otimes\mathbb{I})M(A'\otimes\mathbb{I})\big) = A\, \operatorname{Tr}_B(M)\, A'$ and $\operatorname{Tr}_B\big(M(\mathbb{I}\otimes B)\big) = \operatorname{Tr}_B\big((\mathbb{I}\otimes B)M\big)$.

Proof. All statements are linear in M, so it suffices to check them on $M = |a\rangle\langle a'| \otimes |b\rangle\langle b'|$. (i): $\sum_j \langle j|b\rangle\langle b'|j\rangle = \langle b'|b\rangle$. (ii): both sides equal $\langle b'|b\rangle\langle a'|A|a\rangle$; uniqueness follows since the functionals $X \mapsto \operatorname{Tr}(XA)$, $A \in \mathcal{B}(\mathcal{H}_A)$, separate points. (iii) and (iv) follow directly from (i). □

2.2 Quantum states

Definition 2.2.1: A *density operator* (or *state*) on $\mathcal{H}$ is an operator $\rho \in \mathcal{B}(\mathcal{H})$ with $\rho \geq 0$ and $\operatorname{Tr}\rho = 1$. The set of states is denoted $\mathcal{S}(\mathcal{H})$, and $\mathcal{S}_{\leq}(\mathcal{H})$ denotes the *subnormalized* states, i.e., $\rho \geq 0$ with $\operatorname{Tr}\rho \leq 1$. A state is *pure* if $\rho = |\psi\rangle\langle\psi|$ for a unit vector $|\psi\rangle$, and *mixed* otherwise.

Lemma 2.2.2: Let $\{p_i, |\psi_i\rangle\}_i$ be an *ensemble*, i.e., unit vectors $|\psi_i\rangle$ prepared with probabilities p_i. Then $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ is a density operator. Conversely, every density operator arises in this way (e.g., from its spectral decomposition $\rho = \sum_i \lambda_i |\varphi_i\rangle\langle\varphi_i|$).

Proof. $\langle\varphi|\rho|\varphi\rangle = \sum_i p_i |\langle\varphi|\psi_i\rangle|^2 \geq 0$ and $\operatorname{Tr}\rho = \sum_i p_i \langle\psi_i|\psi_i\rangle = 1$. Conversely, the eigenvalues λ_i of $\rho \geq 0$ are non-negative and sum to $\operatorname{Tr}\rho = 1$. □

Example 2.2.3: The ensembles $\{(\frac{1}{2}, |0\rangle), (\frac{1}{2}, |1\rangle)\}$ and $\{(\frac{1}{2}, |+\rangle), (\frac{1}{2}, |-\rangle)\}$ both give $\rho = \mathbb{I}/2$, the *maximally mixed state*. Since all measurement statistics depend only on ρ, the two ensembles cannot be distinguished. The ensemble $\{(\frac{1}{2}, |0\rangle), (\frac{1}{2}, |+\rangle)\}$ gives

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}.$$

Definition 2.2.4: The *purity* of $\rho \in \mathcal{S}(\mathcal{H})$ is $P(\rho) := \operatorname{Tr}(\rho^2)$.

Lemma 2.2.5: Let $d = \dim \mathcal{H}$. Then $1/d \leq P(\rho) \leq 1$, with $P(\rho) = 1$ if and only if ρ is pure and $P(\rho) = 1/d$ if and only if $\rho = \mathbb{I}/d$.

Proof. Let $\rho = \sum_i \lambda_i |\varphi_i\rangle\langle\varphi_i|$ be the spectral decomposition; then $P(\rho) = \sum_i \lambda_i^2$ with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. Since $\lambda_i^2 \leq \lambda_i$, $P(\rho) \leq 1$ with equality if and only if every $\lambda_i \in \{0, 1\}$, i.e., exactly one $\lambda_i = 1$. By Cauchy–Schwarz in $\mathbb{R}^d$, $1 = (\sum_i \lambda_i)^2 \leq d \sum_i \lambda_i^2$, with equality if and only if all λ_i are equal. □

Lemma 2.2.6 (Bloch sphere): Every qubit state can be written uniquely as

$$\rho = \tfrac{1}{2}\big(\mathbb{I} + r_x X + r_y Y + r_z Z\big), \qquad \vec{r} \in \mathbb{R}^3,\ |\vec{r}| \leq 1.$$

The state is pure if and only if $|\vec{r}| = 1$. Every pure state is, up to a global phase,

$$|\psi\rangle = \cos\tfrac{\theta}{2}|0\rangle + e^{i\phi}\sin\tfrac{\theta}{2}|1\rangle, \qquad 0 \leq \theta \leq \pi,\ 0 \leq \phi < 2\pi,$$

with Bloch vector $\vec{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$.

Proof. $\{\mathbb{I}, X, Y, Z\}$ is a basis of $\mathcal{B}(\mathbb{C}^2)$, orthogonal with respect to $\operatorname{Tr}(A^\dagger B)$, so $\rho = \frac{1}{2}(r_0\mathbb{I} + \vec{r}\cdot\vec{\sigma})$ with $r_0 = \operatorname{Tr}\rho = 1$ and $r_j = \operatorname{Tr}(\rho\sigma_j)$, which are real since ρ and σ_j are Hermitian. Since $(\vec{r}\cdot\vec{\sigma})^2 = |\vec{r}|^2\mathbb{I}$ (by the anticommutation relations), the eigenvalues of $\vec{r}\cdot\vec{\sigma}$ are $\pm|\vec{r}|$ and those of ρ are $\frac{1}{2}(1 \pm |\vec{r}|)$. Hence $\rho \geq 0$ if and only if $|\vec{r}| \leq 1$, and ρ has eigenvalue 0 (i.e., is pure) if and only if $|\vec{r}| = 1$. For the last statement, compute $\langle\psi|\sigma_j|\psi\rangle$ for the given $|\psi\rangle$. □

2.3 Quantum channels

Definition 2.3.1: Let $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ be linear. Then $\mathcal{E}$ is

(i) *trace-preserving* if $\operatorname{Tr}\mathcal{E}(X) = \operatorname{Tr} X$ for all X,

(ii) *positive* if $\mathcal{E}(X) \geq 0$ for all $X \geq 0$,

(iii) *completely positive* if $\mathrm{id}_R \otimes \mathcal{E}$ is positive for every auxiliary system $\mathcal{H}_R$.

A *quantum channel* $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ is a linear, completely positive, trace-preserving (CPTP) map.

Example 2.3.2: The transpose $T(X) := X^T$ on $\mathcal{B}(\mathbb{C}^2)$ is positive (it preserves the spectrum), but not completely positive. Indeed, for $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$,

$$(\mathrm{id} \otimes T)\big(|\Phi^+\rangle\langle\Phi^+|\big) = \frac{1}{2}\sum_{i,j} |i\rangle\langle j| \otimes \big(|i\rangle\langle j|\big)^T = \frac{1}{2}\sum_{i,j} |i\rangle\langle j| \otimes |j\rangle\langle i| = \frac{1}{2}\,\mathrm{SWAP},$$

and SWAP has eigenvalue -1 on $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

Theorem 2.3.1: Kraus representation

A map $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ is a quantum channel if and only if there are operators $K_k \in \mathcal{B}(\mathcal{H}_A, \mathcal{H}_B)$ with $\sum_k K_k^\dagger K_k = \mathbb{I}_A$ such that

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger \qquad \text{for all } \rho.$$

Proof. "⇐": For $X \geq 0$ on $\mathcal{H}_R \otimes \mathcal{H}_A$, $(\mathrm{id} \otimes \mathcal{E})(X) = \sum_k (\mathbb{I} \otimes K_k) X (\mathbb{I} \otimes K_k)^\dagger \geq 0$, and $\mathrm{Tr}\,\mathcal{E}(\rho) = \mathrm{Tr}\big(\sum_k K_k^\dagger K_k \rho\big) = \mathrm{Tr}\,\rho$.

"⇒": Let $\mathcal{H}_R \cong \mathcal{H}_A$ with basis $\{|i\rangle\}$ and $|\Gamma\rangle := \sum_i |i\rangle_R |i\rangle_A$. By complete positivity, the *Choi operator* $J := (\mathrm{id}_R \otimes \mathcal{E})(|\Gamma\rangle\langle\Gamma|)$ is positive, so $J = \sum_k |v_k\rangle\langle v_k|$ for (unnormalized) vectors $|v_k\rangle \in \mathcal{H}_R \otimes \mathcal{H}_B$. For $|\psi\rangle = \sum_i c_i |i\rangle_A$ let $|\bar{\psi}\rangle := \sum_i \bar{c}_i |i\rangle_R$ and define $K_k|\psi\rangle := (\langle\bar{\psi}| \otimes \mathbb{I}_B)|v_k\rangle$, which is linear in $|\psi\rangle$. Since $(\langle\bar{\psi}| \otimes \mathbb{I}_A)|\Gamma\rangle = \sum_i c_i |i\rangle = |\psi\rangle$, and since contracting the system R with $\langle\bar{\psi}|$ commutes with $\mathrm{id}_R \otimes \mathcal{E}$, we get

$$\mathcal{E}(|\psi\rangle\langle\psi|) = (\langle\bar{\psi}| \otimes \mathbb{I}_B)\, J\, (|\bar{\psi}\rangle \otimes \mathbb{I}_B) = \sum_k K_k |\psi\rangle\langle\psi| K_k^\dagger.$$

By linearity this extends to all ρ. Finally, $\mathrm{Tr}\big((\mathbb{I} - \sum_k K_k^\dagger K_k)\rho\big) = 0$ for all ρ implies $\sum_k K_k^\dagger K_k = \mathbb{I}_A$. □

Corollary 2.3.3 (Stinespring dilation): Every channel is of the form $\mathcal{E}(\rho) = \mathrm{Tr}_E(V \rho V^\dagger)$ for an isometry $V : \mathcal{H}_A \to \mathcal{H}_B \otimes \mathcal{H}_E$.

Proof. Set $V|\psi\rangle := \sum_k K_k |\psi\rangle \otimes |k\rangle_E$. Then $V^\dagger V = \sum_k K_k^\dagger K_k = \mathbb{I}_A$ and $\mathrm{Tr}_E(V\rho V^\dagger) = \sum_k K_k \rho K_k^\dagger$. □

Remark 2.3.4: E is the *environment*; in cryptography it is, in the worst case, held by Eve. The *adjoint* map $\mathcal{E}^\dagger(Y) := \sum_k K_k^\dagger Y K_k$, defined by $\mathrm{Tr}(Y\mathcal{E}(X)) = \mathrm{Tr}(\mathcal{E}^\dagger(Y)X)$, is completely positive and *unital*, $\mathcal{E}^\dagger(\mathbb{I}) = \mathbb{I}$. In particular $0 \leq Y \leq \mathbb{I}$ implies $0 \leq \mathcal{E}^\dagger(Y) \leq \mathbb{I}$.

Example 2.3.5: (i) *Unitary channel*: $\mathcal{E}(\rho) = U\rho U^\dagger$; *isometric channel*: $\mathcal{E}(\rho) = V\rho V^\dagger$.

(ii) *Depolarizing channel*: $\mathcal{E}_p(\rho) = (1-p)\rho + p\,\mathbb{I}/2$ on a qubit. Using $\frac{\mathbb{I}}{2} = \frac{1}{4}(\rho + X\rho X + Y\rho Y + Z\rho Z)$ for every qubit state ρ, it has Kraus operators

$$K_0 = \sqrt{1 - \tfrac{3p}{4}}\,\mathbb{I}, \qquad K_j = \sqrt{\tfrac{p}{4}}\,\sigma_j \ (j = x, y, z).$$

It shrinks the Bloch vector, $\vec{r} \mapsto (1-p)\vec{r}$.

(iii) *Bit-flip channel*: $\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$; replacing X by Z gives the *phase-flip channel*.

(iv) *Amplitude damping*: Kraus operators $K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|$ and $K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|$. For $\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}$,

$$\mathcal{E}(\rho) = \begin{pmatrix} \rho_{00} + \gamma\rho_{11} & \sqrt{1-\gamma}\,\rho_{01} \\ \sqrt{1-\gamma}\,\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix}.$$

The excited state decays to $|0\rangle$ with probability γ, and coherences are damped by $\sqrt{1-\gamma}$; the channel is irreversible.

Verification of (ii). The identity follows from the Bloch representation: $\sigma_j\sigma_k\sigma_j = \sigma_k$ if $j = k$ and $-\sigma_k$ otherwise, so $\sum_{j\in\{0,x,y,z\}} \sigma_j\sigma_k\sigma_j = 0$ for $k \in \{x,y,z\}$ and $= 4\mathbb{I}$ for $\sigma_0 = \mathbb{I}$. Hence $\rho + \sum_j \sigma_j\rho\sigma_j = 4 \cdot \frac{1}{2}\mathbb{I} = 2\mathbb{I}$. Then $(1-p)\rho + \frac{p}{4}(\rho + \sum_j \sigma_j\rho\sigma_j) = (1 - \frac{3p}{4})\rho + \frac{p}{4}\sum_j \sigma_j\rho\sigma_j$. □

2.4 Measurements

Definition 2.4.1: A *measurement* with outcome set $\mathcal{X}$ is a family $\{M_x\}_{x\in\mathcal{X}}$, $M_x \in \mathcal{B}(\mathcal{H}_A, \mathcal{H}_B)$, with $\sum_x M_x^\dagger M_x = \mathbb{I}_A$. On input ρ, outcome x occurs with probability $p(x) = \mathrm{Tr}(M_x \rho M_x^\dagger)$, and the post-measurement state is $M_x \rho M_x^\dagger / p(x)$.

Definition 2.4.2: A *projective measurement* is given by a Hermitian operator (an *observable*) $M = \sum_m m P_m$ with spectral projectors P_m, $\sum_m P_m = \mathbb{I}$. Outcome m occurs with probability $p(m) = \mathrm{Tr}(P_m \rho)$, the post-measurement state is $P_m \rho P_m / p(m)$, and the expectation value is $\langle M \rangle_\rho = \sum_m m\, p(m) = \mathrm{Tr}(M\rho)$.

Example 2.4.3: Measuring $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives $+1$ with probability $|\alpha|^2$ and -1 with probability $|\beta|^2$, leaving $|0\rangle$ or $|1\rangle$; $\langle Z\rangle = |\alpha|^2 - |\beta|^2$. Measuring Z on $|\pm\rangle$ gives each outcome with probability $\frac{1}{2}$.

Definition 2.4.4: A *positive operator-valued measure* (POVM) with outcome set $\mathcal{X}$ is a family $\{E_x\}_{x\in\mathcal{X}}$ with $E_x \geq 0$ and $\sum_x E_x = \mathbb{I}$. Outcome x occurs with probability $p(x) = \mathrm{Tr}(E_x \rho)$.

Remark 2.4.5: Every measurement $\{M_x\}$ defines a POVM via $E_x := M_x^\dagger M_x$. Conversely, every POVM can be realized as a projective measurement on a larger Hilbert space (Naimark): with the isometry $V|\psi\rangle := \sum_x \sqrt{E_x}|\psi\rangle \otimes |x\rangle$, one has $\mathrm{Tr}(E_x\rho) = \mathrm{Tr}\big((\mathbb{I}\otimes|x\rangle\langle x|)V\rho V^\dagger\big)$. POVMs suffice whenever only outcome statistics matter.

Lemma 2.4.6: Let $|\psi_1\rangle, |\psi_2\rangle$ be non-orthogonal states. There is no POVM $\{E_1, E_2\}$ with $\langle\psi_1|E_1|\psi_1\rangle = 1$ and $\langle\psi_2|E_2|\psi_2\rangle = 1$.

Proof. Assume such a POVM exists. Then $\langle\psi_1|E_2|\psi_1\rangle = 1 - \langle\psi_1|E_1|\psi_1\rangle = 0$, hence $\|\sqrt{E_2}|\psi_1\rangle\|^2 = 0$. Write $|\psi_2\rangle = \alpha|\psi_1\rangle + \beta|\varphi\rangle$ with $|\varphi\rangle \perp |\psi_1\rangle$ a unit vector; since $\alpha = \langle\psi_1|\psi_2\rangle \neq 0$, $|\beta|^2 = 1 - |\alpha|^2 < 1$. Then $\sqrt{E_2}|\psi_2\rangle = \beta\sqrt{E_2}|\varphi\rangle$ and, using $E_2 \leq \mathbb{I}$,

$$1 = \langle\psi_2|E_2|\psi_2\rangle = |\beta|^2\langle\varphi|E_2|\varphi\rangle \leq |\beta|^2 < 1,$$

a contradiction. □

Example 2.4.7 (Unambiguous discrimination): For the states $|0\rangle$ and $|+\rangle$ consider the POVM

$$E_1 = c\,|1\rangle\langle 1|, \qquad E_2 = c\,|-\rangle\langle -|, \qquad E_3 = \mathbb{I} - E_1 - E_2, \qquad c = \tfrac{\sqrt{2}}{1+\sqrt{2}}.$$

The value of c is the largest one for which $E_3 \geq 0$: the matrix $|1\rangle\langle 1| + |-\rangle\langle -|$ has largest eigenvalue $1 + \frac{1}{\sqrt{2}}$, and $c(1 + \frac{1}{\sqrt{2}}) = 1$. Since $\langle 0|E_1|0\rangle = 0$ and $\langle +|E_2|+\rangle = 0$, outcome 1 proves the state was $|+\rangle$, outcome 2 proves it was $|0\rangle$, and outcome 3 is inconclusive. The success probability is $\langle +|E_1|+\rangle = \langle 0|E_2|0\rangle = c/2 \approx 0.29$.

2.5 Composite systems and entanglement

Definition 2.5.1: Let $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$. The *reduced* (or *marginal*) states are $\rho_A := \mathrm{Tr}_B\, \rho_{AB}$ and $\rho_B := \mathrm{Tr}_A\, \rho_{AB}$. By Lemma 2.1.10(ii), ρ_A reproduces all statistics of measurements on A alone.

Definition 2.5.2: A pure state $|\psi\rangle_{AB}$ is a *product state* if $|\psi\rangle_{AB} = |\varphi\rangle_A \otimes |\chi\rangle_B$, and *entangled* otherwise. A mixed state ρ_{AB} is *separable* if

$$\rho_{AB} = \sum_i p_i\, \rho_A^i \otimes \rho_B^i$$

for a probability distribution $\{p_i\}$ and states ρ_A^i, ρ_B^i, and *entangled* otherwise.

Example 2.5.3 (Bell states): The four states

$$|\Phi^{\pm}\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle \pm |11\rangle\big), \qquad |\Psi^{\pm}\rangle = \tfrac{1}{\sqrt{2}}\big(|01\rangle \pm |10\rangle\big)$$

form an orthonormal basis of $\mathbb{C}^2\otimes\mathbb{C}^2$, the *Bell basis*: orthonormality follows from $\langle ij|kl\rangle = \delta_{ik}\delta_{jl}$, and four orthonormal vectors in a four-dimensional space form a basis. All reduced states equal $\mathbb{I}/2$, e.g. $\operatorname{Tr}_B |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|)$; they are *maximally entangled*. Applying a Pauli operator to the second qubit permutes them:

$$(\mathbb{I}\otimes X)|\Phi^+\rangle = |\Psi^+\rangle, \quad (\mathbb{I}\otimes Z)|\Phi^+\rangle = |\Phi^-\rangle, \quad (\mathbb{I}\otimes Y)|\Phi^+\rangle = i|\Psi^-\rangle.$$

Moreover, $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$ and $|\Psi^-\rangle = -\frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle)$.

Lemma 2.5.4 (Transpose trick): Let $|\Gamma\rangle = \sum_i |ii\rangle$. Then $(A\otimes\mathbb{I})|\Gamma\rangle = (\mathbb{I}\otimes A^T)|\Gamma\rangle$ and $\langle\Gamma|(A\otimes B)|\Gamma\rangle = \operatorname{Tr}(A^T B)$. Consequently, for unit vectors $\vec{a},\vec{b}\in\mathbb{R}^3$,

$$\langle\Phi^+|(\vec{a}\cdot\vec{\sigma})\otimes(\vec{b}\cdot\vec{\sigma})|\Phi^+\rangle = a_x b_x - a_y b_y + a_z b_z, \qquad \langle\Psi^-|(\vec{a}\cdot\vec{\sigma})\otimes(\vec{b}\cdot\vec{\sigma})|\Psi^-\rangle = -\vec{a}\cdot\vec{b}.$$

Proof. $(A\otimes\mathbb{I})|\Gamma\rangle = \sum_{i,j} A_{ji}|j\rangle|i\rangle = \sum_{i,j}|j\rangle\otimes A^T_{ij}|i\rangle = (\mathbb{I}\otimes A^T)|\Gamma\rangle$. Then $\langle\Gamma|(A\otimes B)|\Gamma\rangle = \langle\Gamma|(\mathbb{I}\otimes A^T B)|\Gamma\rangle = \operatorname{Tr}(A^T B)$. With $|\Phi^+\rangle = |\Gamma\rangle/\sqrt{2}$, $X^T = X$, $Y^T = -Y$, $Z^T = Z$ and $\operatorname{Tr}(\sigma_j\sigma_k) = 2\delta_{jk}$ the first formula follows. For $|\Psi^-\rangle$: it is the unique antisymmetric state, and $(U\otimes U)|\Psi^-\rangle = \det(U)|\Psi^-\rangle$ for $U\in\mathrm{U}(2)$. Taking $U=\sigma_j$ (with $\det = -1$) gives $(\sigma_j\otimes\mathbb{I})|\Psi^-\rangle = -(\mathbb{I}\otimes\sigma_j)|\Psi^-\rangle$, hence $\langle\Psi^-|\sigma_j\otimes\sigma_k|\Psi^-\rangle = -\langle\Psi^-|\mathbb{I}\otimes\sigma_j\sigma_k|\Psi^-\rangle = -\frac{1}{2}\operatorname{Tr}(\sigma_j\sigma_k) = -\delta_{jk}$. □

Theorem 2.5.1: Schmidt decomposition

Let $|\psi\rangle_{AB}\in\mathcal{H}_A\otimes\mathcal{H}_B$ be a unit vector. Then there are orthonormal families $\{|i\rangle_A\}$, $\{|i\rangle_B\}$ and $\lambda_i > 0$ with $\sum_i \lambda_i = 1$ such that

$$|\psi\rangle_{AB} = \sum_{i=1}^{r} \sqrt{\lambda_i}\,|i\rangle_A|i\rangle_B.$$

The number r is called the *Schmidt rank*.

Proof. Write $|\psi\rangle = \sum_{j,k} c_{jk}|j\rangle_A|k\rangle_B$ in product bases and take a singular value decomposition $c = UDV$, with U, V unitary and D diagonal with entries $d_i \geq 0$. Then

$$|\psi\rangle = \sum_i d_i \Big(\sum_j U_{ji}|j\rangle_A\Big)\otimes\Big(\sum_k V_{ik}|k\rangle_B\Big) =: \sum_i d_i|i\rangle_A|i\rangle_B,$$

where the new families are orthonormal because U and V are unitary. Set $\lambda_i := d_i^2$ for $d_i > 0$; normalization gives $\sum_i \lambda_i = 1$. □

Corollary 2.5.5: Let $|\psi\rangle_{AB}$ be pure. Then

(i) $\rho_A = \sum_i \lambda_i |i\rangle\langle i|_A$ and $\rho_B = \sum_i \lambda_i |i\rangle\langle i|_B$; in particular they have the same non-zero eigenvalues,

(ii) $|\psi\rangle_{AB}$ is entangled if and only if its Schmidt rank is > 1, if and only if ρ_A is mixed.

Proof. (i) $\operatorname{Tr}_B |i\rangle\langle j|_A\otimes|i\rangle\langle j|_B = \delta_{ij}|i\rangle\langle j|_A$. (ii) A product state has Schmidt rank 1; conversely, rank 1 means $|\psi\rangle = |1\rangle_A|1\rangle_B$. By (i), the rank of ρ_A equals the Schmidt rank. □

Lemma 2.5.6 (Peres–Horodecki criterion): If ρ_{AB} is separable, then its partial transpose $\rho_{AB}^{T_B} := (\mathrm{id}_A\otimes T)(\rho_{AB})$ is positive semidefinite.

Proof. $\rho_{AB}^{T_B} = \sum_i p_i \rho_A^i \otimes (\rho_B^i)^T$, and $(\rho_B^i)^T \geq 0$ since it has the same eigenvalues as ρ_B^i. Hence it is a sum of positive operators. □

Example 2.5.7: $|\Phi^+\rangle\langle\Phi^+|^{T_B} = \frac{1}{2}\mathrm{SWAP}$ has eigenvalue $-\frac{1}{2}$, so $|\Phi^+\rangle$ is entangled (as is every Bell state). For $\dim \mathcal{H}_A \cdot \dim \mathcal{H}_B \leq 6$, positivity of the partial transpose is also sufficient for separability; in general deciding separability is hard.

Definition 2.5.8: A *classical–quantum* (cq) state on $\mathcal{H}_X \otimes \mathcal{H}_B$ is a state of the form

$$\rho_{XB} = \sum_{x \in \mathcal{X}} p(x) \, |x\rangle\langle x|_X \otimes \rho_B^x,$$

where $\{|x\rangle\}$ is an orthonormal basis. It describes a classical random variable X correlated with a quantum system B in state ρ_B^x conditioned on $X = x$. Measuring A of ρ_{AB} with a POVM $\{E_x\}$ and storing the outcome in a register X yields the cq-state with $p(x)\rho_B^x = \mathrm{Tr}_A\big((E_x \otimes \mathbb{I})\rho_{AB}\big)$.

Lemma 2.5.9 (No-signalling): Local operations $\mathcal{E}_A \otimes \mathrm{id}_B$ do not change ρ_B: $\mathrm{Tr}_A\big((\mathcal{E}_A \otimes \mathrm{id}_B)(\rho_{AB})\big) = \rho_B$.

Proof. With Kraus operators K_k of $\mathcal{E}_A$ and Lemma 2.1.10,

$$\mathrm{Tr}_A \sum_k (K_k \otimes \mathbb{I})\rho_{AB}(K_k^\dagger \otimes \mathbb{I}) = \mathrm{Tr}_A\Big(\big(\textstyle\sum_k K_k^\dagger K_k \otimes \mathbb{I}\big)\rho_{AB}\Big) = \mathrm{Tr}_A \, \rho_{AB}. \qquad \square$$

Remark 2.5.10: In particular, Alice's choice of measurement on her half of an entangled state cannot be detected by Bob: entanglement cannot be used to signal.

2.6 The no-cloning theorem

Theorem 2.6.1: No-cloning

There is no unitary U on $\mathcal{H} \otimes \mathcal{H}$ and state $|s\rangle$ such that $U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}$.

Proof. Suppose $U(|\psi\rangle|s\rangle) = |\psi\rangle|\psi\rangle$ and $U(|\varphi\rangle|s\rangle) = |\varphi\rangle|\varphi\rangle$. Taking the inner product of both relations and using $U^\dagger U = \mathbb{I}$ gives $\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$, hence $\langle\psi|\varphi\rangle \in \{0, 1\}$. This fails, e.g., for $|0\rangle$ and $|+\rangle$, where $\langle 0|+\rangle = \frac{1}{\sqrt{2}}$. □

Remark 2.6.1: Only states from a fixed orthonormal basis can be copied. A non-orthogonal set of states (e.g., the four BB84 states, with $|\langle 0|+\rangle| = \frac{1}{\sqrt{2}}$) can neither be copied nor perfectly distinguished (Lemma 2.4.6). This is the basic reason why an eavesdropper cannot obtain information without disturbing the states (cf. Lemma 5.3.1).

2.7 Purification

Definition 2.7.1: A *purification* of $\rho_A \in \mathcal{S}(\mathcal{H}_A)$ is a pure state $|\psi\rangle_{RA} \in \mathcal{H}_R \otimes \mathcal{H}_A$ with $\mathrm{Tr}_R \, |\psi\rangle\langle\psi|_{RA} = \rho_A$.

Lemma 2.7.2: (i) Every $\rho_A = \sum_i p_i |i\rangle\langle i|_A$ has the purification $|\psi\rangle_{RA} = \sum_i \sqrt{p_i} \, |i\rangle_R |i\rangle_A$, with $\dim \mathcal{H}_R = \mathrm{rank} \, \rho_A$.

(ii) If $|\psi\rangle_{RA}$ and $|\psi'\rangle_{R'A}$ are two purifications of ρ_A with $\dim \mathcal{H}_R \leq \dim \mathcal{H}_{R'}$, then there is an isometry $V : \mathcal{H}_R \to \mathcal{H}_{R'}$ with $|\psi'\rangle_{R'A} = (V \otimes \mathbb{I}_A)|\psi\rangle_{RA}$.

Proof. (i) $\operatorname{Tr}_R |\psi\rangle\langle\psi| = \sum_{i,j} \sqrt{p_i p_j}\langle j|i\rangle |i\rangle\langle j| = \rho_A$. (ii) Let $\{|i\rangle_A\}$ be an eigenbasis of ρ_A with eigenvalues p_i. Write $|\psi\rangle = \sum_i |\tilde{r}_i\rangle_R |i\rangle_A$ with $|\tilde{r}_i\rangle := (\mathbb{I}\otimes\langle i|)|\psi\rangle$. Then $\langle\tilde{r}_j|\tilde{r}_i\rangle = \langle i|\rho_A|j\rangle = p_i\delta_{ij}$, so $|\psi\rangle = \sum_{p_i>0}\sqrt{p_i}|r_i\rangle|i\rangle$ with orthonormal $|r_i\rangle$. Similarly $|\psi'\rangle = \sum_{p_i>0}\sqrt{p_i}|r'_i\rangle|i\rangle$. Any isometry V with $V|r_i\rangle = |r'_i\rangle$ (which exists since $\dim\mathcal{H}_R \le \dim\mathcal{H}_{R'}$) does the job. □

Example 2.7.3: Each Bell state is a purification of $\mathbb{I}/2$; by Lemma 2.7.2(ii) they are related by unitaries on the purifying system, as in Example 2.5.3.

Remark 2.7.4: Any extension ρ_{ABE} of ρ_{AB} (i.e., $\operatorname{Tr}_E \rho_{ABE} = \rho_{AB}$) is obtained from a purification $|\psi\rangle_{ABE'}$ by a channel acting on E': purify ρ_{ABE} to $|\varphi\rangle_{ABEF}$; this is a purification of ρ_{AB}, so $|\varphi\rangle = (\mathbb{I}\otimes V)|\psi\rangle$ by Lemma 2.7.2, and $\rho_{ABE} = \operatorname{Tr}_F(\mathbb{I}\otimes V)|\psi\rangle\langle\psi|(\mathbb{I}\otimes V^\dagger)$. Since Eve is not restricted, we always assume that she holds a purification of the state of Alice and Bob: this is the worst case.

2.8 Distance measures

Definition 2.8.1: The *trace norm* of $A \in \mathcal{B}(\mathcal{H})$ is $\|A\|_1 := \operatorname{Tr}|A|$, where $|A| := \sqrt{A^\dagger A}$. The *trace distance* of $\rho,\sigma \in \mathcal{S}(\mathcal{H})$ is

$$D(\rho,\sigma) := \tfrac{1}{2}\|\rho-\sigma\|_1.$$

For Hermitian A with eigenvalues a_i, $\|A\|_1 = \sum_i |a_i|$. The trace norm satisfies the triangle inequality, and $0 \le D(\rho,\sigma) \le 1$.

Lemma 2.8.2: For $\rho,\sigma \in \mathcal{S}(\mathcal{H})$,

$$D(\rho,\sigma) = \max_{0\le\Lambda\le\mathbb{I}} \operatorname{Tr}\big(\Lambda(\rho-\sigma)\big).$$

Consequently, if ρ or σ is prepared with probability $\frac{1}{2}$ each, the optimal probability of guessing correctly which one was prepared is $\frac{1}{2}\big(1 + D(\rho,\sigma)\big)$.

Proof. Decompose $\rho-\sigma = P - N$ with $P, N \ge 0$ supported on orthogonal subspaces (spectral decomposition). Since $\operatorname{Tr}(\rho-\sigma) = 0$, $\operatorname{Tr}P = \operatorname{Tr}N$, so $\|\rho-\sigma\|_1 = \operatorname{Tr}P + \operatorname{Tr}N = 2\operatorname{Tr}P$. For $0 \le \Lambda \le \mathbb{I}$, $\operatorname{Tr}\Lambda(\rho-\sigma) \le \operatorname{Tr}\Lambda P \le \operatorname{Tr}P$, with equality for Λ the projector onto the support of P. A guessing strategy is a POVM $\{\Lambda, \mathbb{I}-\Lambda\}$ (guess "ρ" on outcome Λ); its success probability is $\frac{1}{2}\operatorname{Tr}\Lambda\rho + \frac{1}{2}\operatorname{Tr}(\mathbb{I}-\Lambda)\sigma = \frac{1}{2} + \frac{1}{2}\operatorname{Tr}\Lambda(\rho-\sigma)$. □

Theorem 2.8.1: Monotonicity of the trace distance

Let $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ be CPTP and $\rho,\sigma \in \mathcal{S}(\mathcal{H}_A)$. Then

$$\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_1 \le \|\rho-\sigma\|_1.$$

Proof. By Lemma 2.8.2 there is $0 \le \Lambda \le \mathbb{I}$ with $\frac{1}{2}\|\mathcal{E}(\rho)-\mathcal{E}(\sigma)\|_1 = \operatorname{Tr}\Lambda\mathcal{E}(\rho-\sigma) = \operatorname{Tr}\mathcal{E}^\dagger(\Lambda)(\rho-\sigma)$. Since $0 \le \mathcal{E}^\dagger(\Lambda) \le \mathbb{I}$, Lemma 2.8.2 bounds this by $\frac{1}{2}\|\rho-\sigma\|_1$. □

Definition 2.8.3: The *fidelity* of $\rho,\sigma \in \mathcal{S}(\mathcal{H})$ is

$$F(\rho,\sigma) := \left(\operatorname{Tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}}\right)^2 = \left\|\sqrt{\rho}\sqrt{\sigma}\right\|_1^2.$$

For pure states, $F(|\psi\rangle\langle\psi|,\rho) = \langle\psi|\rho|\psi\rangle$ and $F(|\psi\rangle\langle\psi|,|\varphi\rangle\langle\varphi|) = |\langle\psi|\varphi\rangle|^2$.

Remark 2.8.4: The second expression holds because $|\sqrt{\rho}\sqrt{\sigma}| = \sqrt{\sqrt{\sigma}\rho\sqrt{\sigma}}$ and $\|M\|_1 = \|M^\dagger\|_1$, which also shows $F(\rho,\sigma) = F(\sigma,\rho)$. For $\rho = |\psi\rangle\langle\psi|$, $\sqrt{\rho} = \rho$ and $\rho\sigma\rho = \langle\psi|\sigma|\psi\rangle\rho$, which gives the pure-state formulas. Some authors use the square-root fidelity $\sqrt{F}$; here F always denotes the squared version.

Theorem 2.8.2: Uhlmann

Let $\rho_A, \sigma_A \in \mathcal{S}(\mathcal{H}_A)$ and let $\mathcal{H}_R$ with $\dim \mathcal{H}_R \geq \dim \mathcal{H}_A$. Then

$$F(\rho_A, \sigma_A) = \max_{|\psi_\rho\rangle, |\varphi_\sigma\rangle} |\langle \psi_\rho | \varphi_\sigma \rangle|^2,$$

where the maximum is over all purifications of ρ_A and σ_A in $\mathcal{H}_R \otimes \mathcal{H}_A$. Moreover, for any fixed purification $|\psi_\rho\rangle$ the maximum is attained by some purification $|\varphi_\sigma\rangle$.

Proof. By Lemma 2.7.2 it suffices to take $\mathcal{H}_R \cong \mathcal{H}_A$ (an isometry on R does not change overlaps). Let $|\Gamma\rangle = \sum_i |ii\rangle_{RA}$. The vector $(\mathbb{I} \otimes \sqrt{\rho})|\Gamma\rangle$ purifies ρ, so every purification is $|\psi_\rho\rangle = (U \otimes \sqrt{\rho})|\Gamma\rangle$ and similarly $|\varphi_\sigma\rangle = (V \otimes \sqrt{\sigma})|\Gamma\rangle$ with unitaries U, V. By Lemma 2.5.4,

$$\langle \psi_\rho | \varphi_\sigma \rangle = \langle \Gamma | (U^\dagger V \otimes \sqrt{\rho}\sqrt{\sigma}) | \Gamma \rangle = \operatorname{Tr}\big(W \sqrt{\rho}\sqrt{\sigma}\big),$$

where $W := (U^\dagger V)^T$ is unitary. For any unitary W and any M with polar decomposition $M = W_M |M|$, $|\operatorname{Tr}(WM)| = |\operatorname{Tr}(W W_M |M|^{1/2} |M|^{1/2})| \leq \operatorname{Tr}|M|$ by Cauchy–Schwarz for $(A, B) \mapsto \operatorname{Tr}(A^\dagger B)$, with equality for $W = W_M^\dagger$. Hence the maximum of $|\langle \psi_\rho | \varphi_\sigma \rangle|$ is $\|\sqrt{\rho}\sqrt{\sigma}\|_1 = \sqrt{F(\rho, \sigma)}$. For fixed U, choose V such that $(U^\dagger V)^T = W_M^\dagger$. □

Corollary 2.8.5: (i) $0 \leq F(\rho, \sigma) \leq 1$, with $F = 1$ if and only if $\rho = \sigma$, and $F = 0$ if and only if ρ and σ have orthogonal supports.

(ii) $F(\mathcal{E}(\rho), \mathcal{E}(\sigma)) \geq F(\rho, \sigma)$ for every CPTP map $\mathcal{E}$.

Proof. (i) The upper bound and the case of equality follow from Uhlmann's theorem: $|\langle \psi | \varphi \rangle| = 1$ for unit vectors forces $|\varphi\rangle = e^{i\alpha}|\psi\rangle$, hence $\rho = \sigma$. $F = 0$ if and only if $\sqrt{\rho}\sqrt{\sigma} = 0$, i.e., $\rho\sigma = 0$. (ii) Let V be a Stinespring isometry of $\mathcal{E}$ and $|\psi_\rho\rangle, |\varphi_\sigma\rangle$ optimal purifications. Then $(\mathbb{I}_R \otimes V)|\psi_\rho\rangle$ and $(\mathbb{I}_R \otimes V)|\varphi_\sigma\rangle$ are purifications of $\mathcal{E}(\rho)$ and $\mathcal{E}(\sigma)$ (with purifying system RE) with the same overlap. By Uhlmann's theorem, $F(\mathcal{E}(\rho), \mathcal{E}(\sigma)) \geq |\langle \psi_\rho | \varphi_\sigma \rangle|^2 = F(\rho, \sigma)$. □

Definition 2.8.6: The *purified distance* of $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ is $P(\rho, \sigma) := \sqrt{1 - F(\rho, \sigma)}$. For subnormalized states, F is replaced by the generalized fidelity $\big(\operatorname{Tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}} + \sqrt{(1 - \operatorname{Tr}\rho)(1 - \operatorname{Tr}\sigma)}\big)^2$.

Theorem 2.8.3: Fuchs–van de Graaf inequalities

For $\rho, \sigma \in \mathcal{S}(\mathcal{H})$,

$$1 - \sqrt{F(\rho, \sigma)} \;\leq\; \tfrac{1}{2}\|\rho - \sigma\|_1 \;\leq\; \sqrt{1 - F(\rho, \sigma)} = P(\rho, \sigma).$$

Proof. Upper bound. For pure states $|\psi\rangle, |\varphi\rangle$, $\Delta := |\psi\rangle\langle\psi| - |\varphi\rangle\langle\varphi|$ has rank ≤ 2 and trace 0, so its non-zero eigenvalues are $\pm\lambda$; from $\operatorname{Tr}\Delta^2 = 2 - 2|\langle\psi|\varphi\rangle|^2 = 2\lambda^2$ we get $\frac{1}{2}\|\Delta\|_1 = \lambda = \sqrt{1 - |\langle\psi|\varphi\rangle|^2}$. For mixed states choose purifications with $|\langle \psi_\rho | \varphi_\sigma \rangle|^2 = F(\rho, \sigma)$ (Theorem 2.8.2) and apply Theorem 2.8.1 to the partial trace.

Lower bound. Assume σ invertible (the general case follows by continuity). Let $R := \sqrt{\sigma^{1/2}\rho\sigma^{1/2}}$ and $M := \sigma^{-1/2} R \sigma^{-1/2} \geq 0$, so that $M\sigma M = \rho$. Measure in an eigenbasis $\{|m\rangle\}$ of M with eigenvalues $\mu_m \geq 0$, obtaining distributions $p_m = \langle m|\rho|m\rangle = \mu_m^2 q_m$ and $q_m = \langle m|\sigma|m\rangle$. Then

$$\sum_m \sqrt{p_m q_m} = \sum_m \mu_m q_m = \operatorname{Tr}(M\sigma) = \operatorname{Tr} R = \sqrt{F(\rho, \sigma)}.$$

By Theorem 2.8.1 applied to this measurement and the elementary inequality

$$\sum_m \big(\sqrt{p_m} - \sqrt{q_m}\big)^2 \leq \sum_m \big|\sqrt{p_m} - \sqrt{q_m}\big|\big(\sqrt{p_m} + \sqrt{q_m}\big) = \sum_m |p_m - q_m|,$$

we get $\frac{1}{2}\|\rho - \sigma\|_1 \geq \frac{1}{2}\sum_m |p_m - q_m| \geq \frac{1}{2}\sum_m (\sqrt{p_m} - \sqrt{q_m})^2 = 1 - \sum_m \sqrt{p_m q_m} = 1 - \sqrt{F(\rho, \sigma)}$. □

Corollary 2.8.7: If $\frac{1}{2}\|\rho-\sigma\|_1 \le \varepsilon$, then $F(\rho,\sigma) \ge (1-\varepsilon)^2 \ge 1-2\varepsilon$. If $F(\rho,\sigma) \ge 1-\varepsilon$, then $\frac{1}{2}\|\rho-\sigma\|_1 \le \sqrt{\varepsilon}$.

Proof. The first statement is the lower bound of Theorem 2.8.3 rearranged, the second the upper bound. □

3 Information and entropies

3.1 Classical entropies

Throughout, log denotes the logarithm to base 2, ln the natural logarithm, and $0\log 0 := 0$. Random variables take values in finite sets.

Definition 3.1.1: The *(Shannon) entropy* of a random variable X with distribution p_X on $\mathcal{X}$ is

$$H(X) := -\sum_{x\in\mathcal{X}} p_X(x)\log p_X(x).$$

For $p \in [0,1]$, the *binary entropy* is $h(p) := -p\log p - (1-p)\log(1-p)$.

Example 3.1.2: A fair coin has $H(X) = 1$; a fair eight-sided die has $H(X) = \log 8 = 3$. A deterministic X has $H(X) = 0$.

Remark 3.1.3 (Data compression): $H(X)$ measures the uncertainty about X in bits. Operationally (Shannon's noiseless coding theorem), n i.i.d. copies of X can be compressed into nR bits and recovered with vanishing error as $n\to\infty$ if $R > H(X)$, and not if $R < H(X)$.

Example 3.1.4: Let $X \in \{a,b,c,d\}$ with probabilities $\frac{1}{2},\frac{1}{4},\frac{1}{8},\frac{1}{8}$. Then $H(X) = \frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + 2\cdot\frac{1}{8}\cdot 3 = 1.75$. The prefix-free code $a\mapsto 0$, $b\mapsto 10$, $c\mapsto 110$, $d\mapsto 111$ has expected length 1.75 bits, which is optimal; a fixed-length code needs 2 bits.

Lemma 3.1.5: The binary entropy is symmetric, $h(p) = h(1-p)$, satisfies $h(0) = h(1) = 0$, $h(\frac{1}{2}) = 1$, and is strictly concave:

$$h\big(\lambda p_1 + (1-\lambda)p_2\big) \ge \lambda h(p_1) + (1-\lambda)h(p_2) \qquad (\lambda\in[0,1]).$$

Proof. Symmetry and the values are clear. For $p\in(0,1)$, $h'(p) = \log\frac{1-p}{p}$ and $h''(p) = -\frac{1}{p(1-p)\ln 2} < 0$. □

Definition 3.1.6: Let X, Y be random variables with joint distribution p_{XY}.

(i) The *joint entropy* is $H(XY) := -\sum_{x,y} p_{XY}(x,y)\log p_{XY}(x,y)$.

(ii) The *conditional entropy* is

$$H(X|Y) := \sum_y p_Y(y)\, H(X|Y=y) = -\sum_{x,y} p_{XY}(x,y)\log p_{X|Y}(x|y).$$

(iii) The *mutual information* is $I(X:Y) := H(X) + H(Y) - H(XY)$.

(iv) The *conditional mutual information* is $I(X:Y|Z) := H(X|Z) - H(X|YZ)$.

Lemma 3.1.7 (Chain rules): For random variables $X, Y, Z, X_1, \dots, X_n$,

(i) $H(XY) = H(Y) + H(X|Y)$,

(ii) $H(X_1\cdots X_n) = \sum_{i=1}^n H(X_i|X_1\cdots X_{i-1})$,

(iii) $H(XY|Z) = H(Y|Z) + H(X|YZ)$,

(iv) $I(X : YZ) = I(X : Z) + I(X : Y|Z)$.

Proof. (i) Using $p_{XY}(x,y) = p_Y(y)p_{X|Y}(x|y)$,

$$H(XY) = -\sum_{x,y} p_{XY}(x,y)\big(\log p_Y(y) + \log p_{X|Y}(x|y)\big) = H(Y) + H(X|Y).$$

(ii) by induction on n, applying (i) to X_n and $X_1 \cdots X_{n-1}$. (iii) follows from (i) applied to $H(XYZ) = H(Z) + H(XY|Z)$ and $H(XYZ) = H(YZ) + H(X|YZ)$. (iv) $I(X : YZ) = H(X) - H(X|YZ) = \big(H(X) - H(X|Z)\big) + \big(H(X|Z) - H(X|YZ)\big)$, using $I(X : Y) = H(X) - H(X|Y)$, which follows from (i). □

Example 3.1.8: Let a source output the periodic string $0101\ldots$ or $1010\ldots$, each with probability $\frac{1}{2}$. Each bit X_i is uniform, $H(X_i) = 1$, but $H(X_i|X_{i-1}) = 0$; by the chain rule $H(X_1 \cdots X_n) = 1$.

Definition 3.1.9: The *relative entropy* of two distributions p, q on $\mathcal{X}$ is

$$D(p\|q) := \sum_x p(x) \log \frac{p(x)}{q(x)},$$

with $D(p\|q) := \infty$ if $\operatorname{supp} p \not\subset \operatorname{supp} q$.

Theorem 3.1.1: Gibbs' inequality

$D(p\|q) \geq 0$, with equality if and only if $p = q$.

Proof. Using $\ln t \leq t - 1$ with equality if and only if $t = 1$,

$$-D(p\|q)\ln 2 = \sum_{x \in \operatorname{supp} p} p(x) \ln \frac{q(x)}{p(x)} \leq \sum_{x \in \operatorname{supp} p} \big(q(x) - p(x)\big) \leq 0.$$

Equality requires $q(x) = p(x)$ on $\operatorname{supp} p$ and $\sum_{x \in \operatorname{supp} p} q(x) = 1$, i.e., $p = q$. □

Corollary 3.1.10: Let X, Y, Z be random variables.

(i) $0 \leq H(X) \leq \log|\mathcal{X}|$, with $H(X) = 0$ if and only if X is deterministic and $H(X) = \log|\mathcal{X}|$ if and only if X is uniform.

(ii) $I(X : Y) = D(p_{XY}\|p_X\, p_Y) \geq 0$, with equality if and only if X and Y are independent.

(iii) $H(X|Y) \leq H(X)$ (conditioning does not increase entropy) and $0 \leq H(X|Y)$.

(iv) $H(X_1 \cdots X_n) \leq \sum_i H(X_i)$, with equality if and only if the X_i are independent.

(v) $I(X : Y|Z) = \sum_z p_Z(z)\, I(X : Y|Z = z) \geq 0$.

Proof. (i) $-p\log p \geq 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. With u the uniform distribution, $0 \leq D(p_X\|u) = \log|\mathcal{X}| - H(X)$. (ii) Direct computation:

$$D(p_{XY}\|p_X p_Y) = \sum_{x,y} p_{XY}(x,y)\big(\log p_{XY}(x,y) - \log p_X(x) - \log p_Y(y)\big),$$

which equals $-H(XY) + H(X) + H(Y)$. (iii) $H(X) - H(X|Y) = I(X : Y) \geq 0$; $H(X|Y)$ is an average of entropies. (iv) By the chain rule and (iii), $H(X_1 \cdots X_n) = \sum_i H(X_i|X_{<i}) \leq \sum_i H(X_i)$. (v) Apply (ii) to the conditional distributions. □

Lemma 3.1.11 (Data processing): If $X - Y - Z$ is a Markov chain (i.e., $p_{Z|XY} = p_{Z|Y}$), then $I(X:Z) \le I(X:Y)$ and $H(X|Y) \le H(X|Z)$.

Proof. The Markov property means that X and Z are independent given Y, i.e., $I(X:Z|Y) = 0$. By Lemma 3.1.7(iv) applied in both orders,

$$I(X:Y) + \underbrace{I(X:Z|Y)}_{=0} = I(X:YZ) = I(X:Z) + \underbrace{I(X:Y|Z)}_{\ge 0}.$$

Subtracting both sides from $H(X)$ gives $H(X|Y) \le H(X|Z)$. □

Example 3.1.12 (Binary erasure channel): Let X be a uniform bit and $Y = X$ with probability $1-e$ and $Y = ?$ (erasure) with probability e. Then $H(X|Y=?) = 1$ and $H(X|Y=x) = 0$, so $H(X|Y) = e$ and $I(X:Y) = 1 - e$.

Lemma 3.1.13 (Fano-type bound): Let X, X' be bits with $\Pr[X \ne X'] = \delta$. Then $H(X|X') \le h(\delta)$. More generally, for strings $X^n, X'^n \in \{0,1\}^n$ with $\Pr[X_i \ne X'_i] = \delta$ for all i, $H(X^n|X'^n) \le nh(\delta)$.

Proof. Since X' is known, $X \mapsto X \oplus X'$ is a bijection, so $H(X|X') = H(X \oplus X'|X') \le H(X \oplus X') = h(\delta)$. The second statement follows in the same way with Corollary 3.1.10(iv). □

Example 3.1.14: Let X be a uniform bit and $Y = X \oplus N$ with an independent bit N, $\Pr[N = 1] = \delta$ (binary symmetric channel). Then $H(Y|X) = H(N) = h(\delta)$, $H(Y) = 1$, and $I(X:Y) = 1 - h(\delta)$.

3.2 Quantum entropies

Definition 3.2.1: The *von Neumann entropy* of $\rho_A \in \mathcal{S}(\mathcal{H}_A)$ is

$$H(A)_\rho := -\operatorname{Tr}(\rho_A \log \rho_A) = -\sum_i \lambda_i \log \lambda_i,$$

where λ_i are the eigenvalues of ρ_A. For $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ the *joint entropy* is $H(AB)_\rho := H(\rho_{AB})$.

Example 3.2.2: For the ensemble $\{(\frac{1}{2}, |0\rangle), (\frac{1}{2}, |+\rangle)\}$, $\rho = \frac{1}{4}\begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$ has eigenvalues $\frac{1}{2} \pm \frac{1}{2\sqrt{2}}$, so $H(\rho) = h\big(\frac{1}{2} + \frac{1}{2\sqrt{2}}\big) \approx 0.60 < 1$: although the preparation uses one bit of randomness, the non-orthogonality of the states reduces the entropy.

Definition 3.2.3: The *quantum relative entropy* of $\rho, \sigma \ge 0$ is

$$D(\rho\|\sigma) := \operatorname{Tr}\big(\rho(\log \rho - \log \sigma)\big)$$

if $\operatorname{supp}\rho \subset \operatorname{supp}\sigma$, and $D(\rho\|\sigma) := \infty$ otherwise.

Theorem 3.2.1: Klein's inequality

For $\rho, \sigma \in \mathcal{S}(\mathcal{H})$, $D(\rho\|\sigma) \ge 0$, with equality if and only if $\rho = \sigma$.

Proof. Let $\rho = \sum_i p_i |i\rangle\langle i|$, $\sigma = \sum_j q_j |j'\rangle\langle j'|$ be spectral decompositions and $P_{ij} := |\langle i|j'\rangle|^2$, a doubly stochastic matrix. Then

$$D(\rho\|\sigma) = \sum_i p_i \Big(\log p_i - \sum_j P_{ij} \log q_j\Big) \ge \sum_i p_i \big(\log p_i - \log r_i\big) = D(p\|r) \ge 0,$$

with $r_i := \sum_j P_{ij} q_j$, a probability distribution; we used concavity of log ($\sum_j P_{ij} \log q_j \le \log r_i$) and Theorem 3.1.1. Equality in both steps forces $p = r$ and, by strict concavity, that for each i with $p_i > 0$ all q_j with $P_{ij} > 0$ coincide; together this gives $\rho = \sigma$. The converse is clear. □

Lemma 3.2.4: Let $d = \dim \mathcal{H}_A$.

(i) $0 \leq H(A)_\rho \leq \log d$, with $H(A)_\rho = 0$ if and only if ρ_A is pure and $H(A)_\rho = \log d$ if and only if $\rho_A = \mathbb{I}/d$.

(ii) $H(U\rho U^\dagger) = H(\rho)$ for unitaries (and isometries) U.

(iii) $H(\rho_A \otimes \sigma_B) = H(\rho_A) + H(\sigma_B)$.

Proof. (i) The entropy is the Shannon entropy of the spectrum; use Corollary 3.1.10(i). (ii) The spectrum is invariant. (iii) $\log(\rho \otimes \sigma) = \log \rho \otimes \mathbb{I} + \mathbb{I} \otimes \log \sigma$, so $-\operatorname{Tr}(\rho \otimes \sigma) \log(\rho \otimes \sigma) = H(\rho) + H(\sigma)$. □

Lemma 3.2.5: If $|\psi\rangle_{AB}$ is pure, then $H(A)_\psi = H(B)_\psi$. More generally, if ρ_{ABC} is pure, then $H(AB) = H(C)$, $H(AC) = H(B)$ and $H(BC) = H(A)$.

Proof. By the Schmidt decomposition, the marginals of a pure bipartite state have the same non-zero eigenvalues. For the tripartite case, regard ρ_{ABC} as a bipartite pure state on $(AB) \otimes C$, etc. □

Lemma 3.2.6: For a cq-state $\rho_{XB} = \sum_x p(x)|x\rangle\langle x| \otimes \rho_B^x$,

$$H(XB)_\rho = H(X) + \sum_x p(x)\, H(\rho_B^x).$$

Proof. ρ_{XB} is block-diagonal with blocks $p(x)\rho_B^x$. If ρ_B^x has eigenvalues λ_j^x, then ρ_{XB} has eigenvalues $p(x)\lambda_j^x$ and

$$H(XB) = -\sum_{x,j} p(x)\lambda_j^x \big(\log p(x) + \log \lambda_j^x\big) = H(X) + \sum_x p(x) H(\rho_B^x).$$ □

Definition 3.2.7: For $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ define

(i) the *conditional entropy* $H(A|B)_\rho := H(AB)_\rho - H(B)_\rho$,

(ii) the *coherent information* $I(A\rangle B)_\rho := -H(A|B)_\rho$,

(iii) the *mutual information* $I(A:B)_\rho := H(A)_\rho + H(B)_\rho - H(AB)_\rho$.

Example 3.2.8: For $|\Phi^+\rangle_{AB}$ we have $H(AB) = 0$ and $H(A) = H(B) = 1$, hence $H(A|B) = -1$ and $I(A:B) = 2$. Unlike its classical counterpart, the conditional quantum entropy can be negative, and the mutual information can exceed $\log d_A$. For a cq-state, $H(X|B) = \sum_x p(x)H(\rho_B^x) + H(X) - H(B)$ by Lemma 3.2.6.

Corollary 3.2.9: For $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ and $\sigma_B \in \mathcal{S}(\mathcal{H}_B)$,

(i) $I(A:B)_\rho = D(\rho_{AB} \| \rho_A \otimes \rho_B) \geq 0$, with equality if and only if $\rho_{AB} = \rho_A \otimes \rho_B$ (subadditivity $H(AB) \leq H(A) + H(B)$),

(ii) $D(\rho_{AB} \| \mathbb{I}_A \otimes \sigma_B) = -H(A|B)_\rho + D(\rho_B \| \sigma_B)$; in particular

$$H(A|B)_\rho = -D(\rho_{AB} \| \mathbb{I}_A \otimes \rho_B) = \max_{\sigma_B \in \mathcal{S}(\mathcal{H}_B)} \big(-D(\rho_{AB} \| \mathbb{I}_A \otimes \sigma_B)\big).$$

Proof. Using $\log(X \otimes Y) = \log X \otimes \mathbb{I} + \mathbb{I} \otimes \log Y$ and Lemma 2.1.10(ii),

$$-\operatorname{Tr} \rho_{AB} \log(\rho_A \otimes \rho_B) = H(A) + H(B), \qquad -\operatorname{Tr} \rho_{AB} \log(\mathbb{I}_A \otimes \sigma_B) = -\operatorname{Tr} \rho_B \log \sigma_B.$$

Hence $D(\rho_{AB} \| \rho_A \otimes \rho_B) = -H(AB) + H(A) + H(B)$ and $D(\rho_{AB} \| \mathbb{I} \otimes \sigma_B) = -H(AB) - \operatorname{Tr} \rho_B \log \sigma_B = -H(AB) + H(B) + D(\rho_B \| \sigma_B)$. Now apply Theorem 3.2.1. □

Lemma 3.2.10 (Concavity): $\rho \mapsto H(\rho)$ is concave: $H(\sum_x p(x)\rho_x) \geq \sum_x p(x) H(\rho_x)$.

Proof. For $\omega_{XA} = \sum_x p(x)|x\rangle\langle x| \otimes \rho_x$, subadditivity and Lemma 3.2.6 give

$$H(X) + \sum_x p(x) H(\rho_x) = H(XA) \leq H(X) + H(A) = H(X) + H\Big(\sum_x p(x)\rho_x\Big). \qquad \square$$

Theorem 3.2.2: Bounds on the conditional entropy

For $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$ with $d_A = \dim \mathcal{H}_A$,

(i) $|H(A) - H(B)| \leq H(AB) \leq H(A) + H(B)$ (Araki–Lieb and subadditivity),

(ii) $-\log d_A \leq -H(A) \leq H(A|B) \leq H(A) \leq \log d_A$,

(iii) $I(A:B) \leq 2 \log d_A$.

Proof. (i) The upper bound is Corollary 3.2.9(i). For the lower bound let ρ_{ABC} be a purification. By Lemma 3.2.5 and subadditivity, $H(B) = H(AC) \leq H(A) + H(C) = H(A) + H(AB)$; exchange A and B for the other inequality. (ii) The upper bound is subadditivity. For the lower bound, again with a purification, $H(A|B) = H(AB) - H(B) = H(C) - H(AC) \geq H(C) - H(A) - H(C) = -H(A)$. (iii) $I(A:B) = H(A) - H(A|B) \leq 2H(A)$. $\square$

Corollary 3.2.11: A pure state $|\psi\rangle_{AB}$ is entangled if and only if $H(A|B)_\psi < 0$.

Proof. For pure states, $H(AB) = 0$, so $H(A|B) = -H(B) = -H(A)$, which is negative if and only if ρ_A is mixed, i.e., if and only if $|\psi\rangle$ is entangled. $\square$

Theorem 3.2.3: Monotonicity of relative entropy

For every CPTP map $\mathcal{E}$ and $\rho \in \mathcal{S}(\mathcal{H})$, $\sigma \geq 0$,

$$D\big(\mathcal{E}(\rho) \| \mathcal{E}(\sigma)\big) \leq D(\rho \| \sigma).$$

In particular, $D(\rho_A \| \sigma_A) \leq D(\rho_{AB} \| \sigma_{AB})$.

Proof. See Lindblad (1975) or Wilde, Thm. 11.8.1; it is equivalent to Lieb's concavity theorem. $\square$

Lemma 3.2.12: For $\rho \in \mathcal{S}(\mathcal{H})$, $\sigma, \sigma' \geq 0$ and $c > 0$:

(i) if $\sigma \leq \sigma'$, then $D(\rho\|\sigma) \geq D(\rho\|\sigma')$,

(ii) $D(\rho\|c\sigma) = D(\rho\|\sigma) - \log c$,

(iii) $D(\rho_1 \oplus 0 \| \sigma_1 \oplus \sigma_2) = D(\rho_1\|\sigma_1)$.

Proof. (i) The logarithm is operator monotone, so $\log \sigma \leq \log \sigma'$ (on the relevant supports). (ii) $\log(c\sigma) = \log \sigma + \log c \cdot \mathbb{I}$. (iii) The logarithm acts blockwise and the second block has weight 0. $\square$

Corollary 3.2.13 (Strong subadditivity and data processing): The following holds:

(i) $H(A|BC)_\rho \leq H(A|B)_\rho$ for all ρ_{ABC}, equivalently $H(ABC) + H(B) \leq H(AB) + H(BC)$.

(ii) If $\mathcal{E} : \mathcal{B}(\mathcal{H}_B) \to \mathcal{B}(\mathcal{H}_{B'})$ is CPTP and $\sigma_{AB'} = (\mathrm{id}_A \otimes \mathcal{E})(\rho_{AB})$, then $H(A|B)_\rho \leq H(A|B')_\sigma$, i.e., $I(A\rangle B)_\rho \geq I(A\rangle B')_\sigma$.

(iii) In particular, a measurement on B with classical outcome Y satisfies $H(A|B) \leq H(A|Y)$.

Proof. (ii) By Corollary 3.2.9(ii) and Theorem 3.2.3 applied to $\mathrm{id}_A \otimes \mathcal{E}$,

$$-H(A|B)_\rho = D(\rho_{AB} \| \mathbb{I}_A \otimes \rho_B) \geq D(\sigma_{AB'} \| \mathbb{I}_A \otimes \mathcal{E}(\rho_B)) = -H(A|B')_\sigma,$$

since $\sigma_{B'} = \mathcal{E}(\rho_B)$. (i) is (ii) for the partial trace Tr_C. (iii) is (ii) for the measurement channel $\rho \mapsto \sum_y \mathrm{Tr}(E_y \rho)|y\rangle\langle y|$. □

Corollary 3.2.14: The conditional entropy is concave: for $\rho_{AB} = \sum_x p(x)\rho^x_{AB}$, $H(A|B)_\rho \geq \sum_x p(x) H(A|B)_{\rho^x}$. In particular, if ρ_{AB} is separable, then $H(A|B)_\rho \geq 0$.

Proof. Let $\omega_{ABX} = \sum_x p(x)\rho^x_{AB} \otimes |x\rangle\langle x|$. By Lemma 3.2.6, $H(A|BX)_\omega = H(ABX) - H(BX) = \sum_x p(x) H(A|B)_{\rho^x}$, and $H(A|BX)_\omega \leq H(A|B)_\rho$ by strong subadditivity. For a separable state, each term $\rho^x_A \otimes \rho^x_B$ has $H(A|B) = H(\rho^x_A) \geq 0$. □

Corollary 3.2.15: The quantum mutual information satisfies data processing: $I(A:B)_\rho \geq I(A:B')_\sigma$ for $\sigma_{AB'} = (\mathrm{id} \otimes \mathcal{E})(\rho_{AB})$.

Proof. $I(A:B) = D(\rho_{AB} \| \rho_A \otimes \rho_B) \geq D(\sigma_{AB'} \| \rho_A \otimes \mathcal{E}(\rho_B)) = I(A:B')_\sigma$ by Theorem 3.2.3. □

3.3 One-shot entropies

Definition 3.3.1: Let $\rho_{AB} \in \mathcal{S}_\leq(\mathcal{H}_A \otimes \mathcal{H}_B)$ and $\sigma_B \in \mathcal{S}(\mathcal{H}_B)$. Define

$$H_{\min}(\rho_{AB}|\sigma_B) := -\log \min\{\lambda \geq 0 \mid \rho_{AB} \leq \lambda\, \mathbb{I}_A \otimes \sigma_B\}.$$

The *conditional min-entropy* and *conditional max-entropy* are

$$H_{\min}(A|B)_\rho := \sup_{\sigma_B} H_{\min}(\rho_{AB}|\sigma_B), \qquad H_{\max}(A|B)_\rho := \max_{\sigma_B} \log \left\| \sqrt{\rho_{AB}} \sqrt{\mathbb{I}_A \otimes \sigma_B} \right\|_1^2,$$

where the optimizations are over $\sigma_B \in \mathcal{S}(\mathcal{H}_B)$.

Remark 3.3.2: The definition mirrors Corollary 3.2.9(ii): $H(A|B) = \max_{\sigma_B} \big(-D(\rho_{AB} \| \mathbb{I} \otimes \sigma_B)\big)$, with the relative entropy replaced by the max-relative entropy $D_{\max}(\rho\|\sigma) = \log \min\{\lambda \mid \rho \leq \lambda\sigma\}$. Similarly, $H_{\max}(A|B) = \max_{\sigma_B} \log F(\rho_{AB}, \mathbb{I}_A \otimes \sigma_B)$.

Lemma 3.3.3: Without side information (B trivial),

$$H_{\min}(X) = -\log \max_x p(x), \qquad H_{\max}(X) = 2 \log \sum_x \sqrt{p(x)}.$$

In particular $H_{\min}(X) \leq H(X) \leq H_{\max}(X) \leq \log |\mathcal{X}|$.

Proof. $\rho_X \leq \lambda \mathbb{I}$ if and only if $\lambda \geq \max_x p(x)$, and $\|\sqrt{\rho_X}\|_1 = \sum_x \sqrt{p(x)}$. The inequalities follow from Jensen's inequality: $-\log \max p \leq \sum p \log \frac{1}{p} \leq 2 \log \sum p \sqrt{1/p}$. □

Remark 3.3.4: For a cq-state ρ_{XE},

$$H_{\min}(X|E)_\rho = -\log p_{\mathrm{guess}}(X|E), \qquad p_{\mathrm{guess}}(X|E) := \max_{\{E_x\}} \sum_x p(x) \,\mathrm{Tr}(E_x \rho^x_E),$$

the maximal probability of guessing X by measuring E (König–Renner–Schaffner 2009; the proof uses semidefinite programming duality). Thus $H_{\min}(X|E)$ quantifies how much uniform randomness, independent of E, can be extracted from X (cf. Theorem 4.5.2). $H_{\max}(X|B)$ quantifies how far X can be compressed such that it can be recovered with the help of B.

Lemma 3.3.5: $H_{\min}(A|B)_\rho \le H(A|B)_\rho$ for $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$.

Proof. Let $\rho_{AB} \le \lambda \mathbb{I}_A \otimes \sigma_B$. Since the logarithm is operator monotone, $\log \rho_{AB} \le \log \lambda + \log(\mathbb{I}_A \otimes \sigma_B)$ on $\operatorname{supp} \rho_{AB}$. Taking the expectation in ρ_{AB},

$$-H(AB) \le \log \lambda + \operatorname{Tr} \rho_B \log \sigma_B = \log \lambda - H(B) - D(\rho_B \| \sigma_B) \le \log \lambda - H(B),$$

i.e., $H(A|B) \ge -\log \lambda$. Optimizing over λ and σ_B gives the claim. □

Definition 3.3.6: For $\rho \in \mathcal{S}_\le(\mathcal{H})$ and $\varepsilon \ge 0$, the *ε-ball* around ρ is

$$\mathcal{B}^\varepsilon(\rho) := \{\tilde\rho \in \mathcal{S}_\le(\mathcal{H}) \mid P(\rho, \tilde\rho) \le \varepsilon\}.$$

The *smooth min- and max-entropies* are

$$H^\varepsilon_{\min}(A|B)_\rho := \max_{\tilde\rho \in \mathcal{B}^\varepsilon(\rho)} H_{\min}(A|B)_{\tilde\rho}, \qquad H^\varepsilon_{\max}(A|B)_\rho := \min_{\tilde\rho \in \mathcal{B}^\varepsilon(\rho)} H_{\max}(A|B)_{\tilde\rho}.$$

Theorem 3.3.1: Duality

Let ρ_{ABC} be pure and $\varepsilon \ge 0$. Then $H^\varepsilon_{\min}(A|B)_\rho = -H^\varepsilon_{\max}(A|C)_\rho$.

Proof. See Tomamichel–Colbeck–Renner (2010). For $\varepsilon = 0$ the analogous statement for the von Neumann entropy is elementary: $H(A|B) = H(AB) - H(B) = H(C) - H(AC) = -H(A|C)$ by Lemma 3.2.5. □

Corollary 3.3.7: $H_{\min}(A|B)_\rho \le H(A|B)_\rho \le H_{\max}(A|B)_\rho$.

Proof. The first inequality is Lemma 3.3.5. For the second, let ρ_{ABC} be a purification. By Theorem 3.3.1 and Lemma 3.3.5, $H_{\max}(A|B) = -H_{\min}(A|C) \ge -H(A|C) = H(A|B)$. □

Theorem 3.3.2: Quantum asymptotic equipartition property

For every $\rho_{AB} \in \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$,

$$\lim_{\varepsilon \to 0} \lim_{n \to \infty} \frac{1}{n} H^\varepsilon_{\min}(A^n|B^n)_{\rho^{\otimes n}} = H(A|B)_\rho = \lim_{\varepsilon \to 0} \lim_{n \to \infty} \frac{1}{n} H^\varepsilon_{\max}(A^n|B^n)_{\rho^{\otimes n}}.$$

More precisely, $H^\varepsilon_{\min}(A^n|B^n)_{\rho^{\otimes n}} \ge nH(A|B)_\rho - c_\varepsilon \sqrt{n}$ with c_ε independent of n.

Proof. See Tomamichel–Colbeck–Renner (2009). □

Lemma 3.3.8 (Chain rule for classical side information): Let $\rho_{XCE} = \sum_c \rho^c_{XE} \otimes |c\rangle\langle c|_C$ be a state where C is classical with values in a set $\mathcal{C}$. Then

$$H_{\min}(X|CE)_\rho \ge H_{\min}(X|E)_\rho - \log |\mathcal{C}|,$$

and the same holds for the smooth min-entropy.

Proof. Let $\rho_{XE} \le \lambda \, \mathbb{I}_X \otimes \sigma_E$. Since $0 \le \rho^c_{XE} \le \sum_{c'} \rho^{c'}_{XE} = \rho_{XE}$,

$$\rho_{XCE} = \sum_c \rho^c_{XE} \otimes |c\rangle\langle c| \le \lambda \, \mathbb{I}_X \otimes \sigma_E \otimes \mathbb{I}_C = \lambda |\mathcal{C}| \, \mathbb{I}_X \otimes \Big(\sigma_E \otimes \frac{\mathbb{I}_C}{|\mathcal{C}|}\Big).$$

Hence $H_{\min}(X|CE) \ge -\log \lambda - \log |\mathcal{C}|$; optimize over λ, σ_E. For the smooth version, see Renner, PhD thesis (2005), or Tomamichel (2016). □

3.4 Entropic uncertainty relations

Let ρ_{AB} be a state, and let $\{M_A^x\}$, $\{N_A^y\}$ be two POVMs on A. The post-measurement states are

$$\sigma_{XB} = \sum_x |x\rangle\langle x|_X \otimes \mathrm{Tr}_A\big((M_A^x \otimes \mathbb{I}_B)\rho_{AB}\big), \qquad \tau_{YB} = \sum_y |y\rangle\langle y|_Y \otimes \mathrm{Tr}_A\big((N_A^y \otimes \mathbb{I}_B)\rho_{AB}\big).$$

The *overlap* of the two measurements is

$$c := \max_{x,y} \left\| \sqrt{M_A^x}\sqrt{N_A^y} \right\|_\infty^2 \in (0,1],$$

where $\|\cdot\|_\infty$ is the operator norm. For rank-one projective measurements in bases $\{|x\rangle\}$, $\{|y\rangle\}$, $c = \max_{x,y} |\langle x|y\rangle|^2$.

Example 3.4.1: For the Z- and X-measurements on a qubit, $c = \max |\langle z|x\rangle|^2 = \frac{1}{2}$, i.e., $\log(1/c) = 1$. For two of the three bases X, Y, Z the same holds.

Remark 3.4.2: The usual uncertainty principle seems to fail in the presence of entanglement: for $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$, Bob can predict the outcome of both a Z- and an X-measurement of Alice with certainty. The uncertainty relations below take this into account by allowing a quantum memory B.

Theorem 3.4.1: Tripartite uncertainty relation

Let ρ_{ABE} be a state, and let X, Y denote the outcomes of the two measurements on A. Then

$$H(X|E) + H(Y|B) \geq \log \frac{1}{c}.$$

Proof. We give the proof for rank-one projective measurements in bases $\{|x\rangle\}$ and $\{|y\rangle\}$ (the case needed for QKD); see Berta et al. (2010) and Coles et al. (2012) for the general case. By strong subadditivity it suffices to consider pure ρ_{ABE} (adding a purifying system to E can only decrease $H(X|E)$).

Step 1. Let $V : \mathcal{H}_A \to \mathcal{H}_X \otimes \mathcal{H}_{X'}$, $V|a\rangle := \sum_x \langle x|a\rangle |x\rangle_X |x\rangle_{X'}$, an isometry, and $\omega_{XX'BE} := V\rho_{ABE}V^\dagger$, a pure state. Tracing out X' gives the post-measurement state ω_{XBE}, so $H(X|E)$ is evaluated on ω. By Lemma 3.2.5 (duality for pure states) and Corollary 3.2.9(ii),

$$H(X|E)_\omega = -H(X|X'B)_\omega = D\big(\omega_{XX'B} \,\big\|\, \mathbb{I}_X \otimes \omega_{X'B}\big).$$

Step 2. The map $\mathcal{R}(W) := V^\dagger W V \oplus \mathrm{Tr}\big((\mathbb{I} - VV^\dagger)W\big)$ (with values in $\mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B) \oplus \mathbb{C}$) is CPTP, and $\mathcal{R}(\omega_{XX'B}) = \rho_{AB} \oplus 0$ since ω is supported on the range of V. By Theorem 3.2.3 and Lemma 3.2.12(iii),

$$D\big(\omega_{XX'B} \| \mathbb{I}_X \otimes \omega_{X'B}\big) \geq D\Big(\rho_{AB} \,\Big\|\, V^\dagger(\mathbb{I}_X \otimes \omega_{X'B})V\Big) = D\Big(\rho_{AB} \,\Big\|\, \sum_x |x\rangle\langle x|_A \otimes \omega_B^x\Big),$$

where $\omega_B^x := (\langle x|_{X'} \otimes \mathbb{I})\omega_{X'B}(|x\rangle_{X'} \otimes \mathbb{I}) \geq 0$ with $\sum_x \omega_B^x = \rho_B$. The last equality is the direct computation $V^\dagger(\mathbb{I}_X \otimes W)V = \sum_x |x\rangle\langle x| \otimes \langle x|W|x\rangle$.

Step 3. Apply the measurement channel $\mathcal{P}(W) = \sum_y |y\rangle\langle y|W|y\rangle\langle y|$ on A (Theorem 3.2.3):

$$D\Big(\rho_{AB} \Big\| \sum_x |x\rangle\langle x| \otimes \omega_B^x\Big) \geq D\Big(\tau_{YB} \,\Big\|\, \sum_y |y\rangle\langle y| \otimes \sum_x |\langle x|y\rangle|^2 \omega_B^x\Big) \geq D\big(\tau_{YB} \,\|\, c\,\mathbb{I}_Y \otimes \rho_B\big),$$

using $\sum_x |\langle x|y\rangle|^2 \omega_B^x \leq c \sum_x \omega_B^x = c\rho_B$ and Lemma 3.2.12(i). Finally, by Lemma 3.2.12(ii) and Corollary 3.2.9(ii), $D(\tau_{YB} \| c\,\mathbb{I}_Y \otimes \rho_B) = -H(Y|B)_\tau + \log \frac{1}{c}$. □

Theorem 3.4.2: Uncertainty relation with quantum memory

$$H(X|B)_\sigma + H(Y|B)_\tau \geq \log \frac{1}{c} + H(A|B)_\rho.$$

Proof. Let ρ_{ABE} be a purification of ρ_{AB} and use the notation of the previous proof. Since $\omega_{XX'BE}$ is pure, $H(X|E)_\omega = H(X'B)_\omega - H(XX'B)_\omega = H(X'B)_\omega - H(E)_\rho$. The registers X and X' are identical copies, so $H(X'B)_\omega = H(XB)_\omega$, and $H(E)_\rho = H(AB)_\rho$. Hence

$$H(X|E) = H(XB) - H(AB) = H(X|B)_\sigma - H(A|B)_\rho.$$

Inserting this into Theorem 3.4.1 gives the claim. □

Remark 3.4.3: For $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$ and the Z/X measurements, the right-hand side of Theorem 3.4.2 equals $1 + (-1) = 0$, consistent with Bob predicting both outcomes perfectly. Without memory (B trivial and ρ_A pure), one recovers the Maassen–Uffink relation $H(X) + H(Y) \geq \log(1/c)$. The theorem can be phrased as a guessing game: Bob prepares ρ_{AB} and sends A to Alice, who measures one of two observables and announces which one; Bob then tries to guess the outcome using B.

Theorem 3.4.3: Smooth uncertainty relation

For ρ_{ABE} and $\varepsilon \geq 0$, with X, Y as above,

$$H_{\min}^\varepsilon(X|E) + H_{\max}^\varepsilon(Y|B) \geq \log \frac{1}{c}.$$

For n independent measurements on A^n (each chosen as one of the two measurements) the right-hand side becomes $n \log(1/c)$.

Proof. See Tomamichel–Renner (2011). The proof follows the same steps as for Theorem 3.4.1, using the monotonicity of the max-relative entropy and the duality Theorem 3.3.1 in Step 1. □

Remark 3.4.4: These relations bound Eve's uncertainty $H(X|E)$ in terms of the correlation $H(Y|B)$ between Alice and Bob, which is observable. This is the basis of the modern security proofs in Section 5.5.

4 Quantum key distribution protocols

4.1 Setting

Definition 4.1.1: A *QKD protocol* is a protocol between two parties, Alice and Bob, who have access to

(i) an insecure quantum channel, on which Eve may act arbitrarily (subject only to the laws of quantum mechanics), and

(ii) an authenticated classical channel, which Eve can read but not modify.

At the end of the protocol, Alice and Bob output keys $K_A, K_B \in \mathcal{S} \cup \{\perp\}$, where $\mathcal{S}$ is the key space and $\perp$ means that the protocol aborted.

Remark 4.1.2: A QKD protocol consists of two phases:

(i) *quantum transmission* (distribution and measurement of quantum states, followed by sifting), producing the *raw key*,

(ii) *classical post-processing*: parameter estimation, error correction and privacy amplification (Section 4.5).

One distinguishes *prepare-and-measure* protocols, where Alice prepares states and sends them to Bob, and *entanglement-based* protocols, where a source (in the worst case controlled by Eve) distributes entangled states to Alice and Bob.

4.2 Prepare-and-measure protocols

Definition 4.2.1 (BB84 states): For $a, b \in \{0,1\}$ define

$$|\psi_{00}\rangle = |0\rangle, \quad |\psi_{10}\rangle = |1\rangle, \quad |\psi_{01}\rangle = |+\rangle, \quad |\psi_{11}\rangle = |-\rangle.$$

Here a is the bit value and b the basis ($b = 0$: Z-basis; $b = 1$: X-basis). States from different bases have overlap $|\langle\psi_{a0}|\psi_{a'1}\rangle| = \frac{1}{\sqrt{2}}$, so the four states are not mutually orthogonal and cannot be distinguished perfectly (Lemma 2.4.6).

Example 4.2.2 (Polarization encoding): Photonic implementations use the polarization of single photons: the rectilinear basis $\{|H\rangle, |V\rangle\}$ (horizontal/vertical) corresponds to $\{|0\rangle, |1\rangle\}$, and the diagonal basis $\{|D\rangle, |A\rangle\}$ ($\pm 45°$) corresponds to $\{|+\rangle, |-\rangle\}$. Measurements are realized by (polarizing) beam splitters and single-photon detectors.

Definition 4.2.3 (BB84 protocol): The protocol consists of the following steps.

(i) Alice chooses uniformly random strings $a, b \in \{0,1\}^{4n}$.

(ii) She prepares $|\psi\rangle_A = \bigotimes_{i=1}^{4n} |\psi_{a_i b_i}\rangle$ and sends it to Bob.

(iii) Bob announces that he has received the qubits.

(iv) Bob chooses a random string $b' \in \{0,1\}^{4n}$ and measures the i-th qubit in basis b'_i, obtaining $a' \in \{0,1\}^{4n}$.

(v) *Sifting*: Alice announces b, and both discard all positions with $b_i \neq b'_i$. About $2n$ bits remain.

(vi) *Parameter estimation*: they compare n randomly chosen bits to estimate the error rate and abort if it is above a threshold.

(vii) *Post-processing*: they apply error correction and privacy amplification to the remaining n bits to obtain a key of $m < n$ bits.

Lemma 4.2.4: Assume a noiseless channel and no eavesdropper. If $b'_i = b_i$, then $a'_i = a_i$; if $b'_i \neq b_i$, then a'_i is uniformly random and independent of a_i.

Proof. If $b'_i = b_i$, Bob measures $|\psi_{a_i b_i}\rangle$ in the basis it belongs to and obtains a_i with probability 1. If $b'_i \neq b_i$, outcome a' occurs with probability $|\langle\psi_{a' b'_i}|\psi_{a_i b_i}\rangle|^2 = \frac{1}{2}$ for both values of a', independently of a_i. □

Remark 4.2.5: The order of the steps matters: b must be announced only after Bob has confirmed reception. Otherwise Eve could measure each qubit in the correct basis without disturbing it. The announcement of b itself reveals nothing about a, since a and b are independent. Since $b_i = b'_i$ with probability $\frac{1}{2}$, about $2n$ of the $4n$ positions survive sifting.

Example 4.2.6 (Intercept–resend attack): Eve measures each qubit in a random basis b_i^E and resends the post-measurement state $|\psi_{a_i^E b_i^E}\rangle$. Consider a sifted position ($b'_i = b_i$).

(i) With probability $\frac{1}{2}$, $b_i^E = b_i$: Eve obtains a_i, resends the correct state, and Bob obtains $a'_i = a_i$.

(ii) With probability $\frac{1}{2}$, $b_i^E \neq b_i$: Eve's outcome is uniform, she resends a state of the wrong basis, and Bob's outcome is uniform, i.e., $a'_i \neq a_i$ with probability $\frac{1}{2}$.

The resulting error rate in the sifted key is $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. After the basis announcement Eve knows which bits she measured in the right basis, so she knows half of the sifted bits. If Eve attacks only a fraction f of the qubits, the error rate is $f/4$ and she learns a fraction $f/2$. Hence a full intercept–resend attack is detected by an error rate of 25% in parameter estimation.

Definition 4.2.7 (B92 protocol): Alice chooses a random bit a_i and sends $|0\rangle$ if $a_i = 0$ and $|+\rangle$ if $a_i = 1$. Bob chooses a random bit a'_i and measures in the Z-basis if $a'_i = 0$ and in the X-basis if $a'_i = 1$. If he obtains the outcome -1 (i.e., $|1\rangle$ or $|-\rangle$), he sets $b_i = 1$; if he obtains $+1$ ($|0\rangle$ or $|+\rangle$), he sets $b_i = 0$. He announces b, and both keep only the positions with $b_i = 1$: Alice keeps a_i and Bob keeps $a'_i \oplus 1$.

Lemma 4.2.8: In the B92 protocol (without noise), $b_i = 1$ implies $a_i = a'_i \oplus 1$, and $\Pr[b_i = 1] = \frac{1}{4}$. Hence Alice and Bob share identical key bits on the kept positions.

Proof. If $a_i = a'_i = 0$, Bob measures $|0\rangle$ in the Z-basis and always obtains $+1$; if $a_i = a'_i = 1$, he measures $|+\rangle$ in the X-basis and always obtains $+1$. Hence $b_i = 1$ requires $a_i \neq a'_i$. If $a_i \neq a'_i$, Bob measures in the other basis and obtains -1 with probability $\frac{1}{2}$. Since $a_i \neq a'_i$ with probability $\frac{1}{2}$, $\Pr[b_i = 1] = \frac{1}{4}$. □

Remark 4.2.9: B92 shows that two non-orthogonal states suffice for QKD: Eve cannot distinguish them perfectly (Lemma 2.4.6) without disturbing them. Bob's outcome -1 is exactly an unambiguous identification of the state.

Definition 4.2.10 (Six-state protocol): As BB84, but Alice and Bob choose uniformly among the three mutually unbiased bases X, Y, Z, where the Y-eigenstates are $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. The sifting step keeps a position with probability $\frac{1}{3}$.

Remark 4.2.11: The six states span the full Bloch sphere, whereas the BB84 states lie in the x-z plane. This symmetry restricts Eve more strongly: for a given disturbance she obtains less information (cf. Example 5.3.5), and the six-state protocol tolerates a higher error rate than BB84, at the price of a lower sifting rate.

Definition 4.2.12 (SARG04 protocol): Quantum transmission as in BB84; only sifting differs. For each qubit, Alice announces a pair of states $\{|\psi_{a0}\rangle, |\psi_{a'1}\rangle\}$, one from each basis, one of which is the state actually sent. Her key bit is 0 if the sent state is the Z-basis state of the pair and 1 otherwise. Bob declares the position valid if his outcome is inconsistent with one of the two states (and then knows Alice's bit), and invalid otherwise.

Lemma 4.2.13: Without noise, Bob's valid bits always agree with Alice's, and a position is valid with probability $\frac{1}{4}$.

Proof. By symmetry, let Alice send $|0\rangle$ and announce $\{|0\rangle, |+\rangle\}$ (key bit 0). If Bob measured Z (probability $\frac{1}{2}$), he obtains $|0\rangle$, which is consistent with both states: invalid. If he measured X, he obtains $|+\rangle$ (consistent with both: invalid) or $|-\rangle$ with probability $\frac{1}{2}$; the outcome $|-\rangle$ is impossible for $|+\rangle$, so Bob concludes that the state was $|0\rangle$, i.e., bit 0. Hence a position is valid with probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, and valid conclusions are always correct. □

Remark 4.2.14: Since Alice never announces the basis, SARG04 is more robust than BB84 against photon-number-splitting attacks (Section 5.3): Eve has to store and measure more photons to learn a bit.

4.3 Entanglement-based protocols

Definition 4.3.1: Let A_1, A_2, B_1, B_2 be random variables (or observables) with values in $\{-1, +1\}$. The *CHSH expression* is

$$S := \big|\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle\big|.$$

Lemma 4.3.2 (CHSH inequality): If A_1, A_2, B_1, B_2 are classical random variables on a common probability space, then $S \le 2$.

Proof. For every assignment of values ± 1, $A_1(B_1+B_2)+A_2(B_1-B_2) = \pm 2$, since one of $B_1 \pm B_2$ vanishes and the other equals ± 2. Taking expectations and using linearity, $|\langle A_1B_1\rangle + \langle A_1B_2\rangle + \langle A_2B_1\rangle - \langle A_2B_2\rangle| \le 2$. □

Theorem 4.3.1: Tsirelson's bound

Let A_1, A_2 (on $\mathcal{H}_A$) and B_1, B_2 (on $\mathcal{H}_B$) be Hermitian with $A_i^2 = \mathbb{I}$, $B_j^2 = \mathbb{I}$. Then for every state ρ_{AB},

$$\big|\mathrm{Tr}\big(\rho_{AB}\,\mathcal{X}\big)\big| \le 2\sqrt{2}, \qquad \mathcal{X} := A_1 \otimes (B_1 + B_2) + A_2 \otimes (B_1 - B_2).$$

If $[A_1, A_2] = 0$ (or $[B_1, B_2] = 0$), the bound improves to 2.

Proof. Since $(B_1 \pm B_2)^2 = 2\mathbb{I} \pm \{B_1, B_2\}$, $(B_1 + B_2)(B_1 - B_2) = -[B_1, B_2]$ and $(B_1 - B_2)(B_1 + B_2) = [B_1, B_2]$, expanding gives

$$\begin{aligned}\mathcal{X}^2 &= \underbrace{A_1^2}_{=\mathbb{I}} \otimes (B_1 + B_2)^2 + \underbrace{A_2^2}_{=\mathbb{I}} \otimes (B_1 - B_2)^2 - A_1A_2 \otimes [B_1, B_2] + A_2A_1 \otimes [B_1, B_2] \\ &= 4\,\mathbb{I} - [A_1, A_2] \otimes [B_1, B_2].\end{aligned}$$

Since $\|[A_1, A_2]\|_\infty \le 2\|A_1\|_\infty\|A_2\|_\infty = 2$ and likewise for B, $\|\mathcal{X}\|_\infty^2 = \|\mathcal{X}^2\|_\infty \le 4 + 4 = 8$, and $|\mathrm{Tr}(\rho\mathcal{X})| \le \|\mathcal{X}\|_\infty$. If the commutator vanishes, $\mathcal{X}^2 = 4\mathbb{I}$. □

Example 4.3.3: For $|\Phi^+\rangle$ with $A_1 = Z$, $A_2 = X$, $B_{1,2} = (Z \pm X)/\sqrt{2}$, Lemma 2.5.4 gives $\langle A_1B_1\rangle = \langle A_1B_2\rangle = \langle A_2B_1\rangle = \frac{1}{\sqrt{2}}$ and $\langle A_2B_2\rangle = -\frac{1}{\sqrt{2}}$, so $S = 2\sqrt{2}$: Tsirelson's bound is attained.

Definition 4.3.4 (Ekert protocol (E91)): The protocol consists of the following steps.

(i) A source distributes many copies of $|\Psi^-\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

(ii) On each pair, Alice measures a random observable from

$$A_1 = Z, \quad A_2 = X, \quad A_3 = \tfrac{1}{\sqrt{2}}(Z + X),$$

and Bob a random observable from

$$B_1 = Z, \quad B_2 = \tfrac{1}{\sqrt{2}}(Z - X), \quad B_3 = \tfrac{1}{\sqrt{2}}(Z + X).$$

(iii) They announce their choices. The outcomes of (A_1, B_1) and (A_3, B_3) are perfectly anti-correlated; after Bob flips his bits, they form the sifted key.

(iv) The outcomes of (A_1, B_3), (A_1, B_2), (A_2, B_3), (A_2, B_2) are used to evaluate

$$S = \big|\langle A_1 B_3\rangle + \langle A_1 B_2\rangle + \langle A_2 B_3\rangle - \langle A_2 B_2\rangle\big|.$$

(v) If S is sufficiently close to $2\sqrt{2}$, they perform error correction and privacy amplification.

Lemma 4.3.5: For $|\Psi^-\rangle$, the pairs (A_1, B_1) and (A_3, B_3) are perfectly anti-correlated, and $S = 2\sqrt{2}$. If instead $|\Phi^+\rangle$ is distributed (with the same observables), these pairs are perfectly correlated and still $S = 2\sqrt{2}$. With $|\Phi^+\rangle$ and $B_2 = Z$, $B_3 = X$, one obtains only $S = 2$.

Proof. All observables are of the form $\vec{a} \cdot \vec{\sigma}$ with $\vec{a}$ in the x-z plane. By Lemma 2.5.4, $\langle (\vec{a} \cdot \vec{\sigma}) \otimes (\vec{b} \cdot \vec{\sigma}) \rangle = -\vec{a} \cdot \vec{b}$ for $|\Psi^-\rangle$ and $+\vec{a} \cdot \vec{b}$ for $|\Phi^+\rangle$ (the y-components vanish). For equal directions $\vec{a} = \vec{b}$ this gives ∓ 1, i.e., perfect (anti-)correlation. With $\vec{a}_1 = \hat{z}$, $\vec{a}_2 = \hat{x}$, $\vec{b}_2 = (\hat{z} - \hat{x})/\sqrt{2}$, $\vec{b}_3 = (\hat{z} + \hat{x})/\sqrt{2}$:

$$\vec{a}_1 \cdot \vec{b}_3 = \vec{a}_1 \cdot \vec{b}_2 = \vec{a}_2 \cdot \vec{b}_3 = \tfrac{1}{\sqrt{2}}, \qquad \vec{a}_2 \cdot \vec{b}_2 = -\tfrac{1}{\sqrt{2}},$$

so the CHSH combination equals $\mp \frac{4}{\sqrt{2}} = \mp 2\sqrt{2}$. With $|\Phi^+\rangle$, $B_2 = Z$, $B_3 = X$: $\langle ZX \rangle = \langle XZ \rangle = 0$ and $\langle ZZ \rangle = \langle XX \rangle = 1$, so $S = |0 + 1 + 1 - 0| = 2$. □

Remark 4.3.6: A maximally entangled pure state cannot be correlated with a third system (any extension is $|\Psi^-\rangle\langle\Psi^-| \otimes \rho_E$ by Lemma 2.7.2). Therefore $S = 2\sqrt{2}$ certifies that Eve has no information. Any violation $S > 2$ still allows key generation (Section 6); conversely, if Alice and Bob's state is separable, no secret key can be obtained (Curty–Lewenstein–Lütkenhaus 2004).

Definition 4.3.7 (Entanglement-based BB84): Let C_1, C_2 be classical linear codes defining a CSS code which encodes m qubits into n qubits and corrects up to t errors (Appendix B). The protocol consists of the following steps.

(i) Alice prepares $2n$ pairs in the state $|\Phi^+\rangle^{\otimes 2n}$.

(ii) She randomly selects n of the pairs as *check pairs*.

(iii) She chooses a random string $b \in \{0,1\}^{2n}$ and applies H to her half of the i-th pair whenever $b_i = 1$.

(iv) She sends the second half of each pair to Bob.

(v) Bob announces the reception.

(vi) Alice announces b and the positions of the check pairs.

(vii) Bob applies H to the qubits with $b_i = 1$.

(viii) Alice and Bob measure the check pairs in the Z-basis and compare. If more than t errors occur, they abort.

(ix) Otherwise, they measure the syndromes of $\mathrm{CSS}(C_1, C_2)$ and correct the errors on the remaining n pairs, obtaining (approximately) $|\Phi^+\rangle^{\otimes m}$.

(x) They measure in the Z-basis to obtain the key.

Remark 4.3.8: Applying H before and after the transmission is equivalent to preparing and measuring in the X-basis. In Section 5.4 this protocol is shown to be secure and to reduce to the prepare-and-measure BB84 protocol.

4.4 From prepare-and-measure to entanglement-based protocols

In a prepare-and-measure protocol, Alice draws x according to p_X and sends $|\varphi_x\rangle_B$.

Lemma 4.4.1 (Source replacement): Let

$$|\Phi\rangle_{AB} := \sum_x \sqrt{p_X(x)}\,|x\rangle_A \otimes |\varphi_x\rangle_B,$$

where $\{|x\rangle_A\}$ is an orthonormal basis. If Alice measures A in the basis $\{|x\rangle_A\}$, outcome y occurs with probability $p_X(y)$, and the post-measurement state of B is $|\varphi_y\rangle_B$.

Proof.

$$\langle\Phi|(|y\rangle\langle y|_A \otimes \mathbb{I}_B)|\Phi\rangle = \sum_{x,x'} \sqrt{p_X(x)p_X(x')}\,\underbrace{\langle x|y\rangle}_{\delta_{x,y}}\underbrace{\langle y|x'\rangle}_{\delta_{y,x'}}\langle\varphi_x|\varphi_{x'}\rangle = p_X(y),$$

and $(|y\rangle\langle y|_A \otimes \mathbb{I}_B)|\Phi\rangle/\sqrt{p_X(y)} = |y\rangle_A|\varphi_y\rangle_B$. □

Remark 4.4.2: Thus any prepare-and-measure protocol is equivalent to an entanglement-based one in which Alice keeps A and sends B. Alice's reduced state

$$\rho_A = \mathrm{Tr}_B\,|\Phi\rangle\langle\Phi| = \sum_{x,x'} \sqrt{p_X(x)p_X(x')}\,\langle\varphi_{x'}|\varphi_x\rangle\,|x\rangle\langle x'|$$

is fixed by the protocol and cannot be touched by Eve. In the entanglement-based version (with an untrusted source), Eve controls Alice's system as well, so she is at least as powerful. Hence security of the entanglement-based protocol implies security of the prepare-and-measure protocol. The states $|\varphi_x\rangle$ must be non-orthogonal; otherwise Eve could measure them without disturbance.

Example 4.4.3: For BB84 with a fixed basis b, $|\Phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle|\psi_{0b}\rangle + |1\rangle|\psi_{1b}\rangle)$ is $|\Phi^+\rangle$ for $b = 0$ and $(\mathbb{I}\otimes H)|\Phi^+\rangle$ for $b = 1$. This is exactly the entanglement-based BB84 protocol (Definition 4.3.7), using $(H \otimes \mathbb{I})|\Phi^+\rangle = (\mathbb{I}\otimes H)|\Phi^+\rangle$.

4.5 Classical post-processing

After the quantum phase, Alice and Bob hold partially correlated and partially secret strings. Post-processing consists of parameter estimation (PE), error correction (EC) and privacy amplification (PA). If PE detects too many errors, they abort; otherwise EC makes the strings identical and PA removes Eve's information.

Parameter estimation

Theorem 4.5.1: Serfling's inequality

Let $k_1, \ldots, k_N \in \{0,1\}$ with average $\bar{K} := \frac{1}{N}\sum_i k_i$. Draw a sample of size n without replacement and let $\bar{X}$ denote the sample average. Then for $0 \le \beta \le 1$,

$$\Pr[\bar{X} \ge \bar{K} + \beta] \le \exp\Big(-\frac{2\beta^2 nN}{N - n + 1}\Big).$$

Proof. See Serfling (1974). Compare with Hoeffding's inequality (Theorem A.1.1) for sampling with replacement, $e^{-2n\beta^2}$; sampling without replacement improves the exponent by the factor $N/(N-n+1)$. □

Let the sifted strings of length $N = n + k$ be split into a random test sample of size k and the remaining n bits, $K_A = K_A^k K_A^n$, $K_B = K_B^k K_B^n$. Define the error rates

$$\lambda_k := \tfrac{1}{k}|K_A^k \oplus K_B^k|, \qquad \lambda_n := \tfrac{1}{n}|K_A^n \oplus K_B^n|, \qquad \lambda := \tfrac{1}{N}|K_A \oplus K_B| = \nu\lambda_k + (1-\nu)\lambda_n,$$

where $|\cdot|$ is the Hamming weight and $\nu := k/N$. The protocol aborts if $\lambda_k > \lambda_{\max}$; let $p_{\text{pass}} := \Pr[\lambda_k \le \lambda_{\max}]$.

Corollary 4.5.1: For every $\gamma > 0$,

$$\Pr[\lambda_n \ge \lambda_k + \gamma \mid \lambda_k \le \lambda_{\max}] \le \frac{1}{p_{\text{pass}}} \exp\Big(-\frac{2k^2 n\gamma^2}{(k+1)N}\Big).$$

Proof. By Bayes' rule, $\Pr[\mathcal{A} \mid \mathcal{B}] = \Pr[\mathcal{A} \cap \mathcal{B}]/\Pr[\mathcal{B}] \le \Pr[\mathcal{A}]/\Pr[\mathcal{B}]$, so the left-hand side is at most $\Pr[\lambda_n \ge \lambda_k + \gamma]/p_{\text{pass}}$. Since $\lambda = \nu\lambda_k + (1-\nu)\lambda_n$,

$$\lambda_n \ge \lambda_k + \gamma \iff \nu\lambda_n \ge \nu\lambda_k + \nu\gamma \iff \lambda_n \ge \nu\lambda_k + (1-\nu)\lambda_n + \nu\gamma = \lambda + \nu\gamma.$$

The n remaining bits form a uniformly random sample without replacement of the N bits of $K_A \oplus K_B$, with sample average λ_n and population average λ. Theorem 4.5.1 with $\beta = \nu\gamma = k\gamma/N$ and $N - n + 1 = k + 1$ gives

$$\Pr[\lambda_n \ge \lambda + \nu\gamma] \le \exp\Big(-\frac{2k^2\gamma^2}{N^2} \cdot \frac{nN}{k+1}\Big) = \exp\Big(-\frac{2k^2 n\gamma^2}{(k+1)N}\Big). \qquad \square$$

Remark 4.5.2: The bound is exponentially small in the sample size: if the error rate observed on the sample is small, then with high probability the error rate on the remaining bits is small as well.

Error correction

Example 4.5.3: A simple (inefficient) scheme: Alice picks two random positions and announces them together with the XOR of her two bits. Bob compares with the XOR of his bits. If they differ, both bits are discarded; if they agree, the first bit is kept and the second discarded. Eve learns only the parity, which is compensated by discarding the second bit. Even without errors, at least half of the bits are lost.

Remark 4.5.4: In *error correction* (information reconciliation), Bob computes a guess $\hat{K}_A$ of Alice's string from his string and classical messages sent by Alice. These messages leak leak_{EC} bits to Eve. For an error rate Q, at least $\approx nh(Q) = nH(K_A|K_B)$ bits must be communicated (Slepian–Wolf); practical codes (Cascade, LDPC) achieve $\text{leak}_{\text{EC}} = f\,n\,h(Q)$ with efficiency f slightly above 1.

Definition 4.5.5: A family $\mathcal{F}$ of functions $f : \mathcal{X} \to \mathcal{Z}$ with a probability distribution $p_{\mathcal{F}}$ is *two-universal* if

$$\Pr_{f \sim p_{\mathcal{F}}}[f(x) = f(x')] \le \frac{1}{|\mathcal{Z}|} \qquad \text{for all } x \ne x'.$$

Example 4.5.6: The family of all linear maps $f_M(x) = Mx$, $M \in \mathbb{F}_2^{\ell \times n}$, with M uniform, is two-universal from $\{0,1\}^n$ to $\{0,1\}^\ell$: for $x \ne x'$, write $z := x - x' \ne 0$, say $z_j = 1$. Conditioned on all columns of M except the j-th, Mz is the j-th column plus a fixed vector, hence uniform; so $\Pr[Mz = 0] = 2^{-\ell}$. Smaller families (e.g., Toeplitz matrices, needing only $n + \ell - 1$ random bits) also work.

Lemma 4.5.7 (Verification of error correction): Alice chooses f_{EC} from a two-universal family with output length $\lceil \log(1/\varepsilon_{\mathrm{cor}}) \rceil$ and sends f_{EC} and $f_{\mathrm{EC}}(K_A)$; Bob aborts if $f_{\mathrm{EC}}(\hat{K}_A) \neq f_{\mathrm{EC}}(K_A)$. Then

$$\Pr[K_A \neq \hat{K}_A \text{ and not abort}] \leq \varepsilon_{\mathrm{cor}}.$$

Proof. Since f_{EC} is chosen independently of $(K_A, \hat{K}_A)$, two-universality gives

$$\Pr[f(K_A) = f(\hat{K}_A) \mid K_A \neq \hat{K}_A] \leq 2^{-\lceil \log(1/\varepsilon_{\mathrm{cor}}) \rceil} \leq \varepsilon_{\mathrm{cor}}.$$

Hence $\Pr[K_A \neq \hat{K}_A \wedge f(K_A) = f(\hat{K}_A)] = \Pr[f(K_A) = f(\hat{K}_A) \mid K_A \neq \hat{K}_A] \Pr[K_A \neq \hat{K}_A] \leq \varepsilon_{\mathrm{cor}}$. □

Remark 4.5.8: The check is independent of the error rate and of the error-correcting code; it costs only $\lceil \log(1/\varepsilon_{\mathrm{cor}}) \rceil$ additional bits of leakage.

Privacy amplification

Definition 4.5.9: A function $\mathrm{Ext} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is a (k, ε)-*strong quantum-proof randomness extractor* if for all cq-states ρ_{XE} with $X \in \{0,1\}^n$ and $H_{\min}(X|E) \geq k$, and a uniform seed $Y \in \{0,1\}^d$ independent of XE,

$$\tfrac{1}{2} \left\| \rho_{\mathrm{Ext}(X,Y)YE} - \frac{\mathbb{I}}{2^m} \otimes \rho_Y \otimes \rho_E \right\|_1 \leq \varepsilon.$$

Remark 4.5.10: *Strong* means that the output is independent also of the seed, which is announced publicly. The ideal state $\frac{\mathbb{I}}{2^m} \otimes \rho_Y \otimes \rho_E$ describes a uniform key independent of Y and E. Examples: two-universal hashing (Theorem 4.5.2), Trevisan's extractor.

Theorem 4.5.2: Quantum leftover hash lemma

Let $\rho_{XE} = \sum_x |x\rangle\langle x| \otimes \rho_E^x$ (with $\operatorname{Tr} \rho_E^x = p(x)$) be a cq-state, and let f be chosen (with seed F) uniformly from a two-universal family with output in $\mathcal{Z} = \{0,1\}^\ell$. Then for every $\varepsilon' \geq 0$,

$$\tfrac{1}{2} \left\| \rho_{f(X)FE} - \rho_U \otimes \rho_{FE} \right\|_1 \leq 2\varepsilon' + \tfrac{1}{2} \sqrt{2^{\,\ell - H_{\min}^{\varepsilon'}(X|E)}},$$

where $\rho_U = \mathbb{I}/|\mathcal{Z}|$.

Proof. Step 1 (reduction to a fixed f). The seed register is classical and uniform, so both states are block-diagonal in F, and

$$\tfrac{1}{2} \left\| \rho_{f(X)FE} - \rho_U \otimes \rho_{FE} \right\|_1 = \mathbb{E}_f \, \tfrac{1}{2} \|\Delta_f\|_1, \qquad \Delta_f := \rho_{ZE}^f - \rho_U \otimes \rho_E,$$

where $\rho_{ZE}^f = \sum_z |z\rangle\langle z| \otimes \rho_E^{f,z}$, $\rho_E^{f,z} := \sum_{x : f(x) = z} \rho_E^x$.

Step 2 (from trace norm to a weighted 2-norm). Let σ_E be a state with $\operatorname{supp} \rho_E \subset \operatorname{supp} \sigma_E$ and $\tau := \mathbb{I}_Z \otimes \sigma_E$, so $\operatorname{Tr} \tau = |\mathcal{Z}|$. By Hölder's inequality $\|ABC\|_1 \leq \|A\|_4 \|B\|_2 \|C\|_4$,

$$\|\Delta_f\|_1 = \left\| \tau^{1/4} \left(\tau^{-1/4} \Delta_f \tau^{-1/4} \right) \tau^{1/4} \right\|_1 \leq \sqrt{|\mathcal{Z}|} \left\| \tau^{-1/4} \Delta_f \tau^{-1/4} \right\|_2.$$

By Jensen's inequality, $\mathbb{E}_f \|\Delta_f\|_1 \leq \sqrt{|\mathcal{Z}|} \left(\mathbb{E}_f \operatorname{Tr}\left[(\Delta_f \tau^{-1/2})^2 \right] \right)^{1/2}$.

Step 3 (collision estimate). Write $\langle A, B \rangle_\sigma := \operatorname{Tr}(A \sigma^{-1/2} B \sigma^{-1/2})$, which is ≥ 0 for $A, B \geq 0$. Expanding the square,

$$\operatorname{Tr}\left[(\Delta_f \tau^{-1/2})^2 \right] = \sum_z \langle \rho_E^{f,z}, \rho_E^{f,z} \rangle_\sigma - \frac{2}{|\mathcal{Z}|} \langle \rho_E, \rho_E \rangle_\sigma + \frac{1}{|\mathcal{Z}|} \langle \rho_E, \rho_E \rangle_\sigma,$$

using $\sum_z \rho_E^{f,z} = \rho_E$. The first term equals $\sum_{x,x':f(x)=f(x')} \langle \rho_E^x, \rho_E^{x'} \rangle_\sigma$. Averaging over f with two-universality and the positivity of each term,

$$\mathbb{E}_f \sum_z \langle \rho_E^{f,z}, \rho_E^{f,z} \rangle_\sigma \le \sum_x \langle \rho_E^x, \rho_E^x \rangle_\sigma + \frac{1}{|\mathcal{Z}|} \sum_{x \neq x'} \langle \rho_E^x, \rho_E^{x'} \rangle_\sigma \le \Gamma + \frac{1}{|\mathcal{Z}|} \langle \rho_E, \rho_E \rangle_\sigma,$$

with $\Gamma := \sum_x \langle \rho_E^x, \rho_E^x \rangle_\sigma$. Hence $\mathbb{E}_f \operatorname{Tr}[(\Delta_f \tau^{-1/2})^2] \le \Gamma$.

Step 4 (min-entropy). Let $\rho_{XE} \le \lambda \mathbb{I}_X \otimes \sigma_E$, i.e., $\rho_E^x \le \lambda \sigma_E$ for all x. Then $\sigma_E^{-1/2} \rho_E^x \sigma_E^{-1/2} \le \lambda \mathbb{I}$ and

$$\Gamma = \sum_x \operatorname{Tr}\big(\rho_E^x \, \sigma_E^{-1/2} \rho_E^x \sigma_E^{-1/2}\big) \le \lambda \sum_x \operatorname{Tr} \rho_E^x = \lambda.$$

Choosing $\lambda = 2^{-H_{\min}(X|E)}$ (and the optimal σ_E), Steps 1–4 give $\frac{1}{2} \| \rho_{f(X)FE} - \rho_U \otimes \rho_{FE} \|_1 \le \frac{1}{2} \sqrt{|\mathcal{Z}| \, 2^{-H_{\min}(X|E)}}$, which is the claim for $\varepsilon' = 0$.

Step 5 (smoothing). Let $\tilde{\rho}_{XE} \in \mathcal{B}^{\varepsilon'}(\rho_{XE})$ attain $H_{\min}^{\varepsilon'}(X|E)_\rho$ (it can be chosen classical on X). Then $\frac{1}{2} \| \rho - \tilde{\rho} \|_1 \le P(\rho, \tilde{\rho}) \le \varepsilon'$ (Theorem 2.8.3). Applying hashing (a CPTP map) and the map $\rho_{XE} \mapsto \rho_U \otimes \rho_{FE}$ (also CPTP) to ρ and $\tilde{\rho}$, the triangle inequality and Theorem 2.8.1 give an additional error of at most $\varepsilon' + \varepsilon'$. □

Corollary 4.5.11: The final key $f(X)$ is ε_{sec}-close to ideal whenever

$$\ell \le H_{\min}^{\varepsilon'}(X|E) + 2 - 2 \log \frac{1}{\varepsilon_{\text{sec}} - 2\varepsilon'}.$$

Proof. $\frac{1}{2} \sqrt{2^{\ell - H_{\min}^{\varepsilon'}}} \le \varepsilon_{\text{sec}} - 2\varepsilon'$ is equivalent to $\ell - H_{\min}^{\varepsilon'} \le 2 \log\big(2(\varepsilon_{\text{sec}} - 2\varepsilon')\big) = 2 - 2 \log \frac{1}{\varepsilon_{\text{sec}} - 2\varepsilon'}$. □

Remark 4.5.12: Privacy amplification does not destroy correctness: if $K_A = \hat{K}_A$, then $f_{\text{PA}}(K_A) = f_{\text{PA}}(\hat{K}_A)$. Hence $\Pr[f_{\text{PA}}(K_A) \neq f_{\text{PA}}(\hat{K}_A)] \le \Pr[K_A \neq \hat{K}_A]$. Two-universal hashing thus serves both for verifying EC (short output) and for PA (long output).

5 Security analysis

5.1 Definition of security

Let K_A, K_B be the keys output by the protocol (with $K_A = K_B = \perp$ in case of abort), $p_\perp$ the abort probability, and ρ^{pass} the final state conditioned on not aborting. Security has two parts: *correctness* (Alice's and Bob's keys agree) and *secrecy* (Eve has no information on the key).

Definition 5.1.1 (ε-correctness): A protocol is *ε-correct* if $\Pr[K_A \neq K_B] \le \varepsilon$.

Definition 5.1.2 (ε-secrecy): Let $\rho_{K_A E}^{\text{pass}} = \sum_{k \in \mathcal{S}} p_{K_A}(k) |k\rangle\langle k| \otimes \rho_E^k$. The protocol is *$\varepsilon$-secret* if

$$(1 - p_\perp) \, \tfrac{1}{2} \big\| \rho_{K_A E}^{\text{pass}} - \rho_U \otimes \rho_E \big\|_1 \le \varepsilon, \qquad \rho_U := \frac{1}{|\mathcal{S}|} \sum_{u \in \mathcal{S}} |u\rangle\langle u|.$$

Remark 5.1.3: Let $\rho_{K_A E} = p_\perp |\perp\rangle\langle\perp| \otimes \rho_E^\perp + (1 - p_\perp) \rho_{K_A E}^{\text{pass}}$ be the real state and $\tilde{\rho}_{K_A E} = p_\perp |\perp\rangle\langle\perp| \otimes \rho_E^\perp + (1 - p_\perp) \rho_U \otimes \rho_E^{\text{pass}}$ the ideal one: in case of abort nothing changes, and otherwise the key is replaced by a uniform one independent of Eve. Since both are block-diagonal in the abort flag,

$$\| \rho_{K_A E} - \tilde{\rho}_{K_A E} \|_1 = p_\perp \cdot 0 + (1 - p_\perp) \| \rho_{K_A E}^{\text{pass}} - \rho_U \otimes \rho_E \|_1,$$

which explains the factor $1 - p_\perp$. A protocol which (almost) always aborts is trivially secure.

Definition 5.1.4 (ε-security): The protocol is *ε-secure* if

$$(1-p_\perp)\,\tfrac{1}{2}\big\|\rho^{\text{pass}}_{K_AK_BE}-\rho_{UU}\otimes\rho_E\big\|_1\le\varepsilon,\qquad \rho_{UU}:=\frac{1}{|\mathcal{S}|}\sum_{u\in\mathcal{S}}|u\rangle\langle u|\otimes|u\rangle\langle u|.$$

Remark 5.1.5: By Lemma 2.8.2, ε-security means that no process whatsoever can distinguish the real key from a perfect key (uniform, identical for Alice and Bob, independent of Eve) with advantage larger than ε. By Theorem 2.8.1, this property is preserved under any further processing; ε can be interpreted as the probability that the real protocol deviates from the ideal one. A perfect key of length ℓ assigns probability $2^{-\ell}$ to every string; ε is the weight that has to be moved to turn the real distribution (jointly with Eve) into the ideal one.

Theorem 5.1.1: Correctness and secrecy imply security

If a protocol is ε_{cor}-correct and ε_{sec}-secret, then it is ε-secure with $\varepsilon=\varepsilon_{\text{cor}}+\varepsilon_{\text{sec}}$.

Proof. Write $D(\rho,\sigma)=\frac{1}{2}\|\rho-\sigma\|_1$ and $\rho:=\rho^{\text{pass}}_{K_AK_BE}=\sum_{k_A,k_B}p(k_A,k_B)|k_A\rangle\langle k_A|\otimes|k_B\rangle\langle k_B|\otimes\rho_E^{k_A,k_B}$. Let

$$\gamma:=\sum_{k_A}p(k_A)|k_A\rangle\langle k_A|\otimes|k_A\rangle\langle k_A|\otimes\rho_E^{k_A},\qquad \rho_E^{k_A}:=\sum_{k_B}p(k_B|k_A)\rho_E^{k_A,k_B},$$

be the state where Bob's key is replaced by a copy of Alice's. By the triangle inequality,

$$D(\rho,\rho_{UU}\otimes\rho_E)\le D(\rho,\gamma)+D(\gamma,\rho_{UU}\otimes\rho_E).$$

Second term: γ and $\rho_{UU}\otimes\rho_E$ arise from $\rho^{\text{pass}}_{K_AE}$ and $\rho_U\otimes\rho_E$ by the isometry $|k\rangle\mapsto|k\rangle|k\rangle$, so $D(\gamma,\rho_{UU}\otimes\rho_E)=D(\rho^{\text{pass}}_{K_AE},\rho_U\otimes\rho_E)$. *First term:* $\rho-\gamma$ is block-diagonal in (k_A,k_B). The diagonal blocks are $p(k_A,k_A)\rho_E^{k_A,k_A}-p(k_A)\rho_E^{k_A}=-\sum_{k_B\neq k_A}p(k_A,k_B)\rho_E^{k_A,k_B}$ and the off-diagonal blocks ($k_A\neq k_B$) are $p(k_A,k_B)\rho_E^{k_A,k_B}$. Hence

$$\|\rho-\gamma\|_1\le 2\sum_{k_A\neq k_B}p(k_A,k_B)=2\Pr[K_A\neq K_B\mid\text{pass}].$$

Multiplying by $1-p_\perp$ and using $(1-p_\perp)\Pr[K_A\neq K_B\mid\text{pass}]=\Pr[K_A\neq K_B]\le\varepsilon_{\text{cor}}$ (in case of abort the keys agree) yields

$$(1-p_\perp)D(\rho,\rho_{UU}\otimes\rho_E)\le\varepsilon_{\text{cor}}+\varepsilon_{\text{sec}}.$$ □

Definition 5.1.6 (ε-robustness): A protocol is *ε-robust* if, for the honest implementation without eavesdropper (with the expected noise), the abort probability is at most ε.

Remark 5.1.7: Security alone can be achieved trivially by always aborting; robustness excludes this. Some authors include robustness in the notion of security; here it is an additional criterion.

Lemma 5.1.8 (Sequential composition): Let a key from an ε_1-secure QKD protocol be used in a protocol $\mathcal{P}$ which is ε_2-secure when run with a perfect key. Then the composed protocol is $(\varepsilon_1+\varepsilon_2)$-secure.

Proof. Let $\mathcal{P}$ also denote the CPTP map describing the subsequent protocol (acting on the key and possibly on Eve's system), ρ the real QKD output and $\tilde\rho$ the ideal one. With $\mathcal{I}$ the ideal functionality of $\mathcal{P}$,

$$\tfrac{1}{2}\|\mathcal{P}(\rho)-\mathcal{I}\|_1\le\tfrac{1}{2}\|\mathcal{P}(\rho)-\mathcal{P}(\tilde\rho)\|_1+\tfrac{1}{2}\|\mathcal{P}(\tilde\rho)-\mathcal{I}\|_1\le\varepsilon_1+\varepsilon_2$$

by the triangle inequality and Theorem 2.8.1. □

Remark 5.1.9: Parallel composition is shown similarly. Older security criteria, e.g., bounding the accessible information of Eve on the key, are not composable: Eve may store her quantum system and measure it only after part of the key has been used.

5.2 Assumptions

The following fundamental assumptions underlie all security statements:

(i) *Quantum theory is correct*: it predicts measurement statistics accurately.

(ii) *Quantum theory is complete*: Eve cannot obtain more information than quantum mechanics allows. (Correctness together with the existence of free randomness, e.g., for basis choices, implies completeness; Colbeck–Renner.)

(iii) *Authentic classical communication* is possible: Eve cannot impersonate Alice or Bob.

In addition, the model of the implementation must match the devices. Typical gaps are:

(iv) *Isolation of the labs*: Eve may probe Alice's and Bob's devices through their interface with the channel (e.g., Trojan-horse attacks).

(v) *State preparation*: real sources do not prepare the ideal states (e.g., multi-photon pulses).

(vi) *Measurement devices*: detectors do not implement the ideal POVM, may respond to states outside the intended Hilbert space, have losses, and must be calibrated.

(vii) *Timing*: synchronization of signals; detector dead times.

(viii) *Post-processing*: the leakage estimate must account for all deviations from the ideal model.

Remark 5.2.1: Assumptions (iv)–(viii) are the main gap between theory and practice: deviations of real devices open *side channels*. Device-independent QKD (Section 6) removes them almost entirely; MDI QKD (Section 7.2) removes them for the detectors.

5.3 Eavesdropping strategies

Lemma 5.3.1 (No information without disturbance): Let Eve attach an ancilla $|E\rangle$ and apply a unitary U such that two non-orthogonal states $|\psi\rangle, |\varphi\rangle$ sent by Alice are not disturbed:

$$U|\psi\rangle|E\rangle = |\psi\rangle|E_\psi\rangle, \qquad U|\varphi\rangle|E\rangle = |\varphi\rangle|E_\varphi\rangle.$$

Then $|E_\psi\rangle = |E_\varphi\rangle$, i.e., Eve's ancilla carries no information about which state was sent.

Proof. Taking inner products and using unitarity, $\langle\psi|\varphi\rangle\langle E|E\rangle = \langle\psi|\varphi\rangle\langle E_\psi|E_\varphi\rangle$. Since $\langle\psi|\varphi\rangle \neq 0$ and $\langle E|E\rangle = 1$, $\langle E_\psi|E_\varphi\rangle = 1$, so the unit vectors coincide. □

Remark 5.3.2: If instead the states are disturbed, $U|\psi\rangle|E\rangle = |\psi'\rangle|E_\psi\rangle$ and $U|\varphi\rangle|E\rangle = |\varphi'\rangle|E_\varphi\rangle$, then $\langle\psi|\varphi\rangle = \langle\psi'|\varphi'\rangle\langle E_\psi|E_\varphi\rangle$. To distinguish her ancilla states, Eve needs $|\langle E_\psi|E_\varphi\rangle|$ small, which forces $|\langle\psi'|\varphi'\rangle|$ to be larger than $|\langle\psi|\varphi\rangle|$: the more information Eve gains, the more she disturbs the states (*information–disturbance trade-off*).

Definition 5.3.3: Consider a protocol with N signals $\rho_{A_1}, \ldots, \rho_{A_N}$ sent by Alice. Eve attaches ancillas and applies unitaries; her ancilla state is $\rho_E = \mathrm{Tr}_A\big(U(\rho_A \otimes |E\rangle\langle E|)U^\dagger\big)$.

(i) In an *individual attack*, Eve applies the same unitary U to each signal and its own ancilla, and measures each ancilla separately with the same POVM (before the end of the protocol).

(ii) In a *collective attack*, Eve applies the same U to each signal and its own ancilla, stores the ancillas in a quantum memory, and measures them jointly at any later time, in particular after all classical communication. The state of Alice, Bob and Eve is then $\rho_{ABE}^{\otimes N}$.

(iii) In a *coherent* (or *general*) *attack*, Eve applies a global unitary U_G to $\rho_{A_1}\otimes\cdots\otimes\rho_{A_N}$ and one large ancilla, followed by a global measurement at any time: any strategy allowed by quantum mechanics.

Remark 5.3.4: Individual $\subset$ collective $\subset$ coherent. For protocols which are symmetric under permutations of the rounds, security against collective attacks implies security against coherent attacks, using de Finetti-type theorems or the postselection technique (Remark 5.5.2).

Theorem 5.3.1: Csiszár–Körner

If Alice, Bob and Eve hold i.i.d. classical random variables A, B, E, then with one-way communication from Alice to Bob a secret key can be generated at rate

$$r \geq I(A:B) - I(A:E).$$

Proof. See Csiszár–Körner (1978). The quantum generalization is Theorem 5.5.1. □

Example 5.3.5 (Optimal individual attacks): The most general individual attack on a qubit is

$$U|0\rangle|E\rangle = \sqrt{F}|0\rangle|E_{00}\rangle + \sqrt{1-F}|1\rangle|E_{01}\rangle, \qquad U|1\rangle|E\rangle = \sqrt{F}|1\rangle|E_{10}\rangle + \sqrt{1-F}|0\rangle|E_{11}\rangle,$$

where $F = \langle\psi_{\text{in}}|\rho_B|\psi_{\text{in}}\rangle$ is the fidelity of Bob's state with Alice's state, taken equal for all bases (otherwise Alice and Bob would notice). With the disturbance $D = 1-F$ (the error rate), the optimal attack on BB84 gives (Fuchs et al. 1997)

$$I(A:B) = 1 - h(D), \qquad I(A:E) = 1 - h\Big(\tfrac{1}{2} + \sqrt{D(1-D)}\Big),$$

and for the six-state protocol $I(A:E) = 1 + (1-D)\big(g\log g + (1-g)\log(1-g)\big)$ with $g = \frac{1}{2}\big(1+\frac{\sqrt{D(2-3D)}}{1-D}\big)$. By Theorem 5.3.1, a key can be distilled as long as $I(A:B) > I(A:E)$, i.e., for $D < \frac{1}{2}(1-\frac{1}{\sqrt{2}}) \approx 14.6\%$ (BB84; at this point $\frac{1}{2} + \sqrt{D(1-D)} = 1-D$) and $D \lesssim 15.6\%$ (six-state).

Remark 5.3.6: For coherent attacks on BB84 and the six-state protocol it was found that they do not increase Eve's information per bit, but they increase her probability of guessing the whole message.

Photon-number-splitting attack

Definition 5.3.7: A coherent state with amplitude $\alpha\in\mathbb{C}$ is $|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{k\geq 0}\frac{\alpha^k}{\sqrt{k!}}|k\rangle$, where $|k\rangle$ is the k-photon Fock state. A *weak coherent pulse* with mean photon number $\mu = |\alpha|^2$ is the phase-randomized state

$$\rho_\mu = \int_0^{2\pi}\frac{d\phi}{2\pi}||\alpha|e^{i\phi}\rangle\langle|\alpha|e^{i\phi}| = \sum_{k=0}^{\infty} e^{-\mu}\frac{\mu^k}{k!}|k\rangle\langle k|.$$

Derivation. $||\alpha|e^{i\phi}\rangle\langle|\alpha|e^{i\phi}| = e^{-\mu}\sum_{k,l}\frac{|\alpha|^{k+l}e^{i(k-l)\phi}}{\sqrt{k!\,l!}}|k\rangle\langle l|$, and $\int_0^{2\pi}e^{i(k-l)\phi}\frac{d\phi}{2\pi} = \delta_{kl}$. □

Example 5.3.8: For $\mu = 0.1$: $P(0) = e^{-\mu} \approx 90.5\%$, $P(1) = \mu e^{-\mu} \approx 9.0\%$, $P(k\geq 2) = 1-(1+\mu)e^{-\mu} \approx 0.47\% \approx \mu^2/2$. Vacuum pulses only reduce the rate; single-photon pulses behave like ideal qubits; multi-photon pulses carry several copies of the same polarization.

Example 5.3.9 (PNS attack): Let the channel have single-photon transmittance η. Bob expects a non-vacuum detection with probability $1-e^{-\mu\eta}$. Eve replaces the channel by a lossless one and performs a quantum non-demolition measurement of the photon number (which does not disturb the polarization). Then:

(i) vacuum pulses are forwarded (no information);

(ii) from multi-photon pulses she splits off one photon, stores it, and forwards the rest; after the basis announcement she measures the stored photon in the correct basis and learns the bit without error;

(iii) she blocks a fraction of the single-photon pulses so that Bob's detection rate equals $1-e^{-\mu\eta}$; on the remaining single photons she may perform any other attack.

If $P(k\ge 2)\ge 1-e^{-\mu\eta}$, Eve can block *all* single-photon pulses and learns the entire key without introducing errors. For $\mu=0.1$ this holds if $\eta\le -\ln(1-0.00468)/0.1\approx 0.047$, i.e., for losses $\gtrsim 13.3\,\mathrm{dB}$, which corresponds to about $66\,\mathrm{km}$ of standard fiber ($0.2\,\mathrm{dB/km}$). Roughly, the condition is $\eta\lesssim \mu/2$.

Remark 5.3.10 (Decoy states): The *decoy-state method* (Hwang 2003; Lo–Ma–Chen 2005; Wang 2005) defeats the PNS attack. Alice randomly varies the mean photon number between a signal intensity μ and decoy intensities (e.g., ν, and vacuum). Since Eve only sees the photon number, the *yield* Y_k (probability of a detection given a k-photon pulse) and the error rate e_k do not depend on the intensity. Alice and Bob measure the *gains* and error rates

$$Q_\mu=\sum_{k\ge 0} Y_k e^{-\mu}\frac{\mu^k}{k!},\qquad E_\mu Q_\mu=\sum_{k\ge 0} e_k Y_k e^{-\mu}\frac{\mu^k}{k!}$$

for each intensity. A PNS attack drastically lowers Y_1 relative to Y_k, $k\ge 2$, and is revealed because the gains of different intensities no longer fit a channel with $Y_k\approx 1-(1-\eta)^k$. Qualitatively: a strong decoy (e.g., $\mu_D=0.8$ sent with probability 5%) has many more multi-photon pulses; under a PNS attack Bob observes a much smaller loss for decoy pulses than for signal pulses.

Lemma 5.3.11 (Vacuum + weak decoy bounds): Let $0<\nu<\mu$. Then

$$Y_1\ge Y_1^{L}:=\frac{\mu}{\nu(\mu-\nu)}\Big(Q_\nu e^{\nu}-\frac{\nu^2}{\mu^2}Q_\mu e^{\mu}-\frac{\mu^2-\nu^2}{\mu^2}Y_0\Big),\qquad e_1\le\frac{E_\nu Q_\nu e^{\nu}-e_0 Y_0}{Y_1^{L}\,\nu}.$$

Proof. From the expansions of the gains,

$$Q_\nu e^{\nu}-\frac{\nu^2}{\mu^2}Q_\mu e^{\mu}=Y_0\Big(1-\frac{\nu^2}{\mu^2}\Big)+Y_1\Big(\nu-\frac{\nu^2}{\mu}\Big)+\sum_{k\ge 3}\frac{Y_k}{k!}\nu^2\big(\nu^{k-2}-\mu^{k-2}\big),$$

where the $k=2$ terms cancel. Since $\nu<\mu$, the last sum is ≤ 0; solving for Y_1 gives the first bound. For the second, $E_\nu Q_\nu e^{\nu}=\sum_k e_k Y_k \nu^k/k!\ge e_0 Y_0+e_1 Y_1\nu$. □

Theorem 5.3.2: GLLP key rate

For BB84 with phase-randomized weak coherent pulses, the asymptotic key rate is

$$r\ge q\big\{-Q_\mu f\,h(E_\mu)+Q_1\big[1-h(e_1)\big]\big\},\qquad Q_1=Y_1\mu e^{-\mu},$$

where q is the sifting factor and f the error-correction efficiency. Only single-photon contributions yield key; error correction must be paid for all detected pulses.

Proof. See Gottesman–Lo–Lütkenhaus–Preskill (2004). With decoy states, Q_1 and e_1 are replaced by the bounds of Lemma 5.3.11. □

5.4 Security of BB84

The proof (Shor–Preskill 2000, building on Lo–Chau and Mayers) first shows security of the entanglement-based protocol and then reduces it to the prepare-and-measure protocol.

Security of the entanglement-based version

Lemma 5.4.1: Let $\varepsilon \geq 0$ and let ρ_{AB} be a state of m qubit pairs with

$$F\big(\rho_{AB}, |\Phi^+\rangle\langle\Phi^+|^{\otimes m}\big) \geq 1 - \varepsilon^2.$$

Let K_A, K_B be the m-bit strings obtained by measuring A and B in the computational basis, and let Eve hold a purification of ρ_{AB}. Then there is a state σ_E with

$$\tfrac{1}{2}\big\|\rho_{K_AK_BE} - \rho_{UU} \otimes \sigma_E\big\|_1 \leq \varepsilon.$$

In particular, the key is 2ε-secure.

Proof. Let ρ_{ABE} be a purification. Every purification of the pure state $|\Phi^+\rangle\langle\Phi^+|^{\otimes m}$ has the form $|\Phi^+\rangle\langle\Phi^+|^{\otimes m} \otimes \sigma_E$ with σ_E pure, so by Theorem 2.8.2 there is σ_E with

$$F\big(\rho_{ABE}, |\Phi^+\rangle\langle\Phi^+|^{\otimes m} \otimes \sigma_E\big) = F\big(\rho_{AB}, |\Phi^+\rangle\langle\Phi^+|^{\otimes m}\big) \geq 1 - \varepsilon^2.$$

By Theorem 2.8.3,

$$\tfrac{1}{2}\big\|\rho_{ABE} - |\Phi^+\rangle\langle\Phi^+|^{\otimes m} \otimes \sigma_E\big\|_1 \leq \sqrt{1 - (1-\varepsilon^2)} = \varepsilon.$$

Measuring both halves of $|\Phi^+\rangle^{\otimes m}$ in the computational basis yields ρ_{UU}, since the outcomes are perfectly correlated and uniform. Since this measurement is CPTP, Theorem 2.8.1 implies $\frac{1}{2}\|\rho_{K_AK_BE} - \rho_{UU} \otimes \sigma_E\|_1 \leq \varepsilon$. Tracing out K_AK_B gives $\frac{1}{2}\|\rho_E - \sigma_E\|_1 \leq \varepsilon$, and the triangle inequality yields $\frac{1}{2}\|\rho_{K_AK_BE} - \rho_{UU} \otimes \rho_E\|_1 \leq 2\varepsilon$. □

Lemma 5.4.2 (Bell-basis error detection): A Pauli error on Bob's qubit maps $|\Phi^+\rangle$ to another Bell state: X (*bit flip*) gives $|\Psi^+\rangle$, Z (*phase flip*) gives $|\Phi^-\rangle$, and Y (both) gives $|\Psi^-\rangle$ up to a phase. The projectors onto the states with a bit flip, resp. a phase flip, are

$$\Pi_{\text{bf}} := |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-| = \tfrac{1}{2}(\mathbb{I} - Z\otimes Z), \qquad \Pi_{\text{pe}} := |\Phi^-\rangle\langle\Phi^-| + |\Psi^-\rangle\langle\Psi^-| = \tfrac{1}{2}(\mathbb{I} - X\otimes X).$$

Proof. The first part is Example 2.5.3. $Z\otimes Z$ has eigenvalue $+1$ on $|00\rangle, |11\rangle$ and -1 on $|01\rangle, |10\rangle$, hence $+1$ on $|\Phi^\pm\rangle$ and -1 on $|\Psi^\pm\rangle$; so $\frac{1}{2}(\mathbb{I} - Z\otimes Z)$ is the projector onto $\operatorname{span}\{|\Psi^\pm\rangle\}$. Similarly $X\otimes X$ has eigenvalue $+1$ on $|{+}{+}\rangle, |{-}{-}\rangle$ and -1 on $|{+}{-}\rangle, |{-}{+}\rangle$; writing $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$, $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle - |{-}{-}\rangle)$, $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle + |{-}{+}\rangle)$, $|\Psi^-\rangle = -\frac{1}{\sqrt{2}}(|{+}{-}\rangle - |{-}{+}\rangle)$ shows that $X\otimes X = +1$ on $|\Phi^+\rangle, |\Psi^+\rangle$ and -1 on $|\Phi^-\rangle, |\Psi^-\rangle$. □

Remark 5.4.3: Both projectors are diagonal in the Bell basis and commute, so the bit- and phase-error patterns of all pairs are jointly well-defined classical random variables, and Serfling's inequality (Theorem 4.5.1) applies to them. They can be measured *locally*: $Z\otimes Z$ by comparing Z-outcomes, $X\otimes X$ by comparing X-outcomes. Moreover, $(H\otimes H)|\Phi^+\rangle = |\Phi^+\rangle$ and $HZH = X$, so $(H\otimes H)$ exchanges bit and phase errors. In the entanglement-based BB84 protocol a check pair with $b_i = 0$ therefore tests for a bit flip and one with $b_i = 1$ for a phase flip; since Eve does not know b and the check positions when she acts, she cannot treat check and key pairs, or bit and phase errors, differently.

Proposition 5.4.4: Consider the entanglement-based BB84 protocol (Definition 4.3.7) with a CSS code correcting t bit flips and t phase flips. Let $\varepsilon_{\mathrm{PE}}$ be the probability that the check passes while the key pairs contain more than t bit flips or more than t phase flips. Then, conditioned on passing, the corrected state satisfies $F(\rho_{AB}, |\Phi^+\rangle\langle\Phi^+|^{\otimes m}) \geq 1 - \varepsilon_{\mathrm{PE}}/p_{\mathrm{pass}}$. By Corollary 4.5.1, $\varepsilon_{\mathrm{PE}}$ is exponentially small in n if $t \geq n(\delta + \gamma)$ for the tolerated error fraction δ.

Proof. All operations of Alice and Bob (Hadamards, measurement of the check pairs, CSS syndrome measurements, which are products of $Z \otimes Z$ and $X \otimes X$ on the pairs) commute with a fictitious measurement of the bit- and phase-error pattern in the Bell basis. Performing this measurement first turns the state into a classical mixture over Pauli error patterns, without changing any statistics. For each pattern with at most t bit and t phase errors on the key pairs, the CSS code corrects the errors exactly and outputs $|\Phi^+\rangle^{\otimes m}$. Hence the fidelity conditioned on passing is at least $1 - \Pr[\text{fail} \mid \text{pass}] \geq 1 - \varepsilon_{\mathrm{PE}}/p_{\mathrm{pass}}$. Since the check pairs and their bases are a random sample, the bit (resp. phase) error fraction on the key pairs exceeds the observed fraction by more than γ only with exponentially small probability (Corollary 4.5.1). □

Corollary 5.4.5: The entanglement-based BB84 protocol is ε-secure with $\varepsilon = 2\sqrt{\varepsilon_{\mathrm{PE}}/p_{\mathrm{pass}}}$, which can be made arbitrarily small.

Proof. Combine Proposition 5.4.4 and Lemma 5.4.1 with $\varepsilon^2 = \varepsilon_{\mathrm{PE}}/p_{\mathrm{pass}}$. □

Reduction to the prepare-and-measure version

We modify the entanglement-based protocol step by step, keeping Eve's view unchanged, until we arrive at BB84. We use the CSS codewords

$$|\xi_{x,v,w}\rangle := \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{v \cdot y} |x + y + w\rangle$$

of $\mathrm{CSS}_{v,w}(C_1, C_2)$, which form an orthonormal basis of $(\mathbb{C}^2)^{\otimes n}$ when x ranges over C_1/C_2, v over $\mathbb{F}_2^n/C_2^\perp$ and w over $\mathbb{F}_2^n/C_1$ (Lemma B.2.2).

Step 1: *Check qubits.* The check pairs are only measured in the Z-basis (after the Hadamards). By Lemma 4.4.1, Alice can equivalently prepare n random check bits as $|0\rangle/|1\rangle$ and send them, placing them at random positions among the n halves of $|\Phi^+\rangle^{\otimes n}$.

Step 2: *CSS codes protocol.* Since the basis $\{|\xi_{x,v,w}\rangle\}$ has real coefficients,

$$|\Phi^+\rangle^{\otimes n} = 2^{-n/2} \sum_{x,v,w} |\xi_{x,v,w}\rangle |\xi_{x,v,w}\rangle.$$

When Alice measures her syndromes $(v,\ w)$ and decodes and measures her half (the key $k \leftrightarrow x$), Bob's half collapses to $|\xi_{x,v,w}\rangle$ with (x, v, w) uniformly random (Lemma 4.4.1 in reverse). Hence Alice may instead choose random k, v, w, send the codeword $|\xi_{x_k,v,w}\rangle$ encoding k in $\mathrm{CSS}_{v,w}(C_1, C_2)$, and announce v and w later; Bob decodes and measures. This is the *CSS codes protocol*; it is secure because Eve's view is unchanged.

Step 3: *No phase correction.* The key is obtained by a Z-measurement, so Bob's phase-error correction (which uses only v) does not affect the key and can be omitted. Then v

need not be announced, and Alice effectively sends the average over v:

$$\begin{aligned}\rho_{x,w} &= \frac{1}{2^{k_2}} \sum_v |\xi_{x,v,w}\rangle\langle\xi_{x,v,w}| \\ &= \frac{1}{2^{k_2}|C_2|} \sum_{y,y' \in C_2} \Big(\sum_v (-1)^{v\cdot(y-y')}\Big) |x+y+w\rangle\langle x+y'+w| \\ &= \frac{1}{|C_2|} \sum_{y \in C_2} |x+y+w\rangle\langle x+y+w|,\end{aligned}$$

since for $0 \neq u = y - y' \in C_2$ the character $v \mapsto (-1)^{v\cdot u}$ on $\mathbb{F}_2^n / C_2^\perp$ is non-trivial and sums to 0. This is a mixture of computational basis states: Alice picks random $y \in C_2$ and sends $|x+y+w\rangle$ in the Z-basis.

Step 4: *Classical post-processing.* Since $x \in C_1/C_2$, $y \in C_2$ and $w \in \mathbb{F}_2^n/C_1$ are uniform, $s := x + y + w$ is a uniformly random n-bit string, and $x_k := x + y$ is a uniformly random codeword of C_1. So Alice sends a random string s in the Z-basis (exactly like the check bits) and announces $s - x_k$. Bob measures in the Z-basis, obtains $s + e$, subtracts $s - x_k$ to get $x_k + e$, and corrects to x_k with the classical code C_1 (*error correction*). The key is the coset $x_k + C_2$, i.e., $k_1 - k_2$ bits (*privacy amplification*).

Step 5: *No Hadamards, sifting.* Instead of applying H for $b_i = 1$, Alice prepares the qubit directly in the X-basis and Bob measures in a random basis immediately, without quantum memory. Both keep the positions where the bases agree; this requires starting with $4n$ qubits, and the check positions are chosen after sifting.

Definition 5.4.6 (Secure BB84): The protocol obtained in this way consists of the following steps.

(i) Alice creates $4n$ random bits and encodes each in the Z- or X-basis according to a random string b; she sends the qubits.

(ii) Alice chooses a random $x_k \in C_1$.

(iii) Bob announces reception and measures each qubit in a random basis.

(iv) Alice announces b; both discard the positions with different bases ($\approx 2n$ remain). Alice chooses n of them as check bits and announces their positions.

(v) They compare the check bits and abort if more than t disagree. Otherwise Alice holds $s \in \{0,1\}^n$ and Bob $s + e$.

(vi) Alice announces $s - x_k$; Bob subtracts it and corrects with C_1 to obtain x_k.

(vii) Both compute the coset $x_k + C_2$, which is the key k.

Theorem 5.4.1: Shor–Preskill

The BB84 protocol is secure against coherent attacks. If the bit- and phase-error rates are both Q, the asymptotic secret key rate (per sifted bit) is

$$r = 1 - 2h(Q),$$

which is positive for $Q < 11.0\%$.

Proof. Security follows from the corollary to Proposition 5.4.4 and Steps 1–5, which do not change Eve's view. For the rate: there are CSS codes (Calderbank–Shor, Steane; Gilbert–Varshamov-type existence) such that C_1 corrects a fraction $Q+\gamma$ of bit flips with $k_1/n \to 1-h(Q)$ and $C_2^\perp$ corrects a fraction $Q+\gamma$ of phase flips with $(n-k_2)/n \to 1-h(Q)$. Hence $r = (k_1-k_2)/n \to 1-2h(Q)$. The threshold follows from $h(Q) = \frac{1}{2}$, i.e., $Q \approx 0.110$. □

Remark 5.4.7: The proof covers ideal BB84 (single photons, perfect devices); it does not cover, e.g., PNS attacks (for these, see Theorem 5.3.2). It does not address efficient decoding of C_1, which is needed in practice, and CSS codes are not optimal: with two-way post-processing higher error rates can be tolerated. The proof mixes error correction and privacy amplification, which makes it hard to transfer to other protocols.

5.5 Modern techniques

Theorem 5.5.1: Devetak–Winter

Against collective attacks and with one-way post-processing from Alice to Bob, the asymptotic secret key rate (per round) satisfies

$$r \geq I(A:B) - I(A:E) = H(A|E) - H(A|B),$$

where A, B are the classical outcomes of Alice and Bob in a single round and E is Eve's quantum system.

Proof. For the bound, see Devetak–Winter (2005); it is the quantum version of Theorem 5.3.1, and it also follows from Theorem 5.6.1 together with Theorem 3.3.2 applied to $\rho_{ABE}^{\otimes n}$. The identity: $I(A:B)-I(A:E) = \big(H(A)-H(A|B)\big)-\big(H(A)-H(A|E)\big) = H(A|E)-H(A|B)$. □

Remark 5.5.1: Interpretation: $H(A|B)$ is the cost of error correction per bit, and $H(A|E)$ is the amount of randomness that can be extracted secretly from A.

Remark 5.5.2 (Postselection technique): Let $d = \dim(\mathcal{H}_A \otimes \mathcal{H}_B)$ for a single round. If a permutation-invariant protocol with M rounds is ε-secure against collective attacks, then it is $(M+1)^{d^2-1}\varepsilon$-secure against coherent attacks, provided the key is shortened by $2(d^2-1)\log(M+1)$ bits (Christandl–König–Renner 2009). The overhead is negligible asymptotically. The technique requires a known dimension d.

Corollary 5.5.3: Let Alice measure R (key) or S (test), and Bob R' or S'. Then against collective attacks

$$r \geq \log\frac{1}{c} - H(R|R') - H(S|S'),$$

where c is the overlap of Alice's two measurements.

Proof. By Theorem 5.5.1, $r \geq H(R|E)-H(R|B)$. By Theorem 3.4.1, $H(R|E) \geq \log\frac{1}{c}-H(S|B)$. Hence $r \geq \log\frac{1}{c} - H(S|B) - H(R|B)$. Measuring B does not decrease the conditional entropy (Corollary 3.2.13(iii)), so $H(S|B) \leq H(S|S')$ and $H(R|B) \leq H(R|R')$. □

Remark 5.5.4: The right-hand side involves only quantities that are directly observable by Alice and Bob; no tomography of Eve's system is needed.

Corollary 5.5.5 (BB84 key rate): Let e_Z, e_X be the error rates in the Z- and X-basis. Then, against collective (and hence coherent) attacks,

$$H(Z|E) \geq 1-h(e_X), \qquad r_\infty^{\mathrm{BB84}} \geq 1-h(e_Z)-h(e_X).$$

Proof. For Z/X measurements, $\log(1/c) = 1$. Theorem 3.4.1 and Corollary 3.2.13 give $H(Z|E) \geq 1 - H(X|B) \geq 1 - H(X|X')$. By Lemma 3.1.13, $H(X|X') \leq h(e_X)$ and $H(Z|Z') \leq h(e_Z)$. Apply Corollary 5.5.3. □

Remark 5.5.6: Eve's information on the key basis is bounded by the error rate in the *other* basis: since $H(Z) = 1$ for uniform key bits, $I(Z:E) = H(Z) - H(Z|E) \leq h(e_X)$. By Corollary 3.2.9, $I(Z:E) = D(\rho_{ZE} \| \rho_Z \otimes \rho_E)$, which vanishes, by Theorem 3.2.1, if and only if $\rho_{ZE} = \rho_Z \otimes \rho_E$; hence $e_X \to 0$ implies that Eve is decoupled from the key. For $e_X = e_Z = Q$ we recover the Shor–Preskill rate $1 - 2h(Q)$. The extension from collective to coherent attacks uses the permutation symmetry of the protocol (de Finetti theorem or postselection technique), since in general $\rho_{A_1B_1E_1} \otimes \cdots \otimes \rho_{A_nB_nE_n} \neq \rho_{A_1 \cdots A_n B_1 \cdots B_n E}$.

5.6 Finite-key analysis

Definition 5.6.1 (Generic entanglement-based protocol): The protocol consists of the following steps.

(i) A source distributes M signals; Alice and Bob hold ρ_{AB}^M (in general $\neq \rho_{AB}^{\otimes M}$, to account for coherent attacks) and Eve holds a purification.

(ii) Alice and Bob measure locally and use some rounds for key generation and some for testing.

(iii) *PE*: from the test data and a random sample of the key rounds they estimate the error rate; the raw keys K_A^n, K_B^n have length $n < M$.

(iv) *EC*: Bob computes a guess $\hat{K}_A^n$ from K_B^n and $\mathrm{leak}_{\mathrm{EC}}$ bits from Alice; they verify with a hash of length $\lceil \log(1/\varepsilon_{\mathrm{cor}}) \rceil$ (Lemma 4.5.7) and abort if the hashes differ.

(v) *PA*: they apply a random two-universal hash function with output length ℓ to K_A^n and $\hat{K}_A^n$, obtaining k_A, k_B.

The key rate is $r = \ell/M$.

Lemma 5.6.2: $\lceil \log(1/x) \rceil \leq \log(2/x)$ for all $x \in (0, 1]$.

Proof. $\lceil a \rceil \leq a + 1$ and $\log(1/x) + 1 = \log(2/x)$. □

Theorem 5.6.1: Finite-key security

If

$$\ell \leq H_{\min}^{\varepsilon'}(K_A^n|E) - \mathrm{leak}_{\mathrm{EC}} - \log \frac{2}{\varepsilon_{\mathrm{cor}}} - 2 \log \frac{1}{2\varepsilon_{\mathrm{sec}}},$$

then the protocol is ε-secure with $\varepsilon = \varepsilon_{\mathrm{cor}} + 2\varepsilon' + \varepsilon_{\mathrm{sec}}$.

Proof. Correctness: if the hashes differ, both keys are $\perp$. Otherwise, by Lemma 4.5.7 and since PA preserves equality,

$$\Pr[k_A \neq k_B] \leq \Pr[K_A^n \neq \hat{K}_A^n \wedge \mathrm{pass}] \leq \varepsilon_{\mathrm{cor}}.$$

Secrecy: Eve's total information E_{tot} consists of E, the EC messages and hash C, and the PA seed. The register C takes at most $2^{\mathrm{leak}_{\mathrm{EC}} + \lceil \log(1/\varepsilon_{\mathrm{cor}}) \rceil}$ values, so by Lemma 3.3.8 and Lemma 5.6.2,

$$H_{\min}^{\varepsilon'}(K_A^n|CE) \geq H_{\min}^{\varepsilon'}(K_A^n|E) - \mathrm{leak}_{\mathrm{EC}} - \log \tfrac{2}{\varepsilon_{\mathrm{cor}}}.$$

Theorem 4.5.2 (with E replaced by CE) and the choice of ℓ give

$$\tfrac{1}{2} \| \rho_{k_A E_{\mathrm{tot}}} - \rho_U \otimes \rho_{E_{\mathrm{tot}}} \|_1 \leq 2\varepsilon' + \tfrac{1}{2} \sqrt{2^{\ell - H_{\min}^{\varepsilon'}(K_A^n|CE)}} \leq 2\varepsilon' + \tfrac{1}{2} \sqrt{2^{-2 \log \frac{1}{2\varepsilon_{\mathrm{sec}}}}} = 2\varepsilon' + \varepsilon_{\mathrm{sec}}.$$

The claim follows from Theorem 5.1.1. □

Remark 5.6.3: The main difficulty of any security proof is to lower-bound $H_{\min}^{\varepsilon'}(K_A^n|E)$, since Eve's system is not accessible.

Finite-key analysis of BB84

Let K_A^Z denote Alice's n key bits (measured in Z), and let K_A^X, K_B^X denote the strings Alice and Bob *would* have obtained by measuring these rounds in X. By Theorem 3.4.3 with $\log(1/c) = 1$ per round,

$$H_{\min}^{\varepsilon}(K_A^Z|E) \geq n - H_{\max}^{\varepsilon}(K_A^X|B) \geq n - H_{\max}^{\varepsilon}(K_A^X|K_B^X),$$

where the second step is data processing for the max-entropy. The max-entropy is bounded using the test data. Let λ_k be the error rate observed on k test bits measured in X, λ_n the (hypothetical) X-error rate on the key bits, and $N = n + k$.

Lemma 5.6.4: Let $\gamma > 0$ and

$$\varepsilon := \frac{1}{\sqrt{p_{\text{pass}}}} \exp\Big(-\frac{k^2 n \gamma^2}{(k+1)N}\Big), \qquad \text{equivalently} \qquad \gamma = \sqrt{\frac{(k+1)N}{k^2 n} \ln \frac{1}{\varepsilon\sqrt{p_{\text{pass}}}}}.$$

Then $H_{\max}^{\varepsilon}(K_A^X|K_B^X) \leq n\, h(\lambda_{\max} + \gamma)$ (for $\lambda_{\max} + \gamma \leq \frac{1}{2}$).

Proof. Let P be the joint distribution of $(K_A^X, K_B^X, \lambda_k)$ conditioned on passing, and let Q be P conditioned additionally on the event $\mathcal{G} = \{\lambda_n < \lambda_k + \gamma\}$, i.e., $Q = P \cdot 1_{\mathcal{G}}/P(\mathcal{G})$. Then

$$\begin{aligned} F(P,Q) &= \Big(\sum \sqrt{PQ}\Big)^2 = \Big(\sum_{\mathcal{G}} \frac{P}{\sqrt{P(\mathcal{G})}}\Big)^2 = P(\mathcal{G}) \\ &= \Pr[\lambda_n < \lambda_k + \gamma \mid \lambda_k \leq \lambda_{\max}] \geq 1 - \frac{1}{p_{\text{pass}}} e^{-\frac{2k^2 n\gamma^2}{(k+1)N}} \end{aligned}$$

by Corollary 4.5.1, so $P(P,Q) = \sqrt{1 - F(P,Q)} \leq \varepsilon$. By definition of the smooth max-entropy, $H_{\max}^{\varepsilon}(K_A^X|K_B^X)_P \leq H_{\max}(K_A^X|K_B^X)_Q$. Under Q, the strings differ in at most $w_{\max} := \lfloor n(\lambda_{\max} + \gamma) \rfloor$ positions. For classical distributions $H_{\max}(X|Y) \leq \max_y \log|\operatorname{supp} P_{X|Y=y}|$: for every distribution σ on $\mathcal{Y}$, two applications of Cauchy–Schwarz give

$$\sum_y \sqrt{\sigma(y)} \sum_x \sqrt{P(x,y)} \leq \sum_y \sqrt{\sigma(y)P(y)} \sqrt{|\operatorname{supp} P_{X|Y=y}|} \leq \max_y \sqrt{|\operatorname{supp} P_{X|Y=y}|}.$$

Hence

$$H_{\max}(K_A^X|K_B^X)_Q \leq \log \sum_{w=0}^{w_{\max}} \binom{n}{w} \leq n\, h(\lambda_{\max} + \gamma),$$

using the standard estimate $\sum_{w \leq \theta n} \binom{n}{w} \leq 2^{n h(\theta)}$ for $\theta \leq \frac{1}{2}$. □

Theorem 5.6.2: Finite-key BB84

With ε and γ as in Lemma 5.6.4, the BB84 protocol is $(\varepsilon_{\text{cor}} + 2\varepsilon + \varepsilon_{\text{sec}})$-secure if

$$\ell \leq n\big(1 - h(\lambda_{\max} + \gamma)\big) - \text{leak}_{\text{EC}} - \log \frac{2}{\varepsilon_{\text{cor}}} - 2 \log \frac{1}{2\varepsilon_{\text{sec}}}.$$

Proof. By the smooth uncertainty relation and Lemma 5.6.4, $H_{\min}^{\varepsilon}(K_A^Z|E) \geq n(1 - h(\lambda_{\max} + \gamma))$. Insert this into Theorem 5.6.1 with $\varepsilon' = \varepsilon$. □

Remark 5.6.5: As $n, k \to \infty$ with k/n fixed, $\gamma \to 0$ and $\text{leak}_{\text{EC}}/n \to h(e_Z)$, so $\ell/n \to 1 - h(e_X) - h(e_Z)$, the asymptotic rate of Corollary 5.5.5. For realistic block sizes the corrections are significant; typically 10^5–10^6 signals are needed for a positive key.

6 Device-independent QKD

6.1 Black boxes and Bell inequalities

Definition 6.1.1: A *bipartite box* with input sets $\mathcal{X}$, $\mathcal{Y}$ and output sets $\mathcal{A}$, $\mathcal{B}$ is a conditional probability distribution $p(a,b|x,y) \equiv p_{AB|XY}(a,b|x,y)$: on inputs x (Alice) and y (Bob) it produces outputs a and b. Nothing is assumed about its internal working.

Definition 6.1.2: A box is

(i) *classical* (or *local*) if there is a random variable Λ (shared randomness, "local hidden variable") such that
$$p(a,b|x,y) = \int d\lambda\, p_\Lambda(\lambda)\, p_A(a|x,\lambda)\, p_B(b|y,\lambda),$$

(ii) *quantum* if there are a state ρ_{AB} (of arbitrary dimension) and POVMs $\{M_a^x\}_a$, $\{M_b^y\}_b$ with
$$p(a,b|x,y) = \mathrm{Tr}\big((M_a^x \otimes M_b^y)\rho_{AB}\big),$$

(iii) *non-signalling* if its marginals are well-defined, i.e., for all a,b,x,x',y,y',
$$\sum_b p(a,b|x,y) = \sum_b p(a,b|x,y'), \qquad \sum_a p(a,b|x,y) = \sum_a p(a,b|x',y).$$

Lemma 6.1.3: Classical boxes are quantum, and quantum boxes are non-signalling: $\mathcal{L} \subset \mathcal{Q} \subset \mathcal{NS}$. Every local box is a convex combination of local *deterministic* boxes ($a = f(x)$, $b = g(y)$), so $\mathcal{L}$ is a polytope.

Proof. A local box (with finitely many values of λ, which suffices) is realized by the separable state $\sum_\lambda p(\lambda)|\lambda\rangle\langle\lambda| \otimes |\lambda\rangle\langle\lambda|$ and the POVMs $M_a^x = \sum_\lambda p_A(a|x,\lambda)|\lambda\rangle\langle\lambda|$, $M_b^y = \sum_\lambda p_B(b|y,\lambda)|\lambda\rangle\langle\lambda|$. For a quantum box, $\sum_b p(a,b|x,y) = \mathrm{Tr}\big((M_a^x \otimes \mathbb{I})\rho_{AB}\big)$ does not depend on y. Finally, each $p_A(\cdot|x,\lambda)$ is a convex combination of deterministic functions, and so is the product. □

Remark 6.1.4: The inclusions are strict (CHSH and PR box below). The non-signalling conditions ensure that space-like separated boxes cannot be used to signal, so there is no conflict with relativity.

Definition 6.1.5: A *Bell inequality* is a linear inequality $\sum_{a,b,x,y} s_{xy}^{ab}\, p(a,b|x,y) \le S_{\mathcal{L}}$ satisfied by all local boxes. If a box violates it, the box is not local. (The converse does not hold: satisfying one Bell inequality does not imply locality.) Equivalently, in a *non-local game* with question distribution $q(x,y)$ and winning condition $w(a,b,x,y) \in \{0,1\}$, the winning probability
$$\omega(p) = \sum_{a,b,x,y} q(x,y)\, p(a,b|x,y)\, w(a,b,x,y)$$
is bounded by some $\omega_{\mathrm{c}} < 1$ for local boxes.

Definition 6.1.6 (CHSH game): Inputs $x,y \in \{0,1\}$ are chosen uniformly; Alice and Bob output bits a,b and win if $a \oplus b = x \cdot y$. In ± 1 notation (outcome $(-1)^a$) with correlators $E_{xy} := \langle A_x B_y \rangle$, the CHSH value is $S = E_{00} + E_{01} + E_{10} - E_{11}$.

Lemma 6.1.7: $\omega = \frac{1}{2} + \frac{S}{8} = \frac{4+S}{8}$.

Proof. For fixed (x,y), $\Pr[a \oplus b = 0] = \frac{1}{2}(1 + E_{xy})$ and $\Pr[a \oplus b = 1] = \frac{1}{2}(1 - E_{xy})$. The winning condition requires $a \oplus b = 0$ for $(x,y) \neq (1,1)$ and $a \oplus b = 1$ for $(1,1)$, so $\omega = \frac{1}{4}\sum_{x,y} \frac{1}{2}(1 \pm E_{xy}) = \frac{1}{2} + \frac{1}{8}S$. □

Proposition 6.1.8: The maximal winning probability of the CHSH game is

(i) $\omega_{\mathrm{c}} = \frac{3}{4}$ for classical boxes,

(ii) $\omega_{\mathrm{q}} = \cos^2(\pi/8) = \frac{2+\sqrt{2}}{4} \approx 0.854$ for quantum boxes,

(iii) $\omega_{\mathrm{ns}} = 1$ for non-signalling boxes.

Proof. (i) By convexity it suffices to consider deterministic strategies a_x, b_y. Winning all four rounds would require $a_0 \oplus b_0 = 0$, $a_0 \oplus b_1 = 0$, $a_1 \oplus b_0 = 0$, $a_1 \oplus b_1 = 1$; summing modulo 2 gives $0 = 1$. Hence at most three of four rounds are won, $\omega \leq \frac{3}{4}$, attained by $a = b = 0$ always. (ii) By Theorem 4.3.1, $S \leq 2\sqrt{2}$, i.e., $\omega \leq \frac{1}{2} + \frac{\sqrt{2}}{4}$. This is attained with $|\Phi^+\rangle$, $A_0 = Z$, $A_1 = X$, $B_0 = (Z+X)/\sqrt{2}$, $B_1 = (Z-X)/\sqrt{2}$: by Lemma 2.5.4, $E_{00} = E_{01} = E_{10} = \frac{1}{\sqrt{2}}$ and $E_{11} = -\frac{1}{\sqrt{2}}$. (iii) The *PR box* $p(a,b|x,y) = \frac{1}{2}$ if $a \oplus b = xy$ and 0 otherwise has uniform marginals, hence is non-signalling, and wins with certainty ($S = 4$). □

Remark 6.1.9 (Why device independence): Consider a box with binary inputs and outputs and

$$p(a,b|0,0) = p(a,b|1,1) = \tfrac{1}{2}\delta_{a,b}, \qquad p(a,b|0,1) = p(a,b|1,0) = \tfrac{1}{4}.$$

If Alice and Bob *assume* a two-qubit implementation with $x, y = 0 \leftrightarrow X$ and $x, y = 1 \leftrightarrow Z$, these statistics mean $\langle X \otimes X\rangle = \langle Z \otimes Z\rangle = 1$ and $\langle X \otimes Z\rangle = \langle Z \otimes X\rangle = 0$, which on $\mathbb{C}^2 \otimes \mathbb{C}^2$ is only possible for $|\Phi^+\rangle$ (it is the unique joint $+1$ eigenvector of $X \otimes X$ and $Z \otimes Z$); the key would be secure. However, the same statistics are produced on $\mathbb{C}^4 \otimes \mathbb{C}^4$ by the separable state

$$\rho_{ABE} = \frac{1}{4} \sum_{z_0, z_1 \in \{0,1\}} |z_0 z_1\rangle\langle z_0 z_1|_A \otimes |z_0 z_1\rangle\langle z_0 z_1|_B \otimes |z_0 z_1\rangle\langle z_0 z_1|_E$$

with measurements $x, y = 0 \leftrightarrow Z \otimes \mathbb{I}$ and $x, y = 1 \leftrightarrow \mathbb{I} \otimes Z$ (each party reads out the first or second bit), and Eve holds a perfect copy of the key. Consistently, the CHSH value is $S = 1+0+0-1 = 0$ (in the labelling $E_{00} = E_{11} = 1$, $E_{01} = E_{10} = 0$), so no Bell inequality is violated. In the device-independent setting only a Bell violation certifies security.

Remark 6.1.10: The key observation is *monogamy*: if the statistics violate a Bell inequality, they cannot come from a local box, i.e., the outputs are not predetermined by a variable that Eve could hold. A maximal violation even *self-tests* the devices: it fixes the state and the measurements up to local isometries (Mayers–Yao), so Eve is decoupled.

6.2 DIQKD protocols and security

Definition 6.2.1 (DIQKD protocol): A source (in the worst case controlled by Eve) distributes states to Alice and Bob. Alice chooses among measurements A_0, A_1, A_2 and Bob among B_1, B_2, all with outcomes ± 1.

(i) *Quantum phase*: in n rounds, Alice and Bob use their devices with random inputs. The raw key is generated from rounds with inputs (A_0, B_1); the *quantum bit error rate* is $Q := p(a \neq b \mid A_0, B_1)$.

(ii) *Parameter estimation*: from rounds with inputs in $\{A_1, A_2\} \times \{B_1, B_2\}$ they estimate the CHSH value S, and they estimate Q from a sample of the key rounds. They abort if S and Q do not allow a positive key.

(iii) *Post-processing*: error correction and privacy amplification as before.

A priori, S and Q are independent parameters.

Example 6.2.2 (Honest implementation): Let the source distribute the state obtained from $|\Phi^+\rangle$ by a depolarizing channel,

$$\rho_{AB} = p|\Phi^+\rangle\langle\Phi^+| + (1-p)\frac{\mathbb{I}}{4},$$

with $A_0 = Z$, $A_{1,2} = (Z \pm X)/\sqrt{2}$, $B_1 = Z$, $B_2 = X$. Then

$$S = 2\sqrt{2}\,p, \qquad Q = \frac{1-p}{2}, \qquad \text{hence} \qquad S = 2\sqrt{2}(1-2Q).$$

Indeed, $\langle \vec{a}\cdot\vec{\sigma} \otimes \vec{b}\cdot\vec{\sigma}\rangle_\rho = p\,\vec{a}\cdot\vec{b}$ for vectors in the x-z plane (Lemma 2.5.4; $\mathbb{I}/4$ contributes 0). So $\langle A_0B_1\rangle = p$, i.e., $Q = \frac{1}{2}(1-p)$, and the CHSH combination $\langle A_1B_1\rangle + \langle A_1B_2\rangle + \langle A_2B_1\rangle - \langle A_2B_2\rangle = \frac{p}{\sqrt{2}}(1+1+1+1) = 2\sqrt{2}p$. The security proof does not use this model; it only uses the observed S and Q.

Theorem 6.2.1: Security against collective attacks

Assume that the devices act identically and independently in each round. Then the asymptotic key rate of the DIQKD protocol satisfies

$$r \geq 1 - h(Q) - h\Big(\frac{1+\sqrt{(S/2)^2-1}}{2}\Big).$$

Proof idea. By the Devetak–Winter bound with Bob's string as reference (reverse reconciliation, which is better here since $\chi(A_0:E) \geq \chi(B_1:E)$),

$$r \geq I(A_0:B_1) - \chi(B_1:E), \qquad \chi(B_1:E) = H(\rho_E) - \tfrac{1}{2}\sum_{b_1=\pm1} H(\rho_{E|b_1}),$$

the Holevo quantity of Bob's outcome and Eve's state. One may assume uniform marginals (Alice and Bob flip a random half of their bits by public agreement, which changes neither S nor Q). Then $I(A_0:B_1) = H(A_0) + H(B_1) - H(A_0B_1) = 2 - (1 + h(Q)) = 1 - h(Q)$. The bound $\chi(B_1:E) \leq h\big((1+\sqrt{(S/2)^2-1})/2\big)$ is obtained by reducing to qubits: two binary measurements on each side decompose into a direct sum of qubit blocks (Jordan's lemma), and the optimization reduces to Bell-diagonal two-qubit states (Pironio et al. 2009). The bound is tight: Eve can prepare

$$\rho_{AB} = \tfrac{1+C}{2}|\Phi^+\rangle\langle\Phi^+| + \tfrac{1-C}{2}|\Phi^-\rangle\langle\Phi^-|, \qquad C := \sqrt{(S/2)^2-1},$$

with $A_{1,2} = (Z \pm CX)/\sqrt{1+C^2}$, $B_1 = Z$, $B_2 = X$, and $A_0 = Z$ with probability $1-2Q$ and a uniformly random bit otherwise. Then $Q = 2Q \cdot \frac{1}{2}$ (errors only from the random bit, since both $|\Phi^\pm\rangle$ are perfectly correlated in Z), and with $\langle ZZ\rangle = 1$, $\langle XX\rangle = C$, $\langle ZX\rangle = \langle XZ\rangle = 0$,

$$\frac{(1+C^2)+(1+C^2)}{\sqrt{1+C^2}} = 2\sqrt{1+C^2} = S.$$

For this attack $\chi(B_1:E)$ equals the bound. It is impossible in the device-dependent setting, since the measurements depend on S. □

Corollary 6.2.3: If $S \leq 2$, the bound of Theorem 6.2.1 is never positive.

Proof. At $S = 2$ the last term equals $h(\frac{1}{2}) = 1$, so $r \geq -h(Q)$ gives nothing; for $S < 2$ no quantum-specific bound applies at all, since the statistics are local and Eve may hold the local variable. For $S = 2\sqrt{2}$ the last term vanishes and $r \geq 1 - h(Q)$. □

Example 6.2.4 (Comparison with the device-dependent case): If Alice and Bob trust their devices (qubits, exact measurements), one obtains $\chi(B_1 : E) \leq h\big(Q + \frac{S}{2\sqrt{2}}\big)$. For the honest implementation, $S = 2\sqrt{2}(1-2Q)$, this gives $h(1-Q) = h(Q)$, i.e., the BB84-type rate $1-2h(Q)$, which is positive for $Q \lesssim 11.0\%$. The device-independent rate with $S = 2\sqrt{2}(1-2Q)$ is strictly lower and positive for $Q \lesssim 7.1\%$.

From collective to coherent attacks

Remark 6.2.5: Under the i.i.d. assumption, $K_A^n = K_1 \cdots K_n$ and $E = E_1 \cdots E_n$ with i.i.d. pairs (K_i, E_i), and the chain rule gives

$$H(K_1 \cdots K_n | E_1 \cdots E_n) = \sum_i H(K_i | E_1 \cdots E_n K_1 \cdots K_{i-1}) = \sum_i H(K_i | E_i) = nH(K_1|E_1),$$

since (K_i, E_i) is independent of all other systems. For the smooth min-entropy, Theorem 3.3.2 gives $H_{\min}^{\varepsilon}(K_A^n|E) \geq nH(K_1|E_1) - c_\varepsilon\sqrt{n}$. In both cases the analysis reduces to a single round. But the i.i.d. assumption requires memoryless devices, which is exactly the kind of assumption DIQKD avoids. The de Finetti and postselection techniques require a known dimension and hence do not apply.

Theorem 6.2.2: Entropy accumulation (informal)

Model round i as a channel $\mathcal{M}_i$ mapping a quantum memory R_{i-1} (entangled with Eve's purifying system E) to classical outputs O_i, classical side information S_i (inputs and announcements) and a memory R_i. Let f be an affine function such that for every state σ compatible with the observed winning probability ω of the CHSH game,

$$f(\omega) \leq \min_\sigma H(O_i|S_iR')_{\mathcal{M}_i(\sigma)},$$

where R' purifies the input of round i. Then, conditioned on observing an average winning probability at least ω,

$$H_{\min}^{\varepsilon}(O^n|S^nE) \geq nf(\omega) - O(\sqrt{n}).$$

Proof. See Dupuis, Fawzi and Renner (2020) and Arnon-Friedman et al. (2018). □

Remark 6.2.6: The entropy accumulation theorem (EAT) plays the role of the QAEP for sequential, non-i.i.d. devices: the first term is linear in n and the correction $O(\sqrt{n})$ vanishes after dividing by n. It yields security against coherent attacks with the same asymptotic rate as for collective attacks.

Remark 6.2.7 (Improved protocols): Higher noise tolerance can be obtained, e.g., by

(i) a *random key basis*: key bits are generated from two randomly chosen pairs of measurements, which helps in particular for small Bell violations (Schwonnek et al. 2021),

(ii) *modified Bell inequalities* $S_\alpha = \alpha\langle A_1B_1\rangle + \alpha\langle A_1B_2\rangle + \langle A_2B_1\rangle - \langle A_2B_2\rangle$ with a free weight $\alpha \in \mathbb{R}$ ($\alpha = 1$ is CHSH); for depolarizing noise this raises the tolerable QBER from about 7.15% to 7.34%,

(iii) *noisy preprocessing*: Alice flips each raw key bit with some probability, which hurts Eve more than Bob (Ho et al. 2020).

6.3 Loopholes

A Bell test is only conclusive if (i) no information about one party's input reaches the other party before the output is produced, and (ii) the detection efficiency is sufficiently high.

Remark 6.3.1 (Locality loophole): If a hidden signal can inform Bob's device about Alice's input, local models can reproduce any correlations. In a Bell test this is excluded by space-like separation: the choice of input and the measurement of each party lie outside the backward light cone of the other party's measurement. This requires distances of at least tens of meters, typically bridged with photons. In DIQKD, space-like separation is replaced by the assumption that the laboratories are shielded.

Example 6.3.2 (Detection loophole): Consider two deterministic local strategies with all outputs $+1$, except that in strategy 2 Bob outputs -1 for input B_1. Both have $S = 2$:

$$\text{strategy 1: } 1+1+1-1=2, \qquad \text{strategy 2: } 1-1+1-(-1)=2.$$

Let the source choose a strategy at random, and let Alice's detector click only for input A_0 under strategy 1 and only for input A_1 under strategy 2. If Alice and Bob discard no-click events (*fair sampling*), each correlator is computed only from the favourable strategy, and they observe $\langle A_0B_0\rangle = \langle A_0B_1\rangle = \langle A_1B_0\rangle = 1$, $\langle A_1B_1\rangle = -1$, i.e., $S = 4$, although everything is local. Mixing more strategies can produce any $2 < S \le 2\sqrt{2}$. Treating "no click" as an outcome removes the problem but requires high efficiency.

Lemma 6.3.3: Assign the outcome $+1$ to no-click events. For the maximally entangled state with optimal settings and detection efficiency η on both sides, the observed value is

$$S_\eta = 2\sqrt{2}\,\eta^2 + 2(1-\eta)^2.$$

Hence $S_\eta > 2$ if and only if $\eta > 2/(1+\sqrt{2}) = 2(\sqrt{2}-1) \approx 82.8\%$.

Proof. With probability η^2 both detect (correlator E^{q}_{xy}), with probability $\eta(1-\eta)$ only one detects (correlator equal to that party's marginal, which vanishes for the maximally entangled state), and with probability $(1-\eta)^2$ none detects (correlator 1). Hence $E_{xy} = \eta^2 E^{\mathrm{q}}_{xy} + (1-\eta)^2$, and with signs $(+,+,+,-)$, $S_\eta = \eta^2\, 2\sqrt{2} + (1-\eta)^2(1+1+1-1)$. Then $S_\eta > 2 \iff 2\sqrt{2}\eta^2 + 2 - 4\eta + 2\eta^2 > 2 \iff \eta(2\sqrt{2}+2) > 4$. □

Remark 6.3.4: With partially entangled states the threshold can be lowered to $\eta > 2/3$ (Eberhard 1993). Matter systems (ions, atoms, NV centres) are detected with efficiency close to 1 but are hard to separate far enough. A *heralding* (quantum non-demolition) measurement, which checks the arrival of the photon *before* Bob's input is chosen, removes the transmission losses from the Bell test. Loophole-free Bell tests were achieved in 2015 (Delft, Vienna, Boulder), the Delft experiment via entanglement swapping between NV-centre spins 1.3 km apart. The detection efficiency remains the main obstacle for photonic DIQKD.

7 Practical QKD

7.1 Practical challenges

Remark 7.1.1: The performance of an implementation is judged by

(i) the *distance* over which a key can be generated, limited mainly by photon loss in fiber or free space,

(ii) the *key rate*, which depends crucially on detector efficiency and dead time (classical links carry ~ 100 Gbit/s, QKD reaches Mbit/s),

(iii) the existence of a composable *security proof* against general attacks including finite-size effects, and robustness against side channels,

as well as cost, coexistence with classical traffic in the same fiber, and the need for cooling. The main deviations from the ideal protocols are: *sources* (weak coherent pulses with multi-photon components, Section 5.3), *channels* (loss and noise) and *detectors* (finite efficiency, dark counts, dead time, mismatches, which open side channels).

Lemma 7.1.2 (Fiber transmittance): If a fiber has a loss of l_{dB} dB per km, then a photon is transmitted through length L with probability

$$\eta = 10^{-l_{\mathrm{dB}} L/10}.$$

Proof. A loss of $l_{\%}$ percent corresponds to $l_{\mathrm{dB}} = -10\log_{10}(1 - l_{\%}/100)$, i.e., the transmitted fraction is $1 - l_{\%}/100 = 10^{-l_{\mathrm{dB}}/10}$. Losses in dB add, so the total loss is $l_{\mathrm{dB}} L$ and the transmission probability is $10^{-l_{\mathrm{dB}} L/10}$. □

Example 7.1.3: For standard fiber at 1550 nm, $l_{\mathrm{dB}} \approx 0.2\,\mathrm{dB/km}$: half of the photons are lost after 15 km (3 dB), and $\eta = 10^{-2}$ after 100 km.

Theorem 7.1.1: Repeaterless bound (PLOB)

For a pure-loss channel with transmittance η, the secret key capacity (per channel use) of point-to-point QKD without quantum repeaters is

$$-\log(1-\eta) \approx \frac{\eta}{\ln 2} \approx 1.44\,\eta \qquad (\eta \ll 1).$$

Proof. See Pirandola–Laurenza–Ottaviani–Banchi (2017). The approximation follows from $-\ln(1-\eta) = \eta + O(\eta^2)$. □

Remark 7.1.4: Hence the key rate of any point-to-point protocol scales at best linearly in η, i.e., decays exponentially with distance. Quantum repeaters would overcome this but require quantum memories and error correction.

7.2 Detector side channels and MDI QKD

Detector side-channel attacks

Example 7.2.1 (Time-shift attack): Security proofs assume equal detection efficiencies for the outcomes "0" and "1". In practice the efficiency of each detector depends on the arrival time within the detection window, and the two curves are slightly shifted. Eve shifts the arrival time of each photon to a point A (where detector 0 is more efficient) or B (where detector 1 is), with probabilities chosen such that Bob still sees balanced outcomes. Bob's result is biased towards a value Eve knows, so she gains information without causing errors (Qi et al. 2007; Zhao et al. 2008). This attack is feasible with current technology.

Example 7.2.2 (Detector blinding): Avalanche photodiodes (APDs) detect single photons in *Geiger mode*: the bias voltage is above the breakdown voltage V_{br}, a single photon triggers an avalanche whose current exceeds the threshold I_{th}, and the detector clicks; then the avalanche is quenched and the APD is reset. Bright continuous illumination lowers the voltage below V_{br}, and the APD enters *linear mode*: the current is proportional to the incident power, and the detector clicks only above an optical power threshold P_{th}; single photons are no longer detected. Eve then

(i) measures Alice's states in a random basis,

(ii) resends her result as a bright pulse with power just above P_{th} (in the same basis),

(iii) so that Bob's detector clicks only if his basis coincides with Eve's, and then with Eve's result.

After sifting, Eve and Bob hold identical bits, and by following the public post-processing Eve obtains the full key without introducing errors (Lydersen et al. 2010). Further examples: dead-time attacks and Trojan-horse attacks (bright light reflected from Alice's or Bob's devices reveals internal settings).

Measurement-device-independent QKD

Definition 7.2.3 (MDI QKD): Alice and Bob each prepare phase-randomized weak coherent pulses in a random BB84 polarization state (rectilinear: $0 \mapsto H$, $1 \mapsto V$; diagonal: $0 \mapsto D$, $1 \mapsto A$) and random decoy intensity, and send them to an untrusted relay (Charlie). Charlie lets the pulses interfere on a 50:50 beam splitter, followed by a polarizing beam splitter and two detectors (H, V) in each output port, and announces which detectors clicked. A successful Bell-state measurement (BSM) projects onto $|\Psi^-\rangle$ (clicks $D_{1H}\&D_{2V}$ or $D_{1V}\&D_{2H}$) or $|\Psi^+\rangle$ (clicks $D_{1H}\&D_{1V}$ or $D_{2H}\&D_{2V}$). For each successful BSM with matching bases, Bob flips his bit

(i) always, if they used the rectilinear basis,

(ii) if Charlie announced $|\Psi^-\rangle$, if they used the diagonal basis.

Lemma 7.2.4 (Linear-optics Bell measurement): Let Alice and Bob each send one photon whose polarization is entangled with a virtual qubit, $|\psi\rangle_A = \frac{1}{\sqrt{2}}\big(|H\rangle_A a_H^\dagger + |V\rangle_A a_V^\dagger\big)|0\rangle$ and similarly for Bob with modes b. Let the beam splitter act as $a^\dagger \mapsto (c^\dagger + d^\dagger)/\sqrt{2}$, $b^\dagger \mapsto (c^\dagger - d^\dagger)/\sqrt{2}$ (polarization preserved), where c, d are the two output ports. Then the joint state after the beam splitter is

$$\frac{1}{4}\Big[|HH\rangle\big(c_H^{\dagger 2} - d_H^{\dagger 2}\big) + |VV\rangle\big(c_V^{\dagger 2} - d_V^{\dagger 2}\big)\Big]|0\rangle + \frac{1}{2\sqrt{2}}\Big[|\Psi^+\rangle\big(c_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger\big) - |\Psi^-\rangle\big(c_H^\dagger d_V^\dagger - c_V^\dagger d_H^\dagger\big)\Big]|0\rangle,$$

where $|HH\rangle, \ldots, |\Psi^\pm\rangle$ refer to the virtual qubits. Consequently a click pattern $c_H c_V$ or $d_H d_V$ projects the virtual qubits onto $|\Psi^+\rangle$, a pattern $c_H d_V$ or $c_V d_H$ onto $|\Psi^-\rangle$, each with probability $\frac{1}{4}$; the BSM succeeds with probability $\frac{1}{2}$.

Proof. The global state is $\frac{1}{2}\big(|HH\rangle a_H^\dagger b_H^\dagger + |VV\rangle a_V^\dagger b_V^\dagger + |HV\rangle a_H^\dagger b_V^\dagger + |VH\rangle a_V^\dagger b_H^\dagger\big)|0\rangle$. Using the beam-splitter relation,

$$a_H^\dagger b_H^\dagger \mapsto \tfrac{1}{2}\big(c_H^{\dagger 2} - d_H^{\dagger 2}\big), \qquad a_H^\dagger b_V^\dagger \mapsto \tfrac{1}{2}\big(c_H^\dagger c_V^\dagger - c_H^\dagger d_V^\dagger + d_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger\big),$$

and $a_V^\dagger b_H^\dagger \mapsto \frac{1}{2}\big(c_V^\dagger c_H^\dagger - c_V^\dagger d_H^\dagger + d_V^\dagger c_H^\dagger - d_V^\dagger d_H^\dagger\big)$. Collecting the $|HV\rangle$ and $|VH\rangle$ terms gives $\frac{1}{2}(|HV\rangle + |VH\rangle)(c_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger) + \frac{1}{2}(|VH\rangle - |HV\rangle)(c_H^\dagger d_V^\dagger - c_V^\dagger d_H^\dagger)$, which is the claimed expression. The norm of $(c_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger)|0\rangle$ is $\sqrt{2}$, so each Bell-state term has probability $(\frac{\sqrt{2}}{2\sqrt{2}})^2 = \frac{1}{4}$. Two photons in the same mode ($|HH\rangle$, $|VV\rangle$ terms) reveal both polarizations and are discarded. Two identical photons never leave through different ports with the same polarization (no $c_H d_H$ term): this is the Hong–Ou–Mandel effect. □

Corollary 7.2.5: The sifting rule produces identical bits.

Proof. Measuring the virtual qubits in a basis is equivalent to preparing the corresponding BB84 state, and this measurement commutes with Charlie's detection, so we may use the projection onto $|\Psi^\pm\rangle$. Both $|\Psi^\pm\rangle$ are anti-correlated in the rectilinear basis, so Bob always flips. In the

diagonal basis, $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|DD\rangle - |AA\rangle)$ is correlated and $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|AD\rangle - |DA\rangle)$ is anti-correlated. Equivalently: for $|DA\rangle = \frac{1}{2}(|HH\rangle - |HV\rangle + |VH\rangle - |VV\rangle)$, $\langle\Psi^+|DA\rangle = 0$ and $\langle\Psi^-|DA\rangle = -\frac{1}{\sqrt{2}}$; for $|DD\rangle$, $\langle\Psi^-|DD\rangle = 0$ (HOM effect: identical photons exit together). □

Remark 7.2.6: Charlie's announcement reveals only the parity of Alice's and Bob's bits, which is useless for Eve. Since security does not depend on Charlie's measurement, MDI QKD is immune to all detector side channels; only the sources must be characterized. The protocol is a time-reversed entanglement-based protocol (entanglement swapping). With decoy states (security via GLLP and the decoy method), the asymptotic key rate is

$$r_{\mathrm{MDI}} = Q^{1,1}_{\mathrm{rect}}\big[1 - h(e^{1,1}_{\mathrm{diag}})\big] - Q_{\mathrm{rect}} f(E_{\mathrm{rect}})\, h(E_{\mathrm{rect}}),$$

where $Q^{1,1}_{\mathrm{rect}}$ and $e^{1,1}_{\mathrm{diag}}$ are the gain in the rectilinear basis and the error rate in the diagonal basis for events in which both sent single photons, $Q_{\mathrm{rect}} = \sum_{n,m} Q^{n,m}_{\mathrm{rect}}$ and $E_{\mathrm{rect}} = \sum_{n,m} Q^{n,m}_{\mathrm{rect}} E^{n,m}_{\mathrm{rect}} / Q_{\mathrm{rect}}$ the overall gain and error rate, and $f \gtrsim 1$ the EC inefficiency. In the ideal case $r_{\mathrm{MDI}} = Q^{1,1}_{\mathrm{rect}}$.

Example 7.2.7 (Achievable distance): Let η_{ch} be the channel transmittance, $\eta_{\mathrm{d}} = 50\%$ the detector efficiency and $p_{\mathrm{dark}} = 10^{-5}$ the dark-count probability, and require a noise-to-signal ratio $N/S \leq 10\%$.

(i) *Point-to-point:* $S = \eta_{\mathrm{ch}}\eta_{\mathrm{d}}$, $N = p_{\mathrm{dark}}$, so $\eta_{\mathrm{ch}} \geq p_{\mathrm{dark}}/(0.1\,\eta_{\mathrm{d}}) = 2 \times 10^{-4}$, i.e., $L \leq \frac{10}{0.2}\log_{10}(5 \times 10^3) \approx 185\,\mathrm{km}$.

(ii) *MDI:* each half of the fiber has transmittance $\sqrt{\eta_{\mathrm{ch}}}$ and both photons must be detected, so $S = \eta_{\mathrm{ch}}\eta_{\mathrm{d}}^2$, while $N = p_{\mathrm{dark}}^2 + 2p_{\mathrm{dark}}\sqrt{\eta_{\mathrm{ch}}}\eta_{\mathrm{d}}$ (two dark counts, or one dark count and one genuine click). Solving $N/S = 0.1$ gives $\eta_{\mathrm{ch}} \approx 1.68 \times 10^{-7}$, i.e., $L \approx 339\,\mathrm{km}$.

The intermediate relay handles noise better than a single long channel.

Twin-field QKD

Remark 7.2.8: In *twin-field QKD* (Lucamarini et al. 2018), Alice and Bob generate phase-randomized weak coherent pulses with random phases $\phi_a, \phi_b \in [0, 2\pi)$; the phase interval is divided into M slices $\Delta_k = 2\pi k/M$. A secret bit and basis are encoded in an additional phase. Charlie interferes the pulses on a beam splitter with single-photon detectors at the outputs. After the announcement of the detection, Alice and Bob reveal their phase slices and bases and keep only rounds with matching slices ("twin" fields). The click reveals only the parity of the bits. A large M reduces the QBER, but matching slices occur with probability $1/M$; $M_{\mathrm{opt}} = 16$ in the original analysis.

Since a key event requires only a *single* photon, from either Alice or Bob, which travels only half the distance, the rate scales as $O(\sqrt{\eta})$ instead of $O(\eta)$, so twin-field QKD can beat the repeaterless bound (Theorem 7.1.1). Security against general attacks with $O(\sqrt{\eta})$ scaling was shown for modified versions (Tamaki et al.; Ma et al.; Curty et al., with phase pre-selection). Twin-field QKD inherits measurement-device independence and requires phase stabilization of two independent lasers over long distances.

7.3 Continuous-variable QKD

Gaussian quantum information

Definition 7.3.1: For N bosonic modes with annihilation operators a_j, $[a_j, a_k^\dagger] = \delta_{jk}$, define the quadratures $\hat{q}_j = a_j + a_j^\dagger$, $\hat{p}_j = i(a_j^\dagger - a_j)$ (shot-noise units) and $\hat{r} = (\hat{q}_1, \hat{p}_1, \ldots, \hat{q}_N, \hat{p}_N)$. Then

$[\hat r_j, \hat r_k] = 2i\Omega_{jk}$ with

$$\Omega = \bigoplus_{j=1}^{N} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The *mean* and *covariance matrix* of ρ are $\bar r = \langle \hat r \rangle$ and $V_{jk} = \frac{1}{2}\langle \{\hat r_j - \bar r_j, \hat r_k - \bar r_k\}\rangle$; the diagonal entries are the variances of the quadratures.

Lemma 7.3.2: $[\hat q, \hat p] = 2i$, and consequently $\Delta\hat q\,\Delta\hat p \geq 1$ (the shot noise). A valid covariance matrix satisfies $V + i\Omega \geq 0$.

Proof. $[a+a^\dagger, i(a^\dagger - a)] = i\big([a,a^\dagger] - [a^\dagger, a]\big) = 2i$. The Robertson inequality $\Delta A\,\Delta B \geq \frac{1}{2}|\langle [A,B]\rangle|$ gives $\Delta q\,\Delta p \geq 1$. For the matrix inequality, $\langle (\sum_j \bar z_j \Delta\hat r_j)(\sum_k z_k \Delta\hat r_k)\rangle \geq 0$ for all $z \in \mathbb{C}^{2N}$, and $\Delta\hat r_j \Delta\hat r_k = \frac{1}{2}\{\cdot,\cdot\} + \frac{1}{2}[\cdot,\cdot]$. □

Definition 7.3.3: The *Wigner function* of ρ is a quasi-probability distribution on phase space $\mathbb{R}^{2N}$ whose marginals are the quadrature distributions; it can be negative for non-classical states (e.g., Fock states $|k\rangle$, $k \geq 1$). A state is *Gaussian* if its Wigner function is Gaussian,

$$W(r) = \frac{\exp\big(-\frac{1}{2}(r-\bar r)^T V^{-1}(r-\bar r)\big)}{(2\pi)^N \sqrt{\det V}},$$

so it is fully determined by $\bar r$ and V. Gaussian channels map Gaussian states to Gaussian states.

Lemma 7.3.4: Let $D(\alpha) = e^{\alpha a^\dagger - \bar\alpha a}$ and $|\alpha\rangle = D(\alpha)|0\rangle = e^{-|\alpha|^2/2}\sum_k \frac{\alpha^k}{\sqrt{k!}}|k\rangle$.

(i) $D(\alpha)^\dagger a D(\alpha) = a + \alpha$, hence $D(\alpha)^\dagger \hat q D(\alpha) = \hat q + 2\,\mathrm{Re}\,\alpha$ and $D(\alpha)^\dagger \hat p D(\alpha) = \hat p + 2\,\mathrm{Im}\,\alpha$.

(ii) The vacuum has $\bar r = 0$ and $V = \mathbb{I}_2$; the coherent state $|\alpha\rangle$ has $\bar r = (2\,\mathrm{Re}\,\alpha, 2\,\mathrm{Im}\,\alpha)$ and $V = \mathbb{I}_2$.

(iii) $|\langle\beta|\alpha\rangle|^2 = e^{-|\alpha-\beta|^2}$; in particular, coherent states are never orthogonal.

Proof. (i) By the Baker–Campbell–Hausdorff formula, $e^{-X} a e^X = a + [a, X]$ for $X = \alpha a^\dagger - \bar\alpha a$, since $[a, X] = \alpha$ is a scalar. Then $\hat q \mapsto a + \alpha + a^\dagger + \bar\alpha$ and $\hat p \mapsto i(a^\dagger + \bar\alpha - a - \alpha)$. (ii) $\langle 0|a|0\rangle = 0$ gives $\bar r = 0$. With $a|0\rangle = 0$: $\langle 0|\hat q^2|0\rangle = \langle 0|aa^\dagger|0\rangle = 1$, $\langle 0|\hat p^2|0\rangle = 1$, and $\frac{1}{2}\langle 0|\{\hat q, \hat p\}|0\rangle = \frac{i}{2}\langle 0|(aa^\dagger - aa^\dagger)|0\rangle = 0$. For $|\alpha\rangle$, (i) shifts the mean and leaves the fluctuations $\Delta\hat r$ unchanged. (iii) $\langle\beta|\alpha\rangle = e^{-(|\alpha|^2+|\beta|^2)/2}\sum_k \frac{(\bar\beta\alpha)^k}{k!} = e^{-(|\alpha|^2+|\beta|^2)/2 + \bar\beta\alpha}$, whose modulus squared is $e^{-|\alpha|^2-|\beta|^2+2\,\mathrm{Re}(\bar\beta\alpha)} = e^{-|\alpha-\beta|^2}$. □

Example 7.3.5: *Squeezed states* $|\alpha, s\rangle = D(\alpha)S(s)|0\rangle$ have $V = \mathrm{diag}(e^{-2s}, e^{2s})$. The *two-mode squeezed vacuum* (EPR state) has

$$V = \begin{pmatrix} \nu\mathbb{I}_2 & \sqrt{\nu^2-1}\,Z \\ \sqrt{\nu^2-1}\,Z & \nu\mathbb{I}_2 \end{pmatrix}, \qquad Z = \mathrm{diag}(1,-1),\ \nu \geq 1;$$

its reduced states are thermal with variance ν.

Remark 7.3.6: *Homodyne detection* measures one quadrature: the signal interferes with a strong local oscillator on a 50:50 beam splitter and the difference of the two photocurrents is proportional to $\hat q$ (or $\hat p$ after a $\pi/2$ phase shift); its outcome distribution is the marginal $P(q) = \int W(q,p)\,dp$. *Heterodyne detection* measures both quadratures at the price of one extra unit of noise and needs no random basis choice. The relevant channel is the thermal-loss channel with transmittance T and excess noise ξ, which maps a quadrature variance V to $T(V + \chi)$ with $\chi = (1-T)/T + \xi$. Coherent states in CV QKD carry information in the field quadratures, unlike weak coherent pulses in DV QKD, which approximate single photons.

CV QKD protocols

Definition 7.3.7 (GG02 protocol): The protocol consists of the following steps.

(i) Alice prepares coherent states $|\alpha_1\rangle, \dots, |\alpha_N\rangle$ whose quadrature means are i.i.d. Gaussian with mean 0 and variance V_A, and sends them to Bob.

(ii) Bob measures a randomly chosen quadrature (homodyne) and announces his choice, or both quadratures (heterodyne). Alice keeps the corresponding values x; Bob holds y ($n = N$ or $2N$ real numbers).

(iii) *Reverse reconciliation*: Bob's string is the reference, and Alice corrects her data to match it.

(iv) *Parameter estimation*: they estimate the covariance matrix (i.e., T and ξ) from a disclosed subset (often after error correction).

(v) *Privacy amplification* with two-universal hashing.

Remark 7.3.8: Variants differ in the states (coherent or squeezed), the modulation (Gaussian or discrete, e.g., $|ae^{2\pi ik/4}\rangle$), the detection (homodyne or heterodyne) and the reconciliation (direct or reverse, one- or two-way). Prepare-and-measure and entanglement-based versions are equivalent: Gaussian modulation with variance V_A corresponds to heterodyning one half of an EPR state with $\nu = V_A + 1$.

Lemma 7.3.9: For GG02 with homodyne detection over a thermal-loss channel, with $V := V_A + 1$,

$$I(A:B) = \frac{1}{2}\log\frac{V+\chi}{1+\chi} = \frac{1}{2}\log\Big(1 + \frac{TV_A}{1+T\xi}\Big).$$

Proof. Bob's quadrature is $y = \sqrt{T}\,x + z$, where $x \sim \mathcal{N}(0, V_A)$ is Alice's displacement and z is Gaussian noise independent of x with variance $T\cdot 1 + (1-T)\cdot 1 + T\xi = 1 + T\xi$ (attenuated shot noise, vacuum entering at the loss, excess noise). Hence $V_B = TV_A + 1 + T\xi = T(V+\chi)$ and $V_{B|A} = 1 + T\xi = T(1+\chi)$. For Gaussian variables, $I(A:B) = \frac{1}{2}\log(V_B/V_{B|A})$. □

Security

Theorem 7.3.1: Optimality of Gaussian attacks

For collective attacks and Gaussian-modulated protocols, the asymptotic key rate with reverse reconciliation is

$$r_\infty^{\text{coll}} = \beta I(A:B) - \chi(B:E),$$

where $\beta < 1$ is the reconciliation efficiency, and $\chi(B:E)_\rho \le \chi(B:E)_{\rho_G}$, where ρ_G is the Gaussian state with the same covariance matrix as ρ. In particular, Gaussian attacks are optimal. For direct reconciliation, $\chi(B:E)$ is replaced by $\chi(A:E)$.

Proof. See García-Patrón–Cerf (2006) and Navascués–Grosshans–Acín (2006). □

Remark 7.3.10: The Holevo quantity of a Gaussian state is computed from the symplectic eigenvalues ν_i of the covariance matrices via $H(\rho) = \sum_i g(\nu_i)$, $g(x) = \frac{x+1}{2}\log\frac{x+1}{2} - \frac{x-1}{2}\log\frac{x-1}{2}$. With direct reconciliation over a pure-loss channel, Eve's reflected beam has transmittance $1-T$, so she gets better data than Bob as soon as $T < \frac{1}{2}$: the rate is positive only for losses below 3 dB ("3 dB limit"). Reverse reconciliation removes this limit, because Eve's data are then compared with Bob's, which already contain the loss noise.

Remark 7.3.11: Security against general attacks: for squeezed states, by discretizing the quadratures and using entropic uncertainty relations (Furrer et al.); for coherent states, by a Gaussian de Finetti theorem (Renner–Cirac asymptotically, Leverrier et al. for finite keys). For discrete modulation, Gaussian optimality and the de Finetti symmetry fail; current proofs use semidefinite programs to bound the min-entropy numerically, or map the protocol to qubits.

Remark 7.3.12: *Advantages*: CV QKD uses exactly the states and measurements of the theory, standard telecom components (lasers, balanced homodyne detectors at room temperature), achieves high rates at metropolitan distances, and is compatible with wavelength-division multiplexing. *Challenges*: infinite-dimensional Hilbert spaces and unbounded operators in the proofs, the covariance matrix as a continuous and unbounded figure of merit, efficient reconciliation at low signal-to-noise ratio, and lower noise robustness (no-click events cannot be discarded), which requires very large block sizes.

7.4 Experimental status

Remark 7.4.1: Representative milestones (as of the time of the source text):

(i) decoy-state BB84: 10 Mbit/s at short distance (Yuan et al.), and fiber distances above 400 km with very low rates (Boaron et al. 2018; see the table below);

(ii) satellite-to-ground decoy-state QKD over up to 1200 km with the satellite *Micius* (2017, about 1 kbit/s), and Micius as a trusted relay between ground stations about 7600 km apart (Xinglong–Graz, including a video call encrypted with a one-time pad);

(iii) entanglement-based QKD: a fully connected eight-user city network (2019) and keys over 1120 km between ground stations via Micius without trusted relays (2020, 0.12 bit/s);

(iv) MDI QKD over 404 km of fiber (2016) and twin-field QKD over 509 km (2020);

(v) CV QKD over 202.81 km of fiber (2020).

Distance [km]	QBER [%]	Secret key rate [bit/s]
251.7	0.5	4.9×10^3
302.1	0.4	0.79×10^3
354.5	0.7	62
404.9	1.0	6.5
421.1	2.1	0.25

The rate decreases exponentially with distance until detector noise dominates. Long-distance networks without trusted nodes require quantum repeaters; device-independent QKD requires loophole-free Bell tests with high enough violation and low enough QBER.

A Basic probability theory

A.1 Definitions

Definition A.1.1: A (finite) *probability space* is a finite set Ω with $p : \Omega \to [0,1]$, $\sum_\omega p(\omega) = 1$; events are subsets $\mathcal{A} \subset \Omega$ with $\Pr[\mathcal{A}] = \sum_{\omega \in \mathcal{A}} p(\omega)$. A *random variable* is a map $X : \Omega \to \mathcal{X}$, with distribution $p_X(x) = \Pr[X = x]$. For random variables X, Y the joint distribution is $p_{XY}(x,y) = \Pr[X = x, Y = y]$ and the marginal is $p_X(x) = \sum_y p_{XY}(x,y)$.

Definition A.1.2: For $\Pr[\mathcal{B}] > 0$, the *conditional probability* is $\Pr[\mathcal{A}|\mathcal{B}] := \Pr[\mathcal{A} \cap \mathcal{B}] / \Pr[\mathcal{B}]$, and $p_{X|Y}(x|y) := p_{XY}(x,y)/p_Y(y)$. Events are *independent* if $\Pr[\mathcal{A} \cap \mathcal{B}] = \Pr[\mathcal{A}] \Pr[\mathcal{B}]$; random variables X, Y are independent if $p_{XY}(x,y) = p_X(x) p_Y(y)$ for all x, y, equivalently $p_{Y|X}(y|x) = p_Y(y)$ for all $y \in \mathcal{Y}$ (and all x with $p_X(x) > 0$).

Lemma A.1.3 (Bayes' rule, total probability): For events $\mathcal{A}, \mathcal{B}$ with positive probability,

$$\Pr[\mathcal{A}|\mathcal{B}]\Pr[\mathcal{B}] = \Pr[\mathcal{B}|\mathcal{A}]\Pr[\mathcal{A}].$$

In particular, $\Pr[\mathcal{A}|\mathcal{B}] \le \Pr[\mathcal{A}]/\Pr[\mathcal{B}]$. If $\{\mathcal{B}_i\}$ is a partition of Ω, then $\Pr[\mathcal{A}] = \sum_i \Pr[\mathcal{A}|\mathcal{B}_i]\Pr[\mathcal{B}_i]$.

Proof. Both sides of Bayes' rule equal $\Pr[\mathcal{A}\cap\mathcal{B}]$, and $\Pr[\mathcal{A}\cap\mathcal{B}] \le \Pr[\mathcal{A}]$. The last statement follows from $\Pr[\mathcal{A}] = \sum_i \Pr[\mathcal{A}\cap\mathcal{B}_i]$. □

Lemma A.1.4 (Union bound): $\Pr\left[\bigcup_i \mathcal{A}_i\right] \le \sum_i \Pr[\mathcal{A}_i]$.

Proof. Every ω in the union is counted at least once on the right-hand side. □

Theorem A.1.1: Hoeffding

Let $X_1, \dots, X_n$ be independent with values in $[0,1]$ and $\bar{X} = \frac{1}{n}\sum_i X_i$. Then $\Pr[\bar{X} \ge \mathbb{E}\bar{X} + t] \le e^{-2nt^2}$ for $t \ge 0$.

Proof. For $s > 0$, Markov's inequality gives $\Pr[\bar{X} - \mathbb{E}\bar{X} \ge t] \le e^{-st}\prod_i \mathbb{E}\, e^{s(X_i - \mathbb{E}X_i)/n}$. Hoeffding's lemma, $\mathbb{E}e^{\lambda(Y-\mathbb{E}Y)} \le e^{\lambda^2/8}$ for $Y \in [0,1]$ (by convexity of $e^{\lambda y}$ and an estimate of the resulting log-moment function), gives the bound $e^{-st+s^2/(8n)}$; optimize at $s = 4nt$. □

B Calderbank–Shor–Steane codes

B.1 Classical linear codes

Definition B.1.1: An $[n,k]$ *linear code* is a k-dimensional subspace $C \subset \mathbb{F}_2^n$. A *generator matrix* $G \in \mathbb{F}_2^{n\times k}$ has columns forming a basis of C, so that $C = \{Gu \mid u \in \mathbb{F}_2^k\}$; a *parity-check matrix* $H \in \mathbb{F}_2^{(n-k)\times n}$ has linearly independent rows and satisfies $C = \ker H$. The *syndrome* of $y \in \mathbb{F}_2^n$ is Hy. The *distance* d of C is the minimal Hamming weight of a non-zero codeword; we write $[n,k,d]$.

Example B.1.2 (Repetition code): The $[3,1,3]$ code $C = \{000, 111\}$ has $G = (1\,1\,1)^T$ and

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The single-bit errors e_1, e_2, e_3 have syndromes $He_1 = (1\,0)^T$, $He_2 = (1\,1)^T$, $He_3 = (0\,1)^T$, all distinct and non-zero, so every single error can be located and corrected.

Lemma B.1.3: A code of distance $d \ge 2t+1$ corrects up to t errors: if $y = x + e$ with $x \in C$ and $|e| \le t$, then e is the unique vector of weight $\le t$ with syndrome $Hy = He$.

Proof. $Hy = Hx + He = He$. If $e' \ne e$ with $|e'| \le t$ had the same syndrome, then $e - e' \in C \setminus \{0\}$ with $|e - e'| \le 2t < d$, a contradiction. □

Definition B.1.4: The *dual* of C is $C^\perp := \{y \in \mathbb{F}_2^n \mid x \cdot y = 0 \text{ for all } x \in C\}$, an $[n, n-k]$ code with generator matrix H^T and parity-check matrix G^T. For a subcode $C_2 \subset C_1$, the *cosets* $x + C_2$, $x \in C_1$, partition C_1 into $2^{k_1-k_2}$ classes, and $x + C_2 = x' + C_2$ if and only if $x - x' \in C_2$.

Lemma B.1.5: For a linear code C and $u \in \mathbb{F}_2^n$,

$$\sum_{y\in C}(-1)^{u\cdot y} = \begin{cases} |C| & u \in C^\perp, \\ 0 & u \notin C^\perp. \end{cases}$$

Proof. If $u \notin C^\perp$, there is $y_0 \in C$ with $u \cdot y_0 = 1$, and $y \mapsto y + y_0$ is a bijection of C flipping every sign, so the sum equals its negative. □

Example B.1.6: The $[7, 4, 3]$ Hamming code C has as parity-check matrix the 3×7 matrix whose columns are all non-zero vectors of $\mathbb{F}_2^3$; the syndrome of a single error is the binary representation of its position, so it corrects one error. Its dual is a $[7, 3, 4]$ code with $C^\perp \subset C$.

B.2 Quantum error correction

Definition B.2.1: Let $C_2 \subset C_1$ be $[n, k_2]$ and $[n, k_1]$ codes such that C_1 and $C_2^\perp$ both correct t errors. For $x \in C_1$ define

$$|x + C_2\rangle := \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} |x + y\rangle.$$

This state depends only on the coset $x + C_2$. The quantum code $\mathrm{CSS}(C_1, C_2)$ is the vector space spanned by $\{|x + C_2\rangle\}_{x \in C_1}$. More generally, for $v, w \in \mathbb{F}_2^n$,

$$|\xi_{x,v,w}\rangle := \frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{v \cdot y} |x + y + w\rangle$$

spans the shifted code $\mathrm{CSS}_{v,w}(C_1, C_2)$, which corrects the same errors.

Lemma B.2.2: When x ranges over representatives of C_1/C_2, v over representatives of $\mathbb{F}_2^n/C_2^\perp$, and w over representatives of $\mathbb{F}_2^n/C_1$, the states $|\xi_{x,v,w}\rangle$ form an orthonormal basis of $(\mathbb{C}^2)^{\otimes n}$ with real coefficients. Consequently,

$$|\Phi^+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{j \in \mathbb{F}_2^n} |j\rangle|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{x,v,w} |\xi_{x,v,w}\rangle|\xi_{x,v,w}\rangle.$$

Proof. *Counting:* there are $2^{k_1-k_2} \cdot 2^{k_2} \cdot 2^{n-k_1} = 2^n$ triples. *Orthonormality:* the support of $|\xi_{x,v,w}\rangle$ is the coset $x + w + C_2$ of $\mathbb{F}_2^n/C_2$. If $(x, w) \neq (x', w')$ as classes, then $x + w + C_2 \neq x' + w' + C_2$ (since $x + w - x' - w' \in C_2 \subset C_1$ forces $w = w'$ in $\mathbb{F}_2^n/C_1$ and then $x = x'$ in C_1/C_2), so the supports are disjoint. For equal (x, w), $\langle \xi_{x,v',w} | \xi_{x,v,w} \rangle = \frac{1}{|C_2|} \sum_{y \in C_2} (-1)^{(v-v') \cdot y}$, which is 1 if $v - v' \in C_2^\perp$ and 0 otherwise by Lemma B.1.5. The coefficients are $\pm|C_2|^{-1/2}$, hence real. For the last identity: for any orthonormal basis $\{|e_\alpha\rangle\}$ with real coefficients, $\sum_\alpha |e_\alpha\rangle|e_\alpha\rangle = \sum_{j,k} \big(\sum_\alpha \langle j|e_\alpha\rangle\langle k|e_\alpha\rangle\big) |j\rangle|k\rangle = \sum_j |j\rangle|j\rangle$ by completeness. □

Theorem B.2.1: CSS codes

$\mathrm{CSS}(C_1, C_2)$ is an $[[n, k_1 - k_2]]$ quantum code which corrects arbitrary errors on up to t qubits. Bit-flip errors are corrected with C_1, phase-flip errors with $C_2^\perp$.

Proof. By Lemma B.2.2 (with $v = w = 0$), the $2^{k_1-k_2}$ states $|x + C_2\rangle$ are orthonormal. Every error on t qubits is a linear combination of Pauli errors $X^{e_1} Z^{e_2}$ with $|e_1|, |e_2| \leq t$, so it suffices to correct these. Such an error maps $|x + C_2\rangle$ to $\frac{1}{\sqrt{|C_2|}} \sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y + e_1\rangle$. *Bit flips:* computing the syndrome $H_1(x + y + e_1) = H_1 e_1$ into an ancilla (without disturbing the superposition, since it is the same for all terms) identifies e_1, which is undone by X^{e_1}. *Phase flips:* applying $H^{\otimes n}$ maps $\sum_{y \in C_2} (-1)^{(x+y) \cdot e_2} |x + y\rangle$ to $\frac{1}{\sqrt{2^n |C_2|}} \sum_z \sum_{y \in C_2} (-1)^{(x+y) \cdot (e_2 + z)} |z\rangle = \sqrt{\frac{|C_2|}{2^n}} \sum_{z' \in C_2^\perp} (-1)^{x \cdot z'} |z' + e_2\rangle$ by Lemma B.1.5 (with $z' = z + e_2$). This is a bit-flip error e_2 on a superposition of codewords of $C_2^\perp$, which is corrected with the parity-check matrix of $C_2^\perp$; applying $H^{\otimes n}$ again restores the state. Since the syndrome measurements project onto a definite Pauli error, superpositions of errors are corrected as well. □

Example B.2.3 (Steane code): Taking C_1 the $[7, 4, 3]$ Hamming code and $C_2 = C_1^{\perp}$ (a $[7, 3, 4]$ code, so $C_2^{\perp} = C_1$ corrects one error) gives the $[[7, 1]]$ Steane code, which corrects one arbitrary qubit error.